

From multiplicity of infection to force of infection for sparsely sampled *Plasmodium falciparum* populations at high transmission

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eLife Assessment

The ability to estimate the force of infection for *Plasmodium falciparum* from other more directly measurable epidemiological quantities would contribute to malaria epidemiology. The authors propose a method to accomplish this using genetic data from the var genes of the Pf genome and novel applications of existing methods from queueing theory. After revising the manuscript, this is a **useful** contribution to the field, and the authors provide **solid** evidence to support it.

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Abstract

High multiplicity of infection or MOI, the number of genetically distinct parasite strains co-infecting a single human host, characterizes infectious diseases including *falciparum* malaria at high transmission. This high MOI accompanies high asymptomatic *Plasmodium falciparum* prevalence despite high exposure, creating a large transmission reservoir challenging intervention. High MOI and asymptomatic prevalence are enabled by immune evasion of the parasite achieved via vast antigenic diversity. Force of infection or FOI, the number of new infections acquired by an individual host over a given time interval, is the dynamic sister quantity of MOI, and a key epidemiological parameter for monitoring antimalarial interventions and assessing vaccine or drug efficacy in clinical trials. FOI remains difficult, expensive, and labor-intensive to accurately measure, especially in high-transmission regions, whether directly via cohort studies or indirectly via the fitting of epidemiological models to repeated cross-sectional surveys. We propose here the application of queueing theory to obtain FOI from MOI, in the form of either a two-moment approximation method or Little's Law. We illustrate these two methods with MOI estimates obtained under sparse sampling schemes with the “varcoding” approach. The two methods use infection duration data from naive malaria therapy patients with neurosyphilis. Consequently, they are suitable for FOI inference in subpopulations with a similar immune profile and the highest

vulnerability, for example, infants or toddlers. Both methods are evaluated with simulation output from a stochastic agent-based model, and are applied to an interrupted time-series study from Bongo District in northern Ghana before and immediately after a three-round transient indoor residual spraying (IRS) intervention. The sampling of the simulation output incorporates limitations representative of those encountered in the collection of field data, including under-sampling of *var* genes, missing data, and antimalarial drug treatment. We address these limitations in MOI estimates with a Bayesian framework and an imputation bootstrap approach. Both methods yield good and replicable FOI estimates across various simulated scenarios. Applying these methods to the subpopulation of children aged 1-5 years in Ghana field surveys shows over a 70% reduction in annual FOI immediately post-intervention. The proposed methods should be applicable to geographical locations lacking cohort or cross-sectional studies with regular and frequent sampling but having single-time-point surveys under sparse sampling schemes, and for MOI estimates obtained in different ways. They should also be relevant to other pathogens whose immune evasion strategies are based on large antigenic variation resulting in high MOI.

Introduction

Despite substantial intervention efforts, *falciparum* malaria in high-transmission regions remains a major public health concern, causing mortality among young children and considerable economic burden, particularly in sub-Saharan Africa (World Health Organization, 2023 [↗](#)). It is thus still important to robustly evaluate the effects of intervention efforts in these regions, including on transmission intensity. Force of infection (FOI), defined as the number of new *Plasmodium falciparum* infections acquired by an individual host over a given time interval, is a key metric reflecting the risk of infection and clinical episodes (Mueller et al., 2012 [↗](#)). While other metrics may describe the relationship between transmission intensity and the burden of malaria illness on global or continental scales (Carneiro et al., 2010 [↗](#); Beier et al., 1994 [↗](#)), FOI can relate local variation in malaria burden to transmission (Mueller et al., 2012 [↗](#)). Despite being a key epidemiological parameter for malaria surveillance, FOI is difficult, expensive, and labor-intensive to measure accurately, whether directly through cohort studies or indirectly through the fitting of epidemiological models. As molecular tools for parasite genomics become more readily available, they are contributing to new approaches. In particular, molecular advances are providing a basis for estimating a sister “static” quantity, the multiplicity of infection (MOI), defined as the number of genetically distinct parasite strains co-infecting a single human host (Zhong et al., 2018 [↗](#); Chang et al., 2017 [↗](#); Ruybal-Pesántez et al., 2022 [↗](#); Tiedje et al., 2022 [↗](#); Labbé et al., 2023 [↗](#)). We can therefore ask whether we can go further and on the basis of MOI obtain the dynamical, rate, quantity of force of infection?

Early efforts to directly measure FOI included clearing infections and observing time to reinfection (Macdonald, 1950 [↗](#); Pull and Grab, 1974 [↗](#); Msuya and Curtis, 1991 [↗](#); Alonso et al., 2004 [↗](#)). Molecular approaches now enable the genotyping of individual parasite infections (Mueller et al., 2012 [↗](#); Hofmann et al., 2017 [↗](#)), but differentiating new infections from the temporary absence of an old infection in the peripheral blood and its subsequent re-emergence (Daubersies et al., 1996 [↗](#)) remains challenging due to the low resolution of polymorphic markers and the complex within-host dynamics of malaria infection. Determining molecular FOI remains challenging, labor-intensive, and costly, requiring close long-term monitoring and genotyping of a large cohort.

Alternatives to direct measurements involve cross-sectional surveys with FOI estimated by fitting simple epidemiological models (Bekessy et al., 1976 [↗](#); Muench, 1959 [↗](#); Smith and Vounatsou, 2003 [↗](#); Felger et al., 2012 [↗](#); Mugenyi et al., 2017 [↗](#)). These model-fitting procedures require empirical data sampled regularly and frequently, such as age-stratified large cohorts sampled six times a year, to account for FOI and infection duration heterogeneity (Laishram et al., 2012 [↗](#);

Simpson et al., 2002 [↗](#); Langhorne et al., 2008 [↗](#); Childs and Buckee, 2014 [↗](#); Chang et al., 2016 [↗](#); Ashley and White, 2014 [↗](#)) influenced by the interplay between host immunity and antigen composition of infections (Piper et al., 1999 [↗](#); Molineaux et al., 2002 [↗](#); Doolan et al., 2009 [↗](#); Barry et al., 2007 [↗](#)). Additionally, they may face identifiability issues with model parameters (Mugenyi et al., 2017 [↗](#)). Hence, these indirect measurements share limitations with direct ones.

Due to the described challenges, FOI has not become a readily available epidemiological quantity across geographical locations and times. In contrast, various approaches have been proposed to estimate MOI from clinical samples using size-polymorphic antigenic markers, microsatellites, and panels of biallelic single nucleotide polymorphisms (SNPs) (Felger et al., 1994 [↗](#); Konaté et al., 1999 [↗](#); Anderson et al., 2000 [↗](#); Daniels et al., 2008 [↗](#); Chang et al., 2017 [↗](#)). Because *Plasmodium* parasites reproduce asexually during haploid stages within human hosts (Guttery et al., 2012 [↗](#)), polymorphic genotypes indicate multiclonal infection. As methods for disentangling the molecular complexity of natural parasite infections across various transmission settings emerge and mature, MOI becomes more commonly and easily surveyed across space and time. Although MOI remains one of the most frequently used genetic metrics of parasite transmission (Arnot, 1998 [↗](#); Sondo et al., 2020 [↗](#)), it is by definition a number and not a rate.

A natural correlation exists however between MOI and FOI, mediated by infection duration, offering an opportunity to convert MOI into FOI. This conversion has been challenging due to MOI estimates often being obtained from sparse sampling schemes, like single-time-point surveys at the end of wet (high-transmission) and dry (low-transmission) seasons (Tiedje et al., 2022 [↗](#); Abukari et al., 2019 [↗](#)). Although these MOI estimates have been useful (Earland et al., 2019a [↗](#); Lee et al., 2006 [↗](#)), the sparse sampling scheme has limited their translation into a transmission rate.

In this work, we propose using these MOI estimates for FOI inference with two mathematical modeling frameworks based on queuing theory (Choi et al., 2005 [↗](#); Little and Graves, 2008 [↗](#)). The two methods require infection duration values, for which we relied on data from naive malaria therapy patients with neurosyphilis (Collins and Jeffery, 1999 [↗](#); Maire et al., 2006 [↗](#)). Consequently, our approach is suited for FOI inference in subpopulations with a similar immune profile and the highest vulnerability, for example, infants or toddlers. We evaluate them through numerical simulation of an extended stochastic agent-based model (ABM) (He et al., 2018 [↗](#); Zhan et al., 2024b [↗](#)) for both closed and open systems with constant or seasonal transmission. We consider both homogeneous and heterogeneous transmission, with the latter including a high-risk group of hosts that receives the majority of the infectious bites. We also examine different statistical distributions for the times between local transmission events. We incorporate limitations representative of those encountered in the collection of field data into the sampling of simulation output, including undersampling of *var* genes, missing data, and antimalarial drug treatment. We address these limitations in MOI estimates with a Bayesian framework and an imputation bootstrap approach. Both methods provide good and replicable FOI estimates across simulated scenarios. After validating with simulations, we apply the two methods to empirical data from an interrupted time-series study in Bongo District, northern Ghana, that involved a three-round transient indoor residual spraying (IRS) intervention (Tiedje et al., 2022 [↗](#), 2023 [↗](#)). We focus on children aged 1-5 years whose immune profiles are closer to naive patients than the rest of the population, an aspect we discuss later. We then explore the relationship between FOI and another commonly used surrogate for transmission intensity, the entomological inoculation rate or EIR, defined as the number of infectious bites received by an individual over a given time period (Shaukat et al., 2010 [↗](#)). This relationship underscores the challenges of relating transmission to malaria burden at local scales and achieving substantial reductions in transmission in high-transmission regions.

Results

The Bayesian formulation of the *var*coding method, combined with the bootstrap imputation approach, effectively addresses sampling limitations often encountered in collecting field data for MOI estimates

Because our FOI inference relies on MOI estimates, we first investigate the impact of various sampling limitations on these estimates. We use the Bayesian formulation of the *var*coding method and the bootstrap imputation approach (Materials and Methods) to address sampling limitations often encountered in collecting field data for MOI estimates: under-sampling of *var* genes, missing data, antimalarial drug treatment, and their combination. For this investigation, we utilize simulation output from an agent-based model (ABM) of malaria transmission with known true MOI values (He et al., 2018 [DOI](#); Zhan et al., 2024b [DOI](#)). Key assumptions and processes for the ABM, and the experimental design for simulation output are summarized in Appendix 1-Simulation data, with illustrations in Appendix 1-Figure 5-6.

Our results indicate that MOI estimates obtained using the Bayesian formulation of the *var*coding method and the bootstrap imputation approach closely match true MOI values in most cases. To assess the difference between estimated and true MOI distributions, we use the Cramer-von Mises and Anderson-Darling tests. The Cramer-von Mises test quantifies the sum of the squared differences between cumulative distribution functions, while the Anderson-Darling test, a modification of the former, gives more weight to the tails of distributions. Most p-values are non-significant (> 0.05), indicating insufficient evidence to conclude that the estimated and true distributions differ, with few exceptions in pre-IRS or low-coverage IRS scenarios. The Bayesian formulation of *var*coding tends to underestimate MOI because it does not account for the limited overlap between co-infecting strains, which reduces the number of *var* genes detected per individual. This underestimation bias can be more pronounced in certain high-transmission situations where many hosts have a high true MOI, such as the aforementioned exceptions in pre-IRS or low-coverage IRS scenarios. Consequently, this underestimation in MOI leads to an underestimation of FOI estimates, as described in the next section. Detailed results of both tests are provided in the supplementary file 1-MOImethodsPerformance.xlsx.

The distributions of MOI estimates across different surveys in Ghana and from the simulated outputs are not Poisson-distributed (Appendix 1-Test of deviation from Poisson homogeneity in MOI estimates, supplementary file 2-deviationFromPoissonTest.xlsx) (Potthoff and Whittinghill, 1966 [DOI](#); Lloyd-Smith et al., 2005 [DOI](#)). This deviation suggests that either the arrival of infections deviates from a homogeneous Poisson process, or some latent heterogeneity causes over-dispersion in infection duration. This deviation makes converting MOI to FOI more challenging, but the proposed methods are flexible and applicable (Materials and Methods-Infering FOI from MOI estimates).

The Two-Moment Approximation and Little's Law methods give good and replicable estimates for FOI across various simulated scenarios

We begin with a homogeneous exposure risk scenario for seasonal transmission in a closed system (Appendix 1-Figure 6A-C, and Appendix 1-Simulation data). The times between local transmission events follow a Gamma distribution (Appendix 1-Figure 6C). We infer FOI using true MOI values

and MOI estimates obtained via the Bayesian formulation or the bootstrap imputation approach, accounting for one or all sampling limitations. Details on deriving the confidence intervals are provided in Appendix 1-Confidence intervals for FOI inference.

Across pre-IRS and three IRS coverage levels, the 95% confidence intervals and the bootstrap distributions of FOI estimates are narrow. FOI estimates based on true MOI values, and those corrected for missing data and antimalarial treatment, closely match true FOI values (**Figure 1** [↗](#)). FOI estimates based on MOI estimates corrected for under-sampling of *var* genes and those corrected for all sampling limitations show slight underestimations (**Figure 1** [↗](#)), due to the underestimation of MOI, as described in the previous section.

To quantify the difference between inferred and true FOI values, we check if the true FOI lies within the bootstrap distribution and calculate the relative deviation, which is defined as the true FOI value minus the median of the bootstrap distribution for the estimate, normalized by the true FOI value. These details are in supplementary file 3-FOImethodsPerformance.xlsx.

We continue with a heterogeneous exposure risk scenario in which a high-risk group ($\frac{2}{3}$ of the population) receives approximately 94% of bites, while a low-risk group ($\frac{1}{3}$ of the population) receives the remainder (Appendix 1-Figure 6C). Transmission is seasonal and the system is semi-open (Appendix 1-Figure 6A-B, and Appendix 1-Simulation data). The times between local transmission events are Gamma-distributed (Appendix 1-Figure 6C). As before, FOI estimates across pre-IRS and three IRS coverage levels show narrow 95% confidence intervals and bootstrap distributions close to true FOI values (**Figure 2** [↗](#)), with slight underestimations for FOI estimates based on MOI estimates corrected for under-sampling of *var* genes and those corrected for all sampling limitations.

The performance of the two methods across additional simulated scenarios is shown in Appendix 1-Figure 11-15.

The Two-Moment Approximation and Little's Law methods give replicable FOI estimates for empirical surveys conducted in the Bongo District of northern Ghana

After validating with simulations, we apply the two methods to empirical surveys in the Bongo District of northern Ghana. We first derive their MOI estimates. Due to the high but imperfect detection power of PCR, we assume three levels of sensitivity: 0% (high detectability), 5% (mid detectability), and 10% (low detectability) of PCR-negative individuals carrying infection (Materials and Methods-The under-sampling of infections in empirical surveys).

Antimalarial treatment, sought in response to symptoms or perceived transmission risk, can impact infection duration. Consequently, this may potentially violate the assumption that the two methods use infection duration data from naive malaria therapy patients with neurosyphilis. We address this issue by either excluding treated individuals from the analysis or by discarding their infection status and MOI estimates, instead sampling from non-treated individuals. Since the latter samples non-zero MOIs for these treated and uninfected individuals, it results in an upper bound for FOI estimates. Note that in this latter case, we do not assume that the MOI distribution for treated individuals is the same as that for untreated individuals. Rather, we aim to estimate what their MOI would have been, and consequently, determine what the FOI per individual per year in the combined population would be, had these individuals not received antimalarial treatment. Further details can be found in Materials and Methods-Antimalarial drug treatment of infections in empirical surveys.

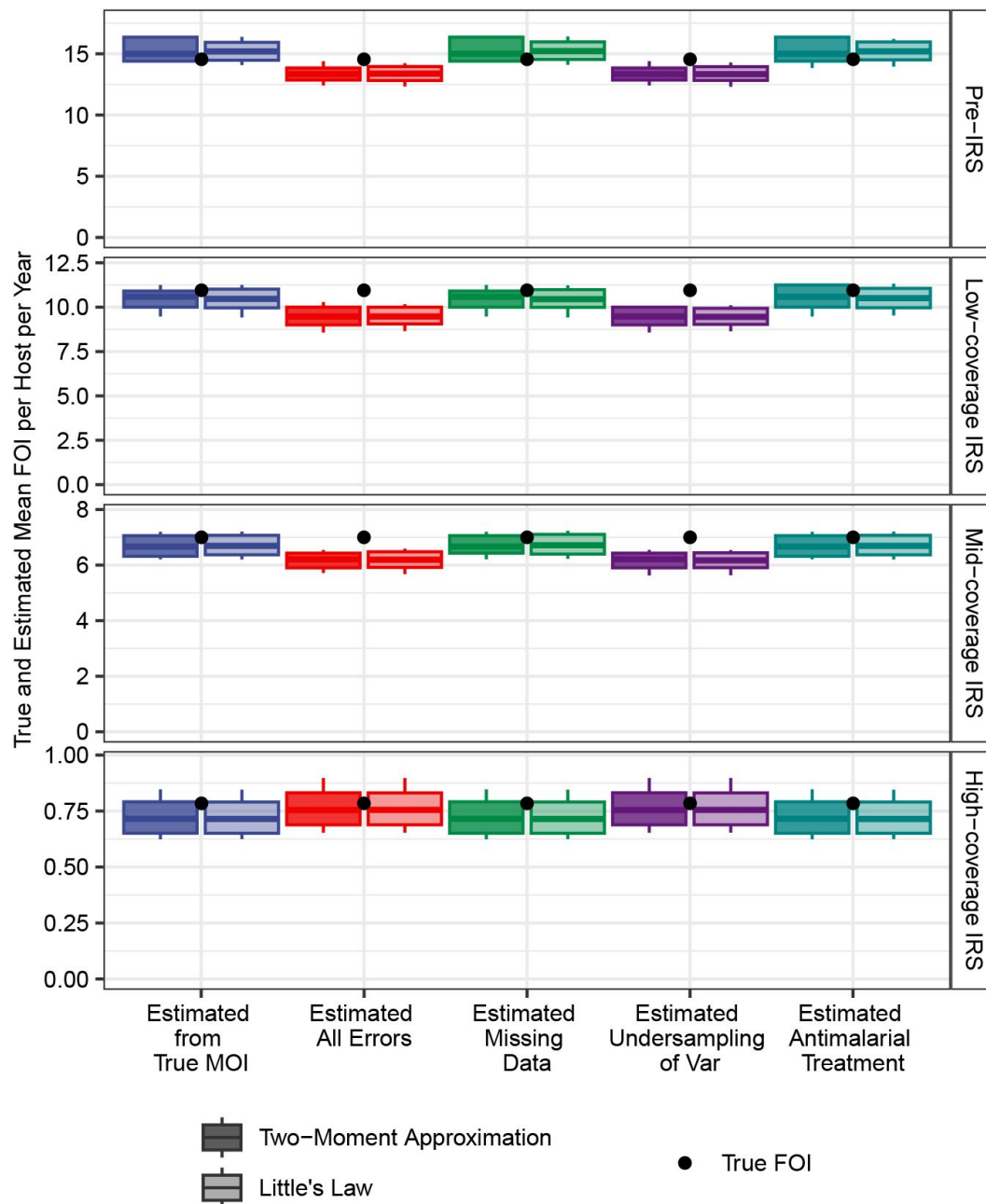


Figure 1.

Confidence intervals for estimated mean FOI values in simulated scenarios of homogeneous exposure risk, before and during IRS interventions at three different coverage levels.

The times between local transmission events follow a Gamma distribution, with seasonal transmission in a closed system. FOI estimates are derived from true MOI values and MOI estimates obtained through the Bayesian formulation or the bootstrap imputation approach correcting for all or individual sampling limitations. The true mean FOI per host per year is computed by dividing the total number of infections acquired by the population by the total number of hosts in the population. Each boxplot shows minimum, 5% quantile, median, 95% quantile, and maximum values.

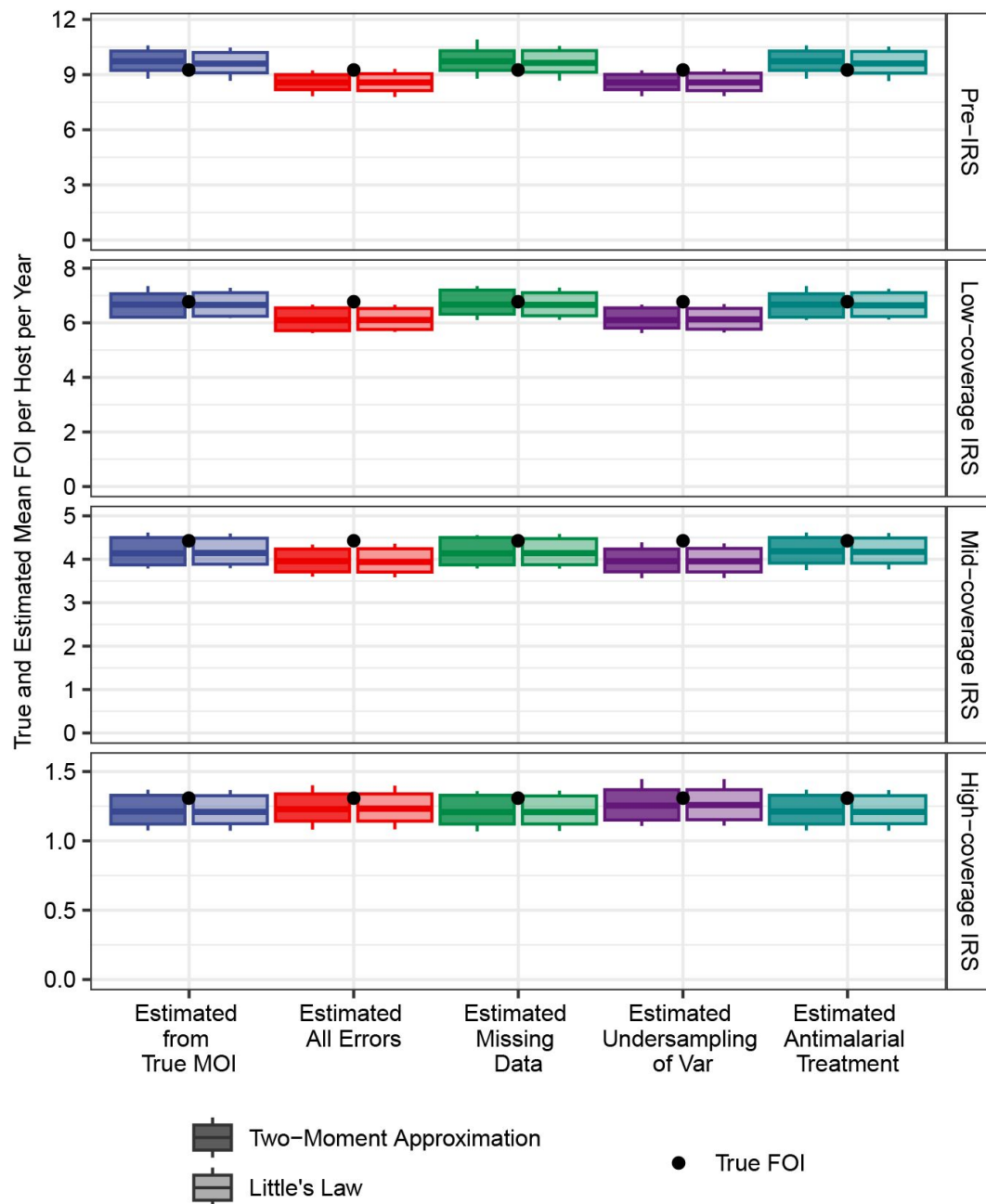


Figure 2.

Confidence intervals for estimated mean FOI values in simulated scenarios of heterogeneous exposure risk, before and during IRS interventions at three different coverage levels.

The times between local transmission events follow a Gamma distribution, with seasonal transmission in a semi-open system. FOI estimates are derived from true MOI values and MOI estimates obtained through the Bayesian formulation of the *varcoding* method or the bootstrap imputation approach correcting for all or individual sampling limitations. The true mean FOI per host per year is computed by dividing the total number of infections acquired by the population by the total number of hosts in the population. Each boxplot shows minimum, 5% quantile, median, 95% quantile, and maximum values.

Next, we apply the two-moment approximation and Little's Law methods to derive FOI estimates from empirical MOI estimates. Both methods yield replicable FOI estimates. The FOI estimates are similar across the three PCR sensitivity levels and the two approaches for handling treated individuals. The 95% confidence intervals and the full sampling distributions from boot-strap analysis are concentrated (**Figure 3**, Appendix 1-Figure 16-18). Notably, there is a significant reduction in FOI, exceeding 70%, indicating that the three-round IRS intervention, although transient, was highly effective.

The inferred FOI and directly measured EIR from the Ghana surveys align with the relationship between these two quantities in previous studies

We plot the measured annual EIR against the estimated annual FOI, and the transmission efficiency (the ratio between FOI and EIR) against the measured annual EIR from previous field studies summarized by (Smith et al., 2010) (**Figure 4**). In these studies, FOI estimation was based on fitting a simple epidemiological model to age-stratified prevalence data from cross-sectional parasitological studies (Smith et al., 2010; Pull and Grab, 1974), while EIR was estimated using various methods such as exit bait collection, human bait collection, pyrethrum spray collection, night bite collection, and outdoor resting collection (Smith et al., 2010). The red line indicates the functional curve fitted to these data points (Materials and Methods-Conversion between FOI and the Entomological Inoculation Rate, or EIR), initially proposed by (Smith et al., 2010).

The data show that mean annual FOI values are consistently below an empirical limit of 20. Due to the highly non-linear relationship between FOI and EIR, there is no single constant factor to convert FOI to EIR (or vice versa) across different settings. In high-transmission regions, transmission efficiency is extremely low: annual EIR can reach a few hundred to a thousand, while annual FOI ranges from about 5 to 10. For future applications, when using our proposed methods to estimate FOI from MOI under sparse sampling schemes, we can rely on this functional curve to convert FOI estimates to corresponding EIR values. There is however an inherently high variance in this conversion. Overall, our paired EIR (measured directly by the entomological team in Ghana (Tiedje et al., 2022)) and FOI estimates align well with previous studies, indicating the consistency of our methods.

The variance inferred by the two-moment approximation method reflects transmission intensity and heterogeneity across individuals

We have focused on presenting the mean FOI in both simulation output and empirical data. The two-moment approximation method also provides estimates for the variance of inter-arrival times of infections. When examining the inferred variance across simulated scenarios, We find that the individual-level mean FOI, when aggregated across the local population, reflects the total number of infections accumulated by that population, regardless of the variance magnitude for FOI.

The inferred variance is most significantly correlated with the inferred mean FOI. Specifically, a smaller mean FOI is associated with a larger variance (Appendix 1-Figure 19). Overall, seasonal runs exhibit slightly greater variance than non-seasonal runs. Runs with heterogeneous transmission (Appendix 1-Figure 6C) have slightly higher variance compared to homogeneous transmission runs. These findings align with expectations, as both seasonality and transmission heterogeneity increase the dispersion of FOI and inter-arrival times of infections.

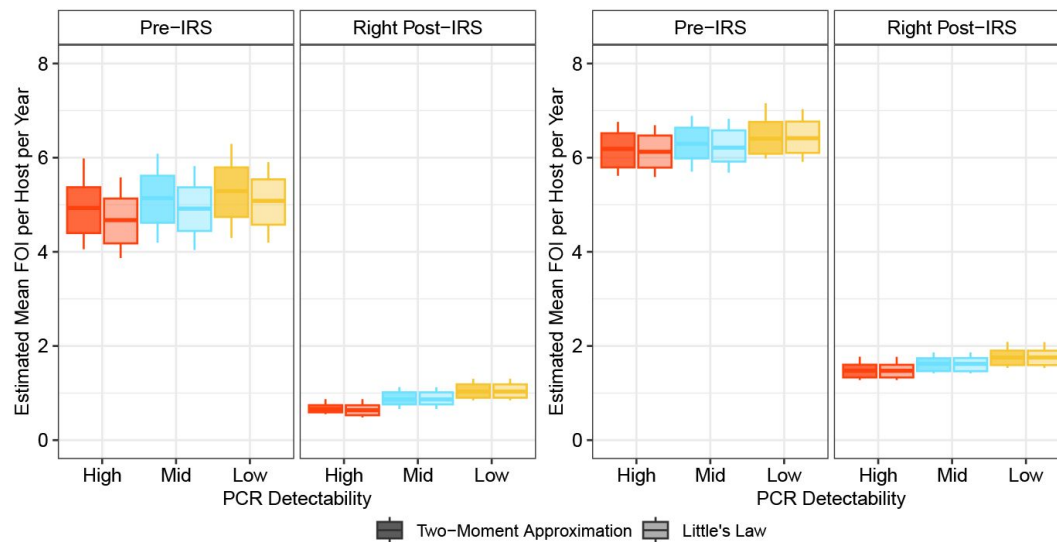


Figure 3.

Confidence intervals for the estimated mean FOI values in Ghana surveys before and immediately after a transient three-round IRS intervention

(A) The estimated FOI values when excluding these treated individuals from the analysis. (B) The estimated FOI values when discarding the infection status and MOI estimates of treated individuals and sampling from non-treated ones. Since this case samples non-zero MOIs for these treated and uninfected individuals, it results in an upper bound for FOI estimates. Each boxplot shows minimum, 5% quantile, median, 95% quantile, and maximum. The value of c is set to 30. FOI estimates with other values of c can be found in Appendix 1-Figure 16-18.

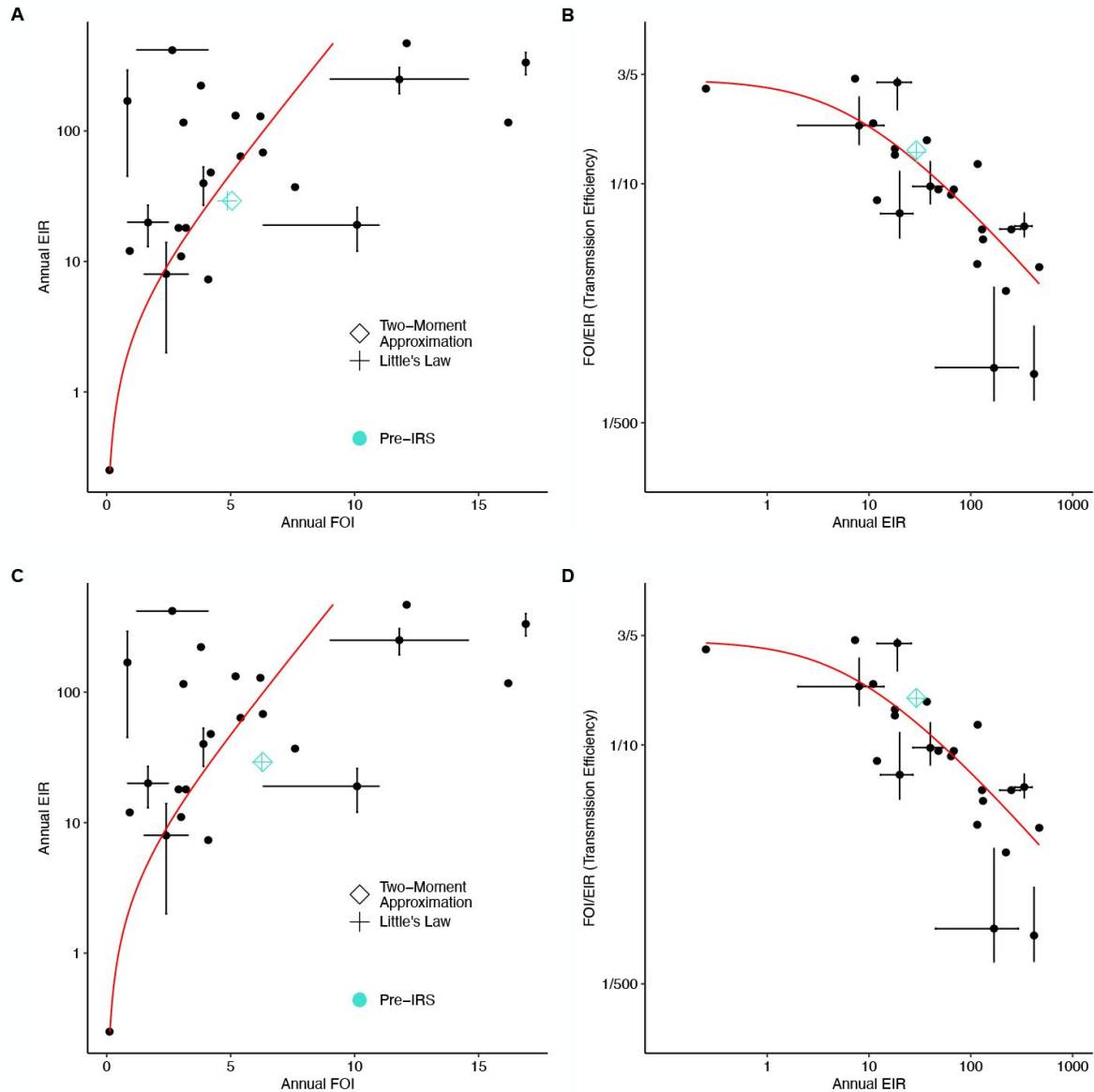


Figure 4.

The saturation in FOI with increasing EIR and their non-linear relationship from previous field studies.

(A) and (B) present our empirical estimates (with $c = 30$) when excluding treated individuals from the analysis. (C) and (D) show our estimates when discarding the infection status and MOI estimates of treated individuals and instead sampling from non-treated ones. Since this case samples non-zero MOIs for these treated and uninfected individuals, it results in an upper bound for FOI estimates. The black points represent paired EIR-FOI values from the literature, as summarized by (Smith et al., 2010), with crosses indicating instances where multiple estimates or ranges were reported or estimated for the same location. The red curve represents the best-fit to these paired EIR-FOI values (Smith et al., 2010). The turquoise hollow diamond and plus represent the Ghana data, showing our FOI estimates using the two methods and the EIR measured in the field by the entomological team (Tiedje et al., 2022).

Discussion

Building on estimates of the multiplicity of infection under sparse sampling schemes, we demonstrate the feasibility of converting these values into estimates of the force of infection. In various simulation scenarios using an extended stochastic agent-based model, the two-moment approximation and Little's Law methods provide good and replicable FOI estimates based on MOI estimates obtained with the Bayesian formulation of the *var*coding method and the bootstrap imputation approach, despite common sampling limitations.

Both methods tend to slightly underestimate FOI due to a bias in underestimating MOI values. The current MOI estimation procedure, specifically the Bayesian formulation of the *var*coding method, does not correct for the limited overlap of *var* genes between co-infecting strains. This limited overlap reduces the number of *var* genes identified per individual, thereby causing the underestimation bias.

Studies have shown parasite co-transmission from single mosquito bites in high-transmission regions (Nkhoma et al., 2020 [DOI](#); Wong et al., 2017 [DOI](#)), predominantly based on clinical or symptomatic infections (Andolina et al., 2021 [DOI](#); Lindblade et al., 2013 [DOI](#)), although asymptomatic cases constitute the majority of the malaria transmission reservoir in these regions. Co-transmitted recombinant parasites, more closely related to each other than parasites from different bites, further reduce the number of identified *var* genes (Wong et al., 2022 [DOI](#); Nkhoma et al., 2020 [DOI](#); Wong et al., 2017 [DOI](#)) per individual. Low or variable parasite densities due to factors like small blood volumes, genomic DNA quality, clinical status, and within host dynamics also affect MOI estimation (Peyerl-Hoffmann et al., 2001 [DOI](#); Bruce et al., 2000 [DOI](#); Okell et al., 2012 [DOI](#); Farnert et al., 1997 [DOI](#); Färnert et al., 2008 [DOI](#); Barry et al., 2021 [DOI](#); Hergott et al., 2024 [DOI](#)). These issues are common to all measures of MOI and direct measures of FOI. We did not correct for these factors when estimating MOI and FOI for the Ghana surveys. In the future, the Bayesian formulation could be extended to account for these confounders in identifying the number of co-infecting strains.

Our proposed methods leverage infection duration data from malaria therapy patients with neurosyphilis who had no prior malaria exposure, making them well-suited for FOI inference in highly vulnerable subpopulations with similarly naive immune profiles. For our Ghana surveys, we focus on children aged 1-5 years, as their immune profiles are closer to those of naive patients than those of older individuals. In the pre-IRS phase of the Ghana surveys, an estimated mean FOI of about 5 per host per year suggests that a 4-year-old child would have experienced around 20 infections, which makes them far from naive. However, the extreme documented diversity of *var* genes (Tiedje et al., 2023 [DOI](#)) means that even with 20 infections, a 4-year-old may have developed immunity to only a small fraction of the total antigenic diversity encoded by these genes. Consequently, they are not as immunologically experienced as it might initially seem. Moreover, studies have shown that long-lived infections in older children and adults can persist for months or even years, including through the dry season. This persistence is driven by high antigenic variation of *var* genes and associated incomplete immunity. Additionally, parasites can skew PfEMP1 expression to produce less adhesive erythrocytes, enhancing splenic clearance, reducing virulence, and maintaining extended periods of sub-clinical parasitemia (Andrade et al., 2024 [DOI](#); Tran et al., 2013 [DOI](#); Zhang and Deitsch, 2022 [DOI](#)). The impact of immunity on infection duration with age for falciparum malaria remains a challenging open question.

We acknowledge, however, the limitations this aspect of the two methods introduces. To reduce misspecification in infection duration and fully utilize our proposed methods, future data collection and sampling could prioritize subpopulations with minimal prior infections and an

immune profile similar to naïve adults, such as infants and toddlers. As these individuals are also the most vulnerable, prioritizing them aligns with the short-term priority of all intervention efforts: to monitor and protect the most vulnerable individuals from severe symptoms and death.

The application of both methods generally requires an evolving time series observed over a long period or many realizations of the same series at a single time. However, empirical surveys often rely on sparse sampling schemes with single-time-point observations per host available. We therefore use the population-level queue length distribution by aggregating MOI estimates across all sampled individuals to approximate the steady-state queue length distribution at the individual level, without explicitly considering individual heterogeneity due to transmission. The estimated FOI represents the FOI experienced by an individual host whose queue length distribution is approximated from the collection of all sampled individuals. Our estimator, combined with demographic information on population size, estimates the total number of *Plasmodium falciparum* infections acquired by all hosts in the population per year. We evaluate the impact of individual heterogeneity due to transmission on FOI inference using simulations. Even for significant heterogeneity across individuals, our methods show performance comparable to that of homogeneous scenarios. Additionally, our methods perform similarly for both non-seasonal and seasonal transmission scenarios. Alternative frameworks such as fitting a non-homogeneous Poisson process or Gamma process to derive time-varying average arrival rates, or simultaneous estimation of the arrival rates and infection duration, cannot be considered due to their requirements for densely sampled time-series data.

After validating our methods with simulations, we applied them to surveys in Bongo District of northern Ghana, a high-transmission endemic region where estimating multiplicity of infection and related FOI has been challenging. The three-round transient IRS intervention proved strong and effective, resulting in a significant reduction in FOI by more than 70%.

Our Ghana surveys lack direct FOI measurements, which prevents us from directly evaluating our methods as we did with simulation outputs. Empirical MOI-FOI pairs from cohort studies are still lacking, and direct FOI measurements are prone to errors due to challenges in differentiating new infections from the temporary absence and re-emergence of old infections. These challenges arise from the low resolution of polymorphic markers used in cohort studies and the complexity of within-host dynamics. Alternative approaches that fit epidemiological models to densely sampled cross-sectional surveys do not provide direct FOI measurements, leaving no available “true” values for benchmarking against fitted FOI estimates. In these approaches, model parameterization relies on capturing certain epidemiological quantities, such as prevalence or incidence, similar to the one done in this work. We selected FOI values that maximize the likelihood of observing given MOI distributions. Additionally, we pair our estimated FOI value for Ghana surveys with independently measured EIR (Tiedje et al., 2022 [\[1\]](#)). We demonstrate alignment between the paired EIR and our FOI estimates with those from previous studies. We acknowledge however that, like these alternative approaches, our validation for field data is indirect and further complicated by high variance in the relationship between EIR and FOI from previous studies.

The FOI estimates obtained through these methods go beyond describing basic malaria epidemiology and evaluating intervention outcomes. FOI for naïve hosts is a fundamental parameter for epidemiological models. FOI of non-naïve hosts is typically a function of their immune status, body size, and the FOI of naïve hosts. Additionally, these FOI estimates can inform process-based models for the population dynamics of complex infectious diseases, serving as priors for parameterizing and validating more complex agent-based or equation-based models, in a way that reduces computational cost, improves efficiency, and minimizes identifiability issues.

A key characteristic of malaria transmission is the saturation in FOI at high transmission, and the highly non-linear relationship between FOI and EIR. While EIR can reach several hundred to a thousand per year, annual FOI typically saturates below 20 (Smith et al., 2010 [DOI](#)). Different choices of field measures of EIR and FOI cannot account for this drastic difference in magnitude. Transmission is highly inefficient in high-transmission regions with high annual EIRs. The difference between these two quantities is mediated primarily by immunity, or within-host dynamics, measurement bias, and heterogeneous transmission (Donovan et al., 2007 [DOI](#); Macdonald, 1950 [DOI](#); John et al., 2005 [DOI](#); Doolan and Martinez-Alier, 2006 [DOI](#)). Mathematical models commonly use the probability of transmission from an infectious mosquito bite to bridge FOI and EIR, as a general parameter encapsulating a variety of processes.

FOI saturation poses significant challenges to intervention efforts in high-transmission endemic regions. In these areas, intervention efforts must dramatically reduce EIR by several orders of magnitude to bring FOI below saturation levels. In other words, high-coverage interventions are needed to achieve and start observing any noticeable impacts on individual exposure risk. Theoretical models suggest a sharp non-linear transition towards sustainable low transmission or elimination, influenced by the high antigenic diversity of *P. falciparum* (de Roos et al., 2023 [DOI](#); Zhan et al., 2024a [DOI](#)). The same molecular information used here to estimate MOI and FOI in Bongo District underlies this diversity. Recent ABM modeling on resurgence dynamics indicates that highly resilient systems can compensate for reduced FOI following intervention through heterogeneity in infection duration and selection for longer-lasting infections (Zhan et al., 2024a [DOI](#)). Consequently, high prevalence and MOI levels can be even maintained during interventions in the regime below FOI saturation.

The proposed methods are applicable to evaluate transmission intensity in many pathogens exhibiting multi-genomic infection due to large antigenic diversity, including those encoding such variation with multigene families (Deitsch et al., 2009 [DOI](#)). Easier estimation of changes in transmission intensity should enhance the efficiency and evaluation of control programs across a broader range of infectious diseases.

Materials and methods

High genetic diversity of *var* and the associated strain structure of limiting similarity

We briefly describe the biology of the malaria parasite *P. falciparum* that underpins our MOI estimation procedure, the Bayesian formulation of the *var*-coding method.

In high-transmission endemic regions, human hosts remain susceptible to malaria re-infection throughout their lifetime (Doolan et al., 2009 [DOI](#)). High asymptomatic prevalence and high MOI result from high transmission rates and incomplete host immunity due to the parasite's high antigenic variation (Deitsch et al., 2009 [DOI](#)). Parasites achieve this variation and evade the immune system by encoding key variant surface antigens (VSAs) using multigene families (Deitsch et al., 2009 [DOI](#)). One important multigene family in the malaria parasite *P. falciparum* is known as *var*, which encodes PfEMP1 (*Plasmodium falciparum* erythrocyte membrane protein 1), the major VSA during the blood stage of infection (Zhang and Deitsch, 2022 [DOI](#); Baruch et al., 1995 [DOI](#); Smith et al., 1995 [DOI](#); Su et al., 1995 [DOI](#)). Each parasite carries 50-60 *var* genes across its chromosomes, encoding different variants of this protein, which are expressed sequentially.

Empirical sequencing of *var* genes focuses on the DBLα tag, a conserved ~450bp region encoding the immunogenic Duffy-binding-like alpha domain of PfEMP1 (Tiedje et al., 2023 [DOI](#); Ruybal-Pesántez et al., 2017, 2022 [DOI](#); Day et al., 2017 [DOI](#)). Bioinformatic analyses of a large database of

exon 1 sequences of *var* genes revealed a predominantly 1-to-1 DBLa-*var* relationship, meaning each DBLa tag typically represents a unique *var* gene (Tan et al., 2023 [DOI](#)). Hereafter, we use DBLa types and *var* genes interchangeably.

In high-transmission endemic regions, local parasite populations exhibit a vast pool of *var* gene variants, ranging from thousands to tens of thousands (Day et al., 2017 [DOI](#); Tiedje et al., 2022 [DOI](#)). These variants are generated primarily through mitotic recombination, but also through meiotic recombination, mutation, and host/mosquito vector migration (Claessens et al., 2014 [DOI](#); Frank et al., 2008 [DOI](#); Freitas-Junior et al., 2000 [DOI](#); Bopp et al., 2013 [DOI](#)). This large pool, combined with negative frequency-dependent selection (NFDS) mediated by hosts' specific immunity, results in the limited overlap of *var* genes among individual repertoires (individual parasite genomes) and isolates (sets of in-dividual parasite genomes co-infecting individual hosts) (Day et al., 2017 [DOI](#); He et al., 2018 [DOI](#)). Major groups of *var* genes are classified based on their 5'-flanking region, called ups, which controls gene expression: upsA and upsB/C (non-upsA) Rask et al. (2010a) [DOI](#). Non-upsA DBLa sequences are ~ 20 times more diverse and less conserved among repertoires than the upsA DBLa sequences. Hence our MOI estimation leverages non-upsA DBLa types, as detailed in the following section.

Bayesian formulation of the “*var*coding” method for MOI estimation

The limited overlap of *var* repertoires allows MOI estimation based on the number of non-upsA DBLa types identified from an isolate. The original *var*coding method assumes a constant repertoire length, i.e., number of non-upsA DBLa types in a parasite genome, to convert the number of types identified in an isolate to the estimated MOI. This method does not account for the measurement error (Appendix 1-Figure 6D) in this length introduced by the under-sampling of *var* genes in an infection. We recently extended this method to a Bayesian formulation that considers this error and provides a posterior distribution of MOI values for each sampled individual (Tiedje et al., 2023 [DOI](#)). We documented the steps of this Bayesian formulation, compared two ways of obtaining population-level MOI distribution (either pooling the maximum a posteriori MOI estimates or calculating a mixture distribution), and examined the impact of different priors (Tiedje et al., 2023 [DOI](#)). In our analyses here, we provide the estimated population-level MOI distribution obtained from a mixture distribution using a uniform prior for individuals.

Empirical surveys from Ghana

We use empirical data from an interrupted time-series study conducted in Bongo District, northern Ghana. This study involves four age-stratified cross-sectional surveys of ~2,000 participants each, conducted between 2012 and 2016. The study assessed the impacts of a transient threeround IRS intervention, combined with long-lasting insecticidal nets (LLINs), on the asymptomatic *P. falciparum* reservoir (Tiedje et al., 2017 [DOI](#), Tiedje et al., 2022 [DOI](#), Tiedje et al., 2023 [DOI](#)). Surveys were conducted at the end of the wet/high-transmission season (i.e., October) or the dry/low-transmission season (i.e., May/June). The study consists of two phases: (1) Pre-IRS: two surveys before the IRS intervention (Survey 1 in October 2012; Survey 2 in May/June 2013); and (2) Immediately post-IRS: two surveys immediately following the three-round IRS intervention (Survey 3 in October 2015; Survey 4 in May/June 2016) (Appendix 1-Figure 6E). Details on the study area/population, study design, malaria control interventions (IRS and LLINs), inclusion/exclusion criteria, data collection/generation procedures, and routine surveys of antimalarial treatment usage in a previous window of time have been previously described (Tiedje et al., 2017 [DOI](#), Tiedje et al., 2022 [DOI](#), Tiedje et al., 2023 [DOI](#)).

The under-sampling of infections in empirical surveys

The empirical MOI estimates in various epidemiological studies, including ours in Bongo District, northern Ghana, rely on microscopy-positive individuals (Tiedje et al., 2023 [↗](#), Tiedje et al., 2022 [↗](#)). Due to microscopy's limited sensitivity, a significant fraction of individuals who carry infections are undetected. A subset of Ghana surveys also include submicroscopic infections detected by PCR (Tiedje et al., 2023 [↗](#), Tiedje et al., 2022 [↗](#)), which is significantly more sensitive and can detect a higher fraction, if not 100%, of individuals with *P. falciparum* infections. Using surveys with both microscopy and PCR detection, we estimate conversion factors of 0.76 for untreated children and 0.67 for antimalarial-treated children aged 1–5 years.

For surveys with both detection methods, we directly calculate the number of microscopy-negative but PCR-positive individuals. For surveys with only microscopy data, we use the estimated conversion factors to estimate the number of microscopy-negative but PCR-positive individuals. Additionally, we account for the high but not exactly known sensitivity of PCR by assuming its detectability ranges from relatively low (10% of all PCR-negative individuals carrying undetected infections) to perfect (none of PCR-negative individuals carrying undetected infections). We calculate the number of individuals with undetected infections by PCR for each sensitivity level.

After estimating the number of individuals with undetected infections (both microscopy-negative but PCR-positive and PCR-negative but infected), we sample from existing MOI estimates of microscopy-positive individuals not under antimalarial treatment (see the following section) to represent the missing MOI data.

Similarly, for individuals who are microscopy-positive but lacking *var* information due to factors like low DNA quality, we sample values from existing MOI estimates of microscopy-positive individuals not under antimalarial treatment to represent the missing MOI data.

We assume that microscopy-negative but PCR-positive children aged 1–5 years and microscopy-positive children aged 1–5 years have similar MOI distributions. This assumption is supported by our analysis of Ghana surveys, which shows no clear relationship between parasitemia levels and MOI (or the number of *var* genes detected within an individual host, on the basis of which our MOI values were estimated) (Appendix 1-Figure 7). We scale the parasitemia levels and the number of non-ups A *var* genes or MOI estimates before performing the regression. Parasitemia levels underlie the difference in detection sensitivity between PCR and microscopy.

This lack of clear relationship can be attributed to several factors. One factor is immune regulation of parasite density, where host immunity may limit parasite density without reducing the diversity of co-infecting strains (Eldh et al., 2019 [↗](#)), leading to individuals with low parasitemia but high MOI. Another factor is asynchronous parasite dynamics, where different parasite clones replicate asynchronously (Färnert et al., 1997 [↗](#)), resulting in varied parasite densities that do not directly correlate with the number of distinct strains present. This could explain why individuals with low parasitemia still exhibit multiple strains. Lastly, competition among parasite strains suppress the growth of individual clones, lowering parasite densities while maintaining high strain diversity, thus reducing the expected correlation between MOI and parasitemia (Sondo et al., 2019 [↗](#); Earland et al., 2019b [↗](#)).

Antimalarial drug treatment of infections in empirical surveys

Individuals may seek and receive antimalarial treatment in response to symptoms or perceived transmission risk. In our surveys from northern Ghana, over 50% of children aged 1–5 years responded that they had received an antimalarial treatment in the previous two-weeks (i.e., participants that reported they were sick, sought treatment, and were provided with an antimalarial treatment) prior to the wet/high-transmission survey before IRS (i.e., Survey 1,

Appendix 1-Figure 6E) (Tiedje et al., 2022 [DOI](#)). This fraction is significantly lower for the dry/low-transmission survey before IRS and the surveys collected immediately after IRS (i.e., Survey 2-4, Appendix 1-Figure 6E) (Tiedje et al., 2022 [DOI](#)).

Disentangling the effect of drug treatment on measurements like infection duration is challenging. Since our methods use infection duration data from naïve malaria therapy patients with neurosyphilis, drug treatment can potentially violate this assumption. We propose two solutions: (1) exclude treated individuals from the analysis; (2) remove treated individuals' samples and use a bootstrap imputation approach based on the remaining population. Specifically, we sample from the MOI estimates of untreated microscopy-positive individuals to represent MOI estimates for treated individuals, which corrects for individuals who have used antimalarial drugs and show either no infection ($\text{MOI} = 0$) or infection ($\text{MOI} > 0$). Hence, this solution provides an upper bound for FOI estimates. Numerical simulations show our bootstrap imputation approach is robust even with a significant fraction of treated individuals, as seen in our Ghana surveys.

The final distribution of MOI estimates at the population level includes values for microscopy-positive individuals, imputed values for individuals with missing MOI information or false negatives, imputed values for treated individuals (for the second solution), and true zeros for uninfected individuals. This distribution is used for FOI inference.

Inferring FOI from MOI estimates

Malaria transmission in relation to queuing theory

In a cohort of individuals acquiring and clearing infections independently, infections follow a homo-geneous Poisson process with a rate equal to the mean FOI, and each infection has an exponentially distributed duration. At equilibrium, MOI is a Poisson variable with a mean equal to the mean FOI divided by the mean clearance rate (Dietz et al., 1974 [DOI](#)). However, the arrival of infections often deviates from a homogeneous Poisson process due to factors like seasonality and heterogeneity in exposure risk. Infection durations can also exhibit higher variance than the mean. For instance, under heterogeneous biting, the MOI distribution is better represented by a negative binomial distribution, which account for higher variance. In this case, there is no straightforward relationship between the parameters of the negative binomial distribution and the mean FOI and clearance rate. We conduct a formal test for deviations from Poisson in simulated and empirical MOI distributions, detailed in Appendix 1-Test of deviation from Poisson homogeneity in MOI estimates and supplementary file 2-deviationFromPoissonTest.xlsx.

Despite these complexities, the qualitative relationship between MOI and FOI is intuitive when information on infection duration is known. Higher FOI values should correspond to higher MOI values. Less variable FOI values should result in narrower or more concentrated MOI distributions, whereas more variable FOI values should lead to more spread-out MOI distributions.

The process of acquiring infectious bites is structurally analogous to stochastic queuing theory, which relates queue length to the intensity of arrivals, priority schedules, and service and waiting times. Modeled using differential equations (with the Kolmogorov equations), queueing systems comprise three main components: queue length, the intensity of arrivals, and service times. Knowing any two allows for the inference of the third.

In malaria transmission, hosts resemble service facilities composed of a collection of servers, with each infection akin to a customer. Just as service facilities have a carrying capacity for the number of customers they can serve simultaneously, hosts have a carrying capacity for blood-stage infections, which limits the maximum number of infections they can harbor (Appendix 1-Figure

8). Empirical MOI estimates provide information on queue length. Hypothetically, knowing the service times, i.e., infection durations, can help infer the intensity of arrivals, i.e., the rate at which hosts acquire infections or FOI.

However, determining infection durations is challenging in endemic areas where multi-genomic infections are common. Popular polymorphic markers often fail to distinguish between co-infecting strains (Argyropoulos et al., 2023 [↗](#)), complicating the tracking of the emergence and clearance of individual strains from the peripheral blood. Complex within-host dynamics further complicate tracking unless daily sampling is conducted, which is impractical in real settings. Frequent ectopic recombination of *var* genes complicates assigning genes to specific chromosomal locations. This difficulty in phasing compromises the integrity of individual strains, making it hard to isolate them and track their first appearance and subsequent clearance in blood over time. Additionally, infection duration varies widely across age groups, geographical locations, and sampling times (Childs and Buckee, 2014 [↗](#); Chang et al., 2016 [↗](#); Ashley and White, 2014 [↗](#); Bretscher et al., 2011 [↗](#)).

We therefore propose focusing on FOI inference in subpopulations with naive or near-naïve immune profiles. Their infection duration can be approximated by that of naive hosts, as seen in a historical medical study of neurosyphilis patients intentionally infected with malaria as a treatment. Between 1940 and 1963, 318 syphilis patients were infected with a single strain of *P. falciparum* (Collins and Jeffery, 1999 [↗](#); Maire et al., 2006 [↗](#)), and data on fever and parasite counts in the blood were recorded. Since these patients had no prior *P. falciparum* infections, the documented infection duration reflects that of naive infections.

In Ghana surveys, we focus on children aged 1-5 years, who have accumulated far fewer infections and less immune memory compared to older individuals, an aspect we discuss in the Discussion section. We treat these children as nearly naive and approximate their duration infection with that of naive hosts. Using their MOI estimates, we infer FOI for these children using the following two methods from queuing theory.

A two-moment approximation for a queue of finite capacity

Analysis of multi-server models is challenging, with exact results available only for specific cases, such as the previously mentioned $M/M/c/k$ models. In these models, M represents exponential inter-arrival and service times, c is the number of servers, and k is the maximum queue capacity, including both customers being served and those waiting. Additional models like $M/G/c/c$ and $GI/M/c/c + r$ queues also require exponential distributions for either inter-arrival or service times, where G and GI represent generic random variables and independent and identically distributed (i.i.d.) generic random variables, respectively. These models can often be too restrictive for realworld scenarios as well.

We examine a two-moment approximation method introduced by Kim and Cha (Choi et al., 2005 [↗](#)). This method considers the $GI/G/c/c + r$ queue, where inter-arrival times (GI) and service times (G) of customers are independent sequences of i.i.d. general random variables A and S , respectively. There are c (≥ 1) identical servers in parallel and r (≥ 0) waiting places.

Let N denote the number of customers in the system at an arbitrary time, and $N^A(N^D)$ denotes the number of customers that an arriving customer finds (that a departing customer leaves behind) in steady state. Customers who arrive to find $c + r$ customers in the system depart immediately,

leaving those $c + r$ customers behind. Let P_n , P_n^A and P_n^D denote the probabilities that $N = n$, $N^A = n$, and $N^D = n$, respectively, for $0 \leq n \leq c + r$. These probabilities are expressed in terms of the following quantities:

$$\begin{aligned} a &= E(A) = \frac{1}{\lambda} \\ b &= E(S) \\ a_n^D &= E[A_n^D], 0 \leq n \leq c + r \\ b_n^A &= E[S_n^A], 0 \leq n \leq c + r \\ b_n^D &= E[S_n^D], 0 \leq n \leq c + r \end{aligned}$$

Where A_n^D , $0 \leq n \leq c + r$, is the residual inter-arrival time at the departure instant of a customer who leaves behind n customers in the system, and $S_n^A(S_n^D)$, $1 \leq n \leq c + r$, is the residual service time of a randomly chosen busy server at the arrival instant (the departure instant) of a customer who finds (leaves behind) n customers in the system. From these definitions, we have $a_{c+r}^D = a$ and $b_{c+r}^D = b_{c+r}^A$. We set $b_0^A = b_0^D = 0$. We assume that all the above quantities are well defined and finite.

Using Theorems 4.3.19 and 4.3.43 from Franken et al (Heyman, 1985), the steady-state queue-length distribution can be derived:

$$\begin{aligned} P_n^A &= P_n^D = P_0^A \prod_{i=0}^{n-1} \frac{\lambda_i}{\mu_{i+1}}, 1 \leq n \leq c + r \\ P_n &= P_n^A \gamma_n, 0 \leq n \leq c + r \end{aligned}$$

Where

$$\frac{1}{\mu_i} = \begin{cases} b - i(a - a_{i-1}^D) + (i - 1)(b_{i-1}^A - b_{i-1}^D), 1 \leq i \leq c \\ -c(a - a_{i-1}^D) + b_{i-1}^A + (c - 1)(b_{i-1}^A - b_{i-1}^D), c + 1 \leq i \leq c + r \end{cases}$$

$$\frac{1}{\lambda_i} = \begin{cases} (i + 1)(a_{i+1}^D + b_{i+1}^A - b_{i+1}^D), 0 \leq i \leq c - 2 \\ ca_{i+1}^D + b_{i+1}^A - b + (c - 1)(b_{i+1}^A - b_{i+1}^D), c - 1 \leq i \leq c + r - 2 \\ ca, i = c + r - 1 \end{cases}$$

$$\gamma_i = \begin{cases} \lambda a_0^D, i = 0 \\ \lambda [\mu_i \frac{(a - a_{i-1}^D)}{\lambda_{i-1}} + a_i^D], 1 \leq i \leq c + r \end{cases}$$

And, by normalization, $\sum_{n=0}^{c+r} P_n^A = 1$:

$$P_0^A = (1 + \sum_{n=1}^{c+r} \prod_{i=0}^{n-1} \frac{\lambda_i}{\mu_{i+1}})^{-1}$$

Since quantities a_n^D , b_n^A , and b_n^D are difficult to compute in general, Kim and Cha propose an approximation to the exact expression which replaces these unknown arrival- and departure-average quantities by their corresponding (well-known) time-average counterparts, which are exact for exponential inter-arrival and service times. That is:

$$a_n^D \approx a_R = \frac{E[A^2]}{2E[A]} = \frac{(1 + c_A^2)a}{2}, 0 \leq n \leq c + r - 1$$

$$b_n^A(b_n^D) \approx b_R = \frac{E[S^2]}{2E[S]} = \frac{(1 + c_S^2)b}{2}, 1 \leq n \leq c + r - 1$$

where $c_X^2 = \frac{\text{Var}[X]}{(E[X])^2}$ is the square coefficient of variation (SCV) of a random variable X with distribution function F .

Therefore, a two-moment approximation for the steady-state queue-length distribution is:

$$\tilde{P}_n^A = \tilde{P}_n^D = \tilde{P}_0^A \prod_{i=0}^{n-1} \frac{\tilde{\lambda}_i}{\tilde{\mu}_{i+1}}, 1 \leq n \leq c + r$$

$$\tilde{P}_n = \tilde{P}_n^A \tilde{\gamma}_n, 0 \leq n \leq c + r$$

Where

$$\frac{1}{\tilde{\mu}_i} = \begin{cases} b - i(a - a_R), 1 \leq i \leq c \\ -c(a - a_R) + b_R, c + 1 \leq i \leq c + r \end{cases}$$

$$\frac{1}{\tilde{\lambda}_i} = \begin{cases} (i + 1)a_R, 0 \leq i \leq c - 2 \\ ca_R + b_R - b, c - 1 \leq i \leq c + r - 2 \\ ca, i = c + r - 1 \end{cases}$$

$$\tilde{\gamma}_i = \begin{cases} \lambda a_R, i = 0 \\ \lambda[\tilde{\mu}_i \frac{(a - a_R)}{\tilde{\lambda}_{i-1}} + a_R], 1 \leq i \leq c + r - 1 \\ \lambda[\tilde{\mu}_i \frac{(a - a_R)}{\tilde{\lambda}_{i-1}} + a], i = c + r \end{cases}$$

And, by normalization, $\sum_{n=0}^{c+r} \tilde{P}_n^A = 1$:

$$\tilde{P}_0^A = (1 + \sum_{n=1}^{c+r} \prod_{i=0}^{n-1} \frac{\tilde{\lambda}_i}{\tilde{\mu}_{i+1}})^{-1}$$

Likelihood formulation and parameter estimation

We vary the mean and variance parameters for inter-arrival times across wide ranges (supplementary file 4-meanAndVarianceParams.xlsx). For each mean and variance combination, we calculate the steady-state queue length distribution, i.e., the probability density distribution of MOI, using the two-moment approximation method. The goal is to identify the parameter combination that minimizes the negative log-likelihood (or maximizes the likelihood) of observed MOI distributions from simulated outputs or Ghana surveys:

$$\underset{\mu, \sigma}{\text{argmin}} - \ln L(m; \mu, \sigma)$$

with the likelihood defined as follows:

$$L(m; \mu, \sigma) = \prod_{i=1}^N P(m_i; \mu, \sigma)$$

where N is the number of individual hosts, m_i is the MOI estimate for individual i , m is the vector of MOI estimates for all hosts, μ and σ represent the mean and variance parameters, respectively, and P is the steady state queue length distribution from the two-moment approximation method with specific mean and variance parameter values, as defined in the previous section.

The shape of the negative log likelihood for both simulated outputs and Ghana surveys is concave upwards around the trough, signifying a clear minimum point (Appendix 1-Figure 9). We tested the impact of different grid value choices on the FOI inference results by refining the grid to include more points, ensuring the FOI inference results are consistent. Specifically, we reduce the grid width for the mean parameter to half and a quarter of the original width, and for the variance parameter to half, a quarter, an eighth, and a sixteenth of the original width. The FOI inference results remain either unchanged or within a 1% deviation from those based on the original grid width (Appendix 1-Figure 10).

Details for deriving confidence intervals for the estimated parameters are provided in Appendix 1-Confidence intervals for FOI inference.

Choice of c and r for the $GI/G/c/c + r$ queue in the two-moment approximation method

When applying the two-moment approximation method, values for the number of parallel servers (c) and waiting places (r) need to be specified. Since MOI is defined exclusively for blood-stage infections, r is set to 0 by default. The parameter c corresponds to the carrying capacity of blood-stage infections. The maximum MOI observed in empirical data from Bongo, based on the *var*coding method, is 20. Certain factors which reduce the number of *var* genes identified in an individual, and thus affect MOI estimation, are not explicitly accounted for in the current MOI estimation (see Discussion), so the actual carrying capacity could be higher. For simplicity, we assume the value of c to be 30 in the simulation. Provided that c is kept consistent across simulations and the two-moment approximation method, this choice should not affect FOI inference. For empirical surveys from Bongo, we set c to 25, 30, 40, and 60 to systematically investigate its impact on FOI inference results. The FOI inference results are similar across these values (Figure 3 and Appendix 1-Figure 16-18).

In general, the choice of c depends on the maximum MOI observed in specific empirical datasets. To account for factors that cause underestimation of MOI, c should be set higher than the observed maximum MOI. Since Bongo District of northern Ghana is a high-transmission endemic region, we expect the range of its c to be applicable to other empirical datasets.

The mean arrival rate of infection from Little's Law

The second method is known as Little's Law (Little and Graves, 2008), which describes a relationship between the three main components of queuing systems. This law states that the average number of items in a queuing system L equals the average arrival rate λ multiplied by the average waiting time of an item in the system, W . Reformulating Little's Law for malaria transmission, the average arrival rate of infection λ equals the average number of blood-stage infections present in an individual L divided by the average duration of infection W .

$$\lambda = \frac{L}{W} \quad (1)$$

The relationship is simple and general, holding true regardless of the number of servers (carrying capacity of blood-stage infections in hosts), the service time distribution (infection duration distribution), the distribution of inter-arrival times, the order of service, or the queue structure.

Population-level MOI distribution for approximating time-series observation of MOI per host or many realizations at the same sampling time per host

Both the two-moment approximation and Little's Law methods require either the steady-state queue length distribution (MOI distribution) or the moments of this distribution. Obtaining such a steady-state queue length distribution necessitates either densely tracked time-series observations per host or many realizations at the same sampling time per host. However, under sparse sampling schemes, we only have two one-time-point observations per host: one at the end of wet/high-transmission and another at the end of dry/low-transmission. This is typically the case for empirical data, although numerical simulations could circumvent this limitation and generate such output.

Nonetheless, we have a population-level queue length distribution from both simulation outputs and empirical data by aggregating MOI estimates across all sampled individuals. We use this distribution to represent and approximate the steady-state queue length distribution at the individual level, not explicitly considering individual heterogeneity due to transmission. We assess the impact of individual heterogeneity due to transmission on FOI inference using simulation outputs (Appendix 1-Figure 6C). The performance of our methods across various simulated scenarios is reported and discussed in the Results and Discussion sections.

Conversion between FOI and the Entomological Inoculation Rate, or EIR

As an indirect proxy for transmission intensity, malariologists typically measure EIR by counting the number of infectious bites a human host receives within a fixed time interval (Shaukat et al., 2010 [\[1\]](#)). EIR is considered a standard metric of malaria transmission. Although both FOI and EIR reflect transmission intensity, FOI directly concerns detectable blood-stage infections, while EIR pertains to human-infectious vector contact rates. FOI is defined as the rate at which a host acquires infections, with the focus specifically on blood-stage strains for the following reason. Only blood-stage infections are detectable in all direct measures of FOI. Quantities used in indirect model-fitting approaches for estimating FOI are also based on or reflect these blood-stage strains/infections. Only these blood-stage strains/infections are transmissible to other individuals, impacting disease dynamics.

Studies comparing annual *P. falciparum* EIR and FOI estimates from age-stratified prevalence in cross-sectional parasitological studies have found significantly different magnitudes for these two quantities (Smith et al., 2010 [\[2\]](#)). The number of blood-stage infections per infectious bite (FOI/EIR) is referred to as transmission efficiency. Multiple studies indicate that malaria transmission is inefficient in high-intensity settings, and the reasons for this have been debated. Potential causes include heterogeneous biting, immunity, and measurement bias.

We utilize a functional curve with empirically derived parameters under the assumption of heterogeneous transmission, describing the highly non-linear relationship between reported EIR-FOI pairs (Smith et al., 2010 [\[2\]](#)). The functional curve, with FOI h , EIR E , and the corresponding parameters $b = 0.55$, $\alpha = 4.6$ and $t = 43$ days, is as follows:

$$h = \frac{\log(1 + \alpha b E t)}{\alpha t} \quad (2)$$

The EIR-FOI pairs (Smith et al., 2010 [↗](#)) and the functional curve provide a basis for converting between these two quantities. EIR data for a subset of surveys in Bongo District, northern Ghana, were obtained (Tiedje et al., 2022 [↗](#)). Combined with FOI estimates from our two proposed methods, we generate an EIR-FOI pair for empirical surveys in Bongo District. This enables us to evaluate whether our EIR-FOI pair aligns with historical data and the functional curve with the best-fit parameter values.

Data and code availability

The sequences utilized in this study are publicly available in GenBank under BioProject Number PRJNA 396962. All additional data associated with this study can be found in the main text, supplementary information, or obtained upon reasonable request from the authors. The simulation code and analysis scripts, please see <https://github.com/qzhan321/FOI> [↗](#).

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Additional files

supplementary file 1 [↗](#)

supplementary file 2 [↗](#)

supplementary file 3 [↗](#)

supplementary file 4 [↗](#)

supplementary file 5 [↗](#)

supplementary file 6 [↗](#)

supplementary file 7 [↗](#)

Appendix

The measurement error

We incorporate a measurement error model (Appendix 1-Figure 6D) into the sampling of simulation outputs and MOI estimation for both simulation outputs and empirical data. This model accounts for the under-sampling or imperfect detection of *var* genes in the field. By sub-sampling *var* genes per strain, we reduce the number of distinct *var* genes available per host. This model is based on the repertoire size distribution, derived from molecular sequences of infections expected to be monoclonal (MOI = 1), i.e., hosts infected by a single *P. falciparum* strain as they had 45 or fewer non-upsA DBLa types. These molecular sequences were collected during six cross-sectional surveys conducted from 2012 to 2016 in Bongo District, northern Ghana (Appendix 1-Figure 6E) (Tiedje et al., 2023 [DOI](#), Tiedje et al., 2022 [DOI](#); Pilosof et al., 2019 [DOI](#)).

Simulation data

An extended *var* model

Overview

To evaluate the performance of two proposed queuing theory methods for FOI inference across different transmission settings, we use a computational model, (He et al., 2018 [DOI](#); Zhan et al., 2024b [DOI](#)) detailed in this section and illustrated in Appendix 1-Figure 5. This model is an agent-based (individual-based), discrete-event, and continuous-time stochastic system in which all known possible future events are stored in a single event queue along with their putative times, which may be fixed or drawn from a probability distribution with a certain rate. Event rates are chosen based on malaria epidemiology literature, field studies, and in vitro/in vivo values. When an event occurs, it can cause the addition or removal of future events in the queue or the modification of their rates, resulting in a recalculation of putative times. This approach is implemented using the next-reaction method (Gibson and Bruck, 2000 [DOI](#)), an optimization of the Gillespie first-reaction method (Gillespie, 1976 [DOI](#)).

Individual human hosts die and are immediately replaced with newborns who have no immunity. The age structure of the human host population follows a truncated exponential distribution with a mean age of 30 years and a maximum age of 80 years. Individual infections and the immune history of each host are tracked. Evolutionary mechanisms, such as mitotic/ectopic recombination and mutation, are explicitly modeled.

At the beginning of each simulation, a small number of hosts are randomly selected and infected with distinct parasite genomes, assembled from a pool of *var* genes, to initiate local transmission. Mosquito vectors are not explicitly represented as agents in the model; instead, we consider an effective contact rate (referred to as the transmission rate, which under some assumptions is effectively equivalent to vectorial capacity), which determines the times of local transmission events. At these times, a donor and a recipient host are randomly selected, and successful transmission occurs only if the donor has blood-stage infections and the liver stage of the recipient is below the specified carrying capacity. The times between transmission events can follow various distributions, such as exponential and Gamma.

Var repertoire and gene structure

We assume a parasite repertoire size of 45, based on the median number of non-upsA DBLa sequences identified in our 3D7 laboratory isolate (Ruybal-Pesántez et al., 2022 [DOI](#); Tiedje et al., 2022 [DOI](#)). The classification of *var* genes into upsA and non-upsA (upsB and upsC) types is based on their semi-conserved upstream promoter sequences (ups) (Gardner et al., 2002 [DOI](#); Lavstsen et al., 2003 [DOI](#); Kraemer et al., 2007 [DOI](#); Rask et al., 2010b [DOI](#)). Although each parasite carries both types of

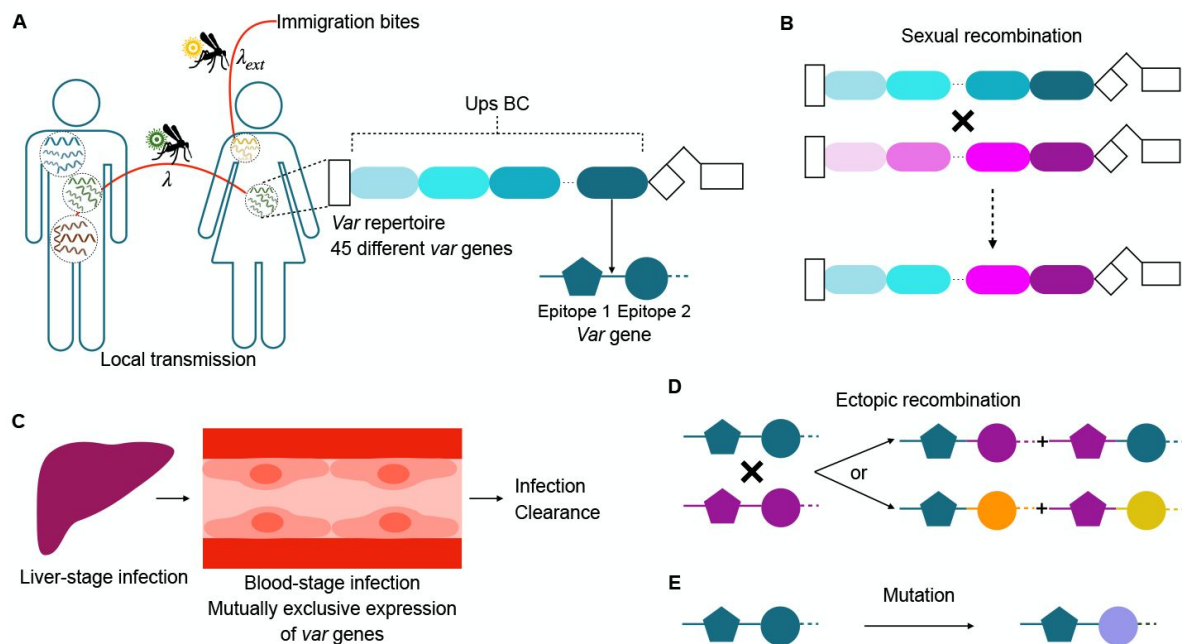


Figure 5.

Agent-based model for falciparum malaria transmission.

(A) The stochastic model tracks infection history and specific immune memory of individual hosts to variant surface antigens encoded by *var* genes. At transmission events, a donor and a recipient host are randomly selected. Transmission occurs if the donor host has blood-stage infections and the recipient host has not reached carrying capacity of infections in its liver. Each parasite genome in the donor host is transmitted to a mosquito with a probability of $1/(\text{number of genomes})$ multiplied by the transmissibility of the currently expressed gene. Each parasite genome carries 45 *var* genes, with each gene represented by a linear combination of two epitopes (depicted by different shapes), with many possible variants each (alleles, depicted by different colors). (B) During the sexual stage within mosquitoes, different parasite genomes can exchange *var* genes through meiotic recombination, generating novel recombinant repertoires. The recipient host can receive either recombinant genomes or original genomes. (C) When a repertoire is successfully transmitted to a recipient host, it undergoes a 7-day dormant liver stage before entering the blood stage, where *var* genes are sequentially expressed. If the host has no immunity against both epitopes of a *var* gene, the total duration of expression of the gene lasts 7 days. Immunity against one of the two epitopes shortens the duration of expression by half. Complete immunity against both epitopes results in immediate clearance of the gene product. Infection ends when all *var* genes in the repertoires are expressed. (D) During the asexual blood stage of infection, *var* genes within the same genome can swap their two epitope alleles through mitotic (ectopic) recombination, generating new epitopes with a certain probability. (E) *Var* genes can also mutate their epitopes to create new genes.

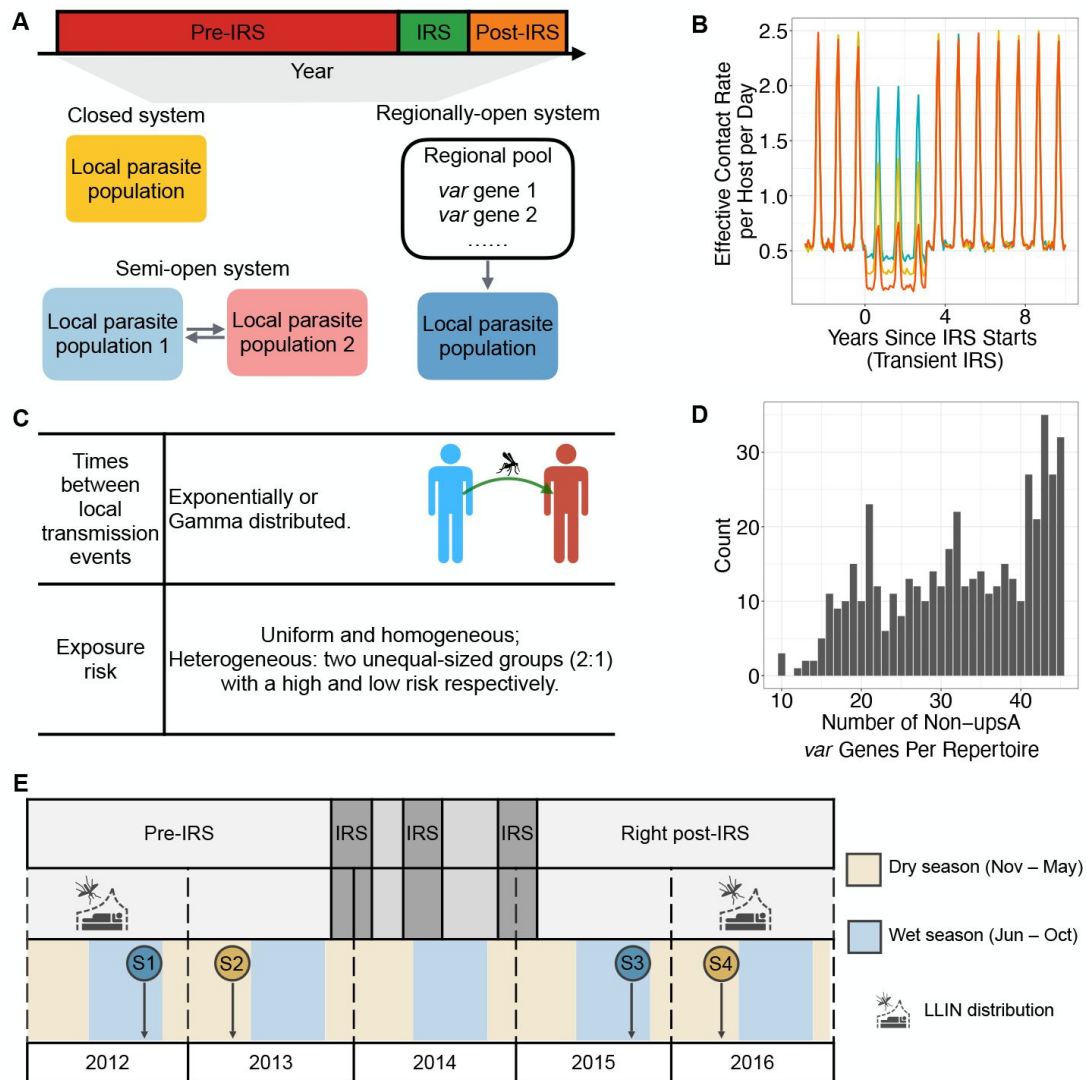


Figure 6.

(A) Each simulation comprises three stages: a “pre-IRS” period where local transmission reaches a semi-stationary state, followed by a three-year “IRS” intervention period (transient IRS) which reduces transmission rate, and a “post-IRS” period where transmission rates return to original levels. After transmission initialization, closed systems do not receive migrant genomes from the regional pool. Semi-open systems explicitly model two local populations connected by migration. Regionally-open systems continually receive migrant genomes from the regional pool throughout the simulation. (B) Transmission intensity or effective contact rate varies over time, across the pre-, during, and post-intervention periods. We simulate three levels of perturbation coverage, approximately 20% (low-coverage IRS), 40-45% (mid-coverage IRS), and 65-75% (high-coverage IRS) reduction in transmission. (C) We examine different statistical distributions for times between local transmission events: exponential and Gamma. We consider homogeneous and heterogeneous exposure risks. In the latter, $\frac{2}{3}$ of the population are high-risk, receiving approximately 94% of all bites, while the remaining population receives the rest. (D) The measurement error is depicted as a histogram showing the number of non-upsA (i.e., upsB and upsC) DBL α types per repertoire from putatively “monoclonal” infections, characterized by having 45 or fewer non-upsA DBL α types. These sequences were collected during six cross-sectional surveys conducted from 2012 to 2016 in Bongo District. This measurement error represents under-sampling or imperfect detection of *var* genes. (E) The study consists of four age-stratified cross-sectional surveys in Bongo District, Ghana, conducted at the end of wet/high-transmission seasons (blue circles) and dry/low-transmission seasons (gold circles). Two phases are covered: (1) Pre-IRS: Survey 1 (S1) in October 2012 and Survey 2 (S2) in May/June 2013; (2) Right post-IRS: Survey 3 (S3) in October 2015 and Survey 4 (S4) in May/June 2016. IRS was implemented with widespread LLIN usage distributed between 2010-2012 and again in 2016 (Tiedje et al., 2023; Gogue et al., 2020).

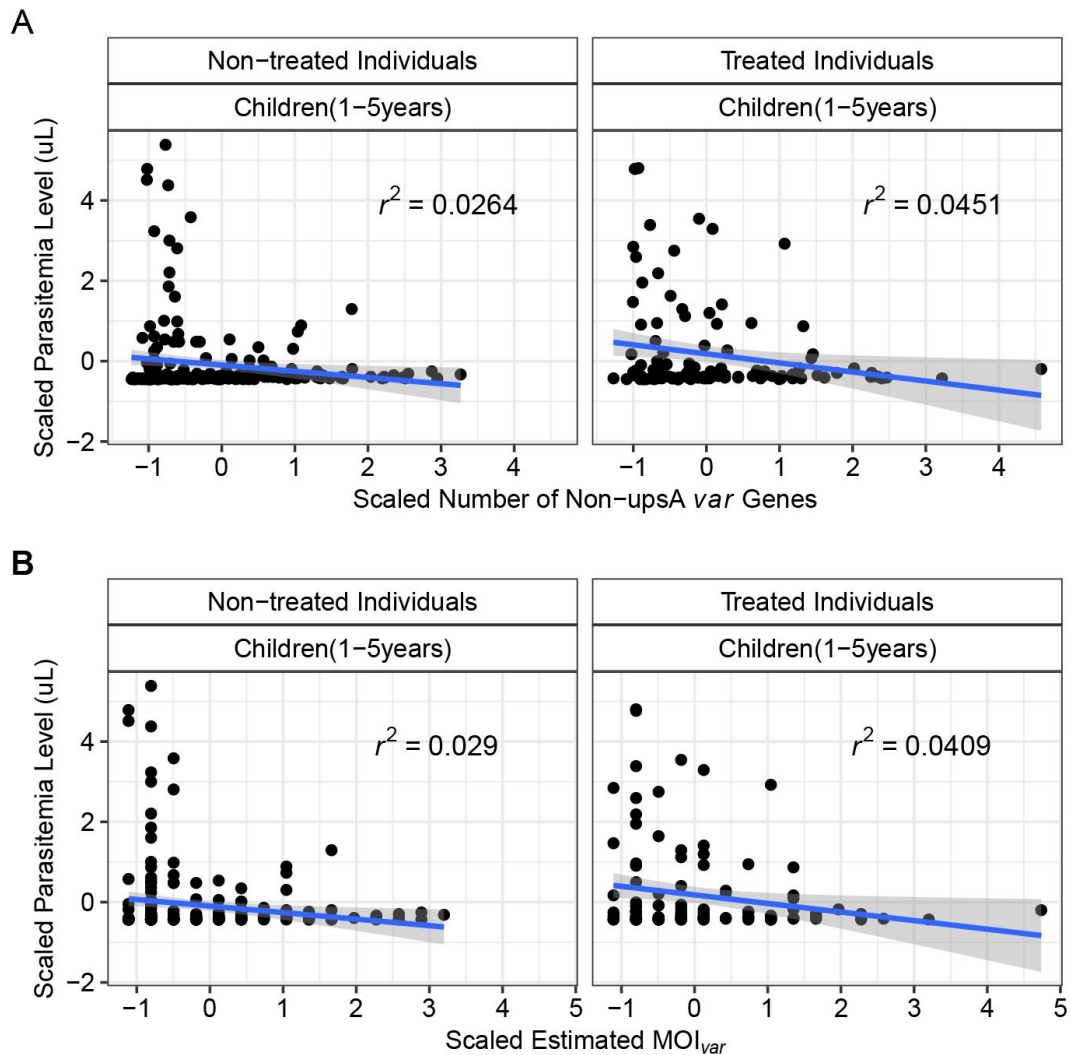


Figure 7.

The relationship between the parasitemia level of the individual (measured in uL) and (A) the number of non-upsA *var* genes per isolate/individual, or (B) MOI estimates from the Bayesian formulation of the *var*coding method. There is a lack of association between the x-axis and y-axis variables among both untreated and antimalarial drug-treated individuals. We scale the parasitemia levels and the number of non-ups A *var* genes or MOI estimates before performing the regression.

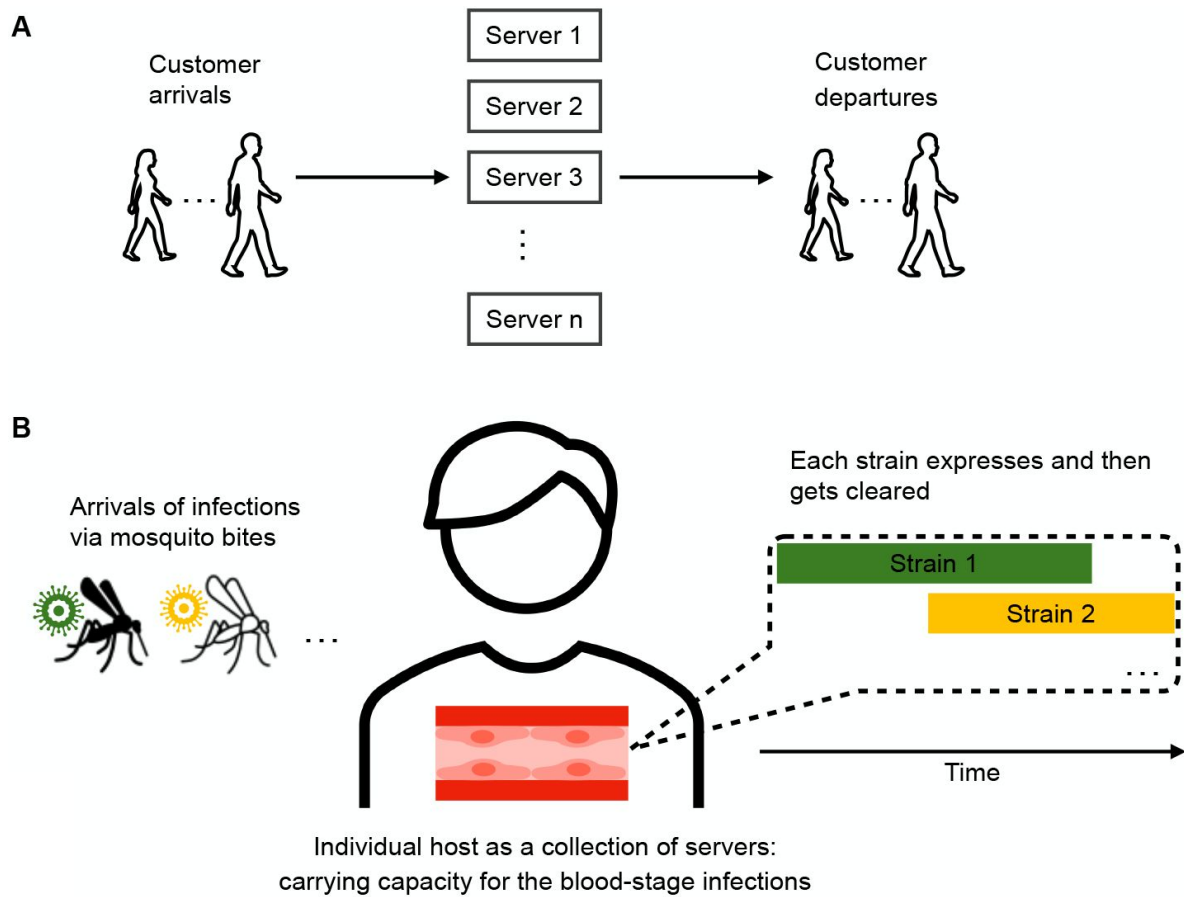


Figure 8.

Schematic illustration of (A) systems in queueing theory and (B) malaria transmission.

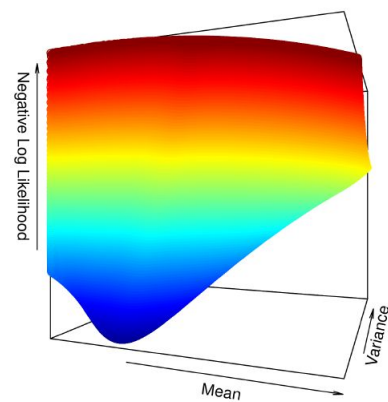
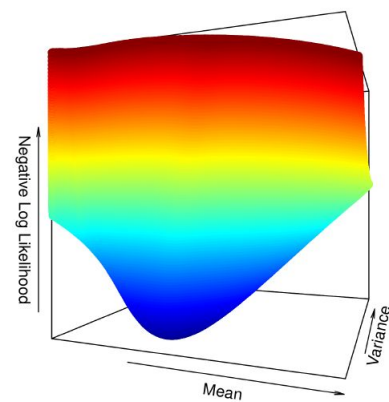
A**B**

Figure 9.

The shape of the negative log likelihood for (A) a simulation run (pre-IRS) with Gamma-distributed times between local transmission events in a seasonal, semi-open system with heterogeneous exposure risk, and (B) Ghana pre-IRS surveys (Survey 1 and 2) with $c = 30$ and mid PCR detectability. We remove the infinite and extremely large values of the negative log likelihood, and plot the rest to improve visualization.

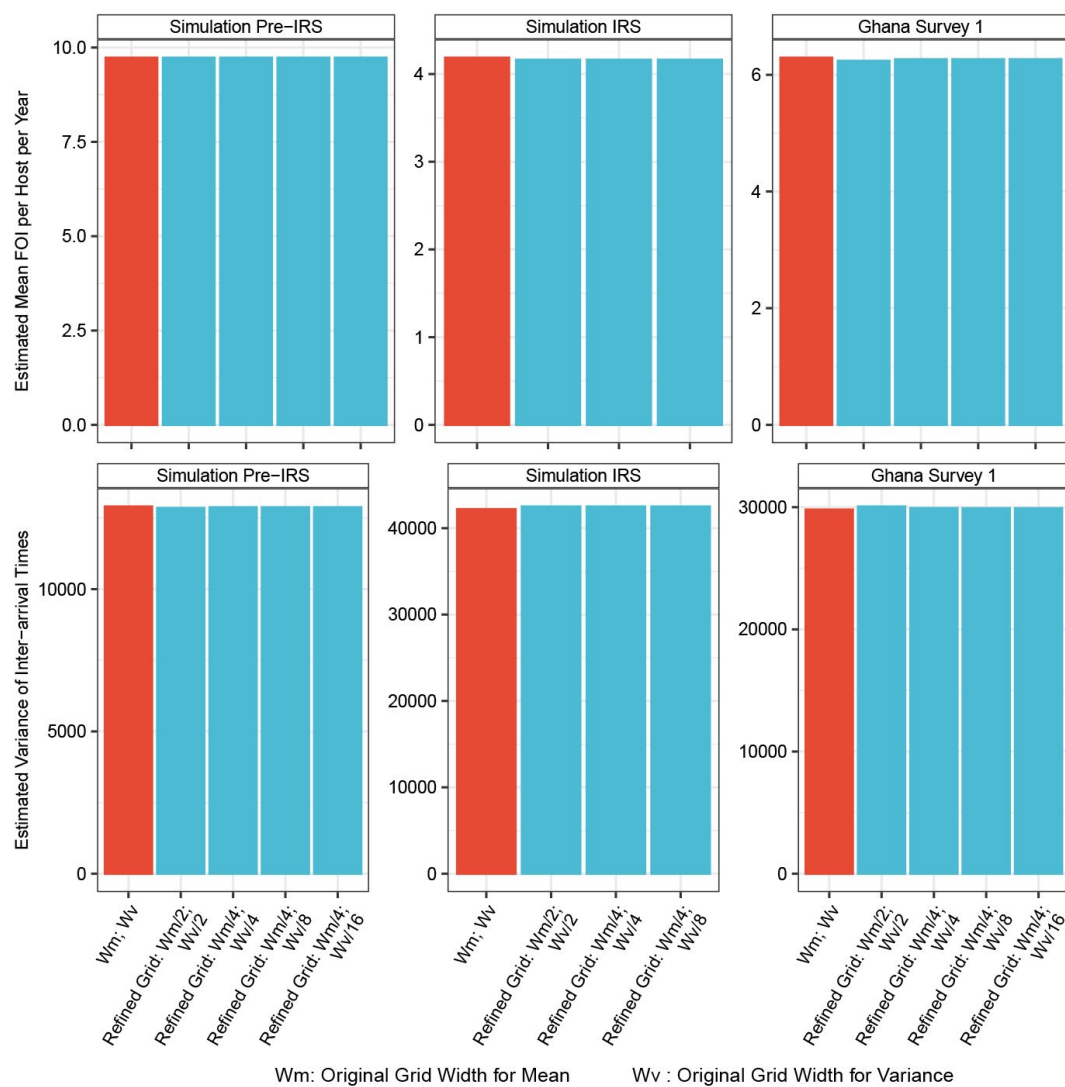


Figure 10.

The impact of grid value choices on the results of FOI inference in either simulated outputs or Ghana data.

By further reducing the grid width to include more combinations of the mean and variance values of inter-arrival times, the FOI inference results remain either unchanged or deviate by no more than 1% from those based on the original grid width.

var genes in a fairly constant proportion (Ruybal-Pesántez et al., 2017; Tiedje et al., 2023; Ruybal-Pesántez et al., 2022; Buckee and Recker, 2012; Rask et al., 2010b), we focus on the more diverse and less conserved non-upsA sequences for MOI estimation. Despite functional differences, the groupings do not necessarily correlate with function, as there is within-group functional heterogeneity and cross-group functional similarity (Claessens et al., 2012; Kaestli et al., 2006; Rottmann et al., 2006). The mechanism behind the fairly constant proportion of these groups in empirical samples across different times and locations remains unclear, so we simplify by considering only the non-upsA type.

Each gene is modeled as a linear combination of two epitopes (alleles), based on the empirical description of the two hypervariable subregions in the *var* tag region amplified from field isolates (Larremore et al., 2013).

Ectopic recombination

We model ectopic recombination among genes within the same genome during the asexual stage inside the human host. This process is a major mechanism of *var* gene diversification, occurring during both sexual and asexual stages (Claessens et al., 2014). For simplicity, we focus on the asexual stage, where two randomly selected genes from a strain undergo recombination. The breakpoint location is chosen randomly. Under normal recombination, alleles of the two genes are swapped with a probability of creating new alleles. Under conversion, the second gene remains unchanged. In this implementation, we assume all events result in normal recombination rather than gene conversion.

Newly recombined genes have a probability of being functional (i.e., viable), influenced by the similarity of their parental genes and the breakpoint locations (Drummond et al., 2005):

$$P = \rho^{\frac{x(d-x)}{d-1}}$$

Where ρ represents recombination tolerance, d is the genetic distance between the two parental genes, and x is the genetic distance between the offspring gene and one of the two parental genes.

Non-functional offspring genes replace their parental ones. As they do not express and get deactivated immediately, they shorten the infection duration of the strain they constitute, thereby reducing its fitness.

Meiotic recombination

Meiotic recombination occurs between strains during sexual replication inside the mosquito vector. While we do not explicitly model mosquitoes, we represent meiotic recombination between genomes at the time of a transmission event.

When multiple strains, denoted by m ($m > 1$), co-infect a donor host, a Bernoulli trial is conducted for each strain to determine if it will be transmitted via the contact event, with the success probability equal to its transmissibility. Each strain's transmissibility depends on the currently expressed gene's transmissibility. Co-infection reduces the transmissibility of each strain by a factor of m . Only a subset n of these strains, where $n \leq m$, is selected. In nature, this subset would co-infect a mosquito vector.

To simulate meiotic recombination, each of the n strains which are to be transmitted from the mosquito vector to a recipient human host is obtained by drawing two parental strains from the pool of n with replacement. If the two parental strains are the same, the original strain is transmitted. If different, they recombine, and the recombinant strain is transmitted. The probabilities of transmitting the original or recombinant strain are $\frac{1}{n}$ and $1 - \frac{1}{n}$, respectively.

Given that orthologous gene pairs between two parental strains are often unknown, we implement meiotic recombination by randomly selecting genes from the pooled genes of the two strains. This assumption is reasonable as physical locations of *var* genes can be mobile through ectopic recombination and gene conversions. The resulting offspring strains share some fraction of their *var* genes with the parental strains.

Within-host dynamics

Each strain is individually tracked through its entire life cycle, encompassing the liver stage and asexual blood stage in the human host, and the sexual stage in the mosquito. As we do not explicitly model mosquitoes, we delay the expression of each strain in the recipient host by 7 days to account for the sexual stage, e.g., the time required for gametocytes to develop into sporozoites within mosquitoes. Additionally, we delay the expression of each strain by another 7 days to account for the liver stage, e.g., the time required for parasites to be released as merozoites into the bloodstream to invade red blood cells. Hence, after a total of 14 days, the active asexual bloodstage and the expression of the *var* repertoire begin.

The *var* genes in an infection are expressed sequentially. During repertoire expression, the host is considered infectious with the active strain, and only these blood-stage strains are transmissible to another host. The deactivation rate of the genes is controlled by the host's specific immunity. When one gene is actively expressed, host immunity "checks" whether it has seen any of the two epitopes in its infection history. If both epitopes have been previously encountered, the gene is deactivated immediately. The duration of the active period of a gene is therefore proportional to the number of unseen epitopes of the gene. Once deactivated, the host adds the deactivated two epitopes to its immunity memory. The next new gene from the repertoire then becomes immediately active. The strain is cleared from the host when the whole repertoire of *var* genes is depleted. Hence, the total infection duration of a particular repertoire is proportional to the number of unseen epitopes by the infected host aggregated across all individual *var* genes of that repertoire. Immunity to a certain epitope wanes over time (COLLINS et al., 1968 [DOI](#); Collins et al., 1964 [DOI](#)), requiring re-exposure to maintain it.

We consider carrying capacities for both liver- and blood-stage infection, adopting a value of 30 for both based on reasons detailed in Materials and Methods-Choice of c and r for the $GI/G/c/c + r$ queue in the two-moment approximation method ?. When the number of liver-stage strains reaches the specified carrying capacity, the host no longer receives additional infections if selected as the recipient host for transmission events. Similarly, when the number of blood-stage strains reaches the carrying capacity, liver-stage strains are not released into the bloodstream and fail to transition to the blood-stage, effectively being lost.

Details of the parameters and their values are summarized in the supplementary file 5-simParams.xlsx.

Experimental designs

Each simulation runs for either 200 years (closed systems) or 150 years (open systems) to reach a semi-stationary state before introducing transmission-reducing interventions. We simulate malaria dynamics with parameters representative of high-transmission endemic regions, considering both constant and seasonal transmission, and across different spatial configurations, including closed, semi-open, and regionally-open systems.

We consider indoor residual spraying (IRS), which involves applying insecticide to the internal walls and ceilings of homes (World Health Organization, 2015 [DOI](#)) in the field. IRS effectively reduces the mosquito population, thereby decreasing the transmission rate. We simulate three

temporary IRS interventions with varying coverages: low (reducing transmission by around 20%), mid (reducing transmission by around 40-45%), and high (reducing transmission by around 65-75%). Each IRS intervention lasts for three years.

Seasonality is represented as a scaling constant multiplied by a temporal vector of 360 days, which represents the daily number of mosquitoes over a year. This temporal vector (Pilosof et al., 2019 [\[4\]](#)) is derived from a deterministic model of mosquito population dynamics (White et al., 2011 [\[5\]](#)). The model, originally developed for *Anopheles gambiae*, includes a set of ordinary differential equations describing the dynamics of four mosquito stages: eggs, larvae, pupae, and adults. Seasonality is implemented via density dependence at the egg and larva stages as a function of rainfall (availability of breeding sites). For our purposes, the values of the effective contact rate are more critical than the absolute number of daily mosquitoes. Essentially, we have a basic vector that represents the number of mosquitoes throughout the year and a scaling constant that encapsulates all other parameters related to vectorial capacity. The product of this temporal vector and the scaling constant results in the effective contact rate.

In closed systems, migration is discontinued after the initial seeding of local transmission from a regional pool of *var* genes. In semi-open systems, two individual populations are explicitly coupled via migration. Regionally-open systems involve a local population with migration from a regional pool, which acts as a proxy for regional parasite diversity, i.e., diversity from the aggregated individual populations in the region. Because each parasite genome is a repertoire with a given number of *var* genes, migrant genomes are assembled by randomly sampling *var* genes from the regional pool.

Transmission can be homogeneous or heterogeneous across individual hosts. For heterogeneity, we consider two groups of human hosts: a high risk group, which receives approximately 94% of the bites, and a low risk group, which receives the remaining fraction. The high-risk group is twice as large as the low-risk group (2:1 ratio).

Test of deviation from Poisson homogeneity in MOI estimates

Many studies have explored whether a count dataset deviates significantly from a homogeneous Poisson distribution. We use the Potthoff-Whittinghill “index of dispersion” test (Potthoff and Whittinghill, 1966 [\[6\]](#); Lloyd-Smith et al., 2005 [\[7\]](#)), which is asymptotically locally most powerful against the negative binomial alternative. For a dataset X with N elements, the statistic is $\frac{(N-1) \cdot \text{var}(X)}{\text{mean}(X)}$ and its asymptotic distribution is chi-squared with $N - 1$ degrees of freedom. The p-value is the cumulative density of the chi-squared ($N - 1$) distribution to the right of the test statistic, representing the probability that the observed variance could have arisen by chance from a Poisson distribution.

We summarize the p-values of the Potthoff-Whittinghill “index of dispersion” test in supplementary file 2-deviationFromPoissonTest.xlsx. The p-values are close to 0, indicating a significant deviation from a Poisson distribution for both simulated outputs and empirical surveys in Bongo District. Few exceptions occur in high-coverage IRS scenarios in the simulated outputs, where prevalence is very low and most infections are mono-clonal.

Reducing stochastic impact in sampling processes

To mitigate the effects of stochasticity in various sampling processes within the analysis, we perform 200 realizations, including the incorporation of the measurement error model into the simulation output. Each realization results in a slightly different collection of subsampled *var* genes per host, potentially leading to minor variations in individual MOI estimates. To impute MOI estimates for treated individuals or those with missing data, we conduct 200 samplings from

available non-treated individuals. We then calculate a final population-level MOI distribution, weighted across these 200 realizations or samplings. This approach reduces the impact of extreme single-sampling variations on MOI estimates and FOI inference.

Confidence intervals for FOI inference

Non-parametric bootstrap

Bootstrap datasets are generated by resampling with replacement from the original populationlevel MOI distribution, i.e., the collection of individual MOI estimates, using individual-level MOI as the unit of the bootstrap sampling. We run 200 bootstrap replicates, as this number has been tested and shown to achieve a coefficient of variation comparable to that obtained with a higher number of replicates (Efron and Tibshirani, 1994 [\[1\]](#)). For each bootstrap dataset, we determine the maximum likelihood estimates for FOI. The 200 replicates thus produce a bootstrap sampling distribution for FOI estimates. We calculate the skewness of these distributions. The majority fall within the range of -0.5 to 0.5, with a few exceptions falling within the range of 0.5-0.75. Therefore, we consider them fairly symmetric and do not apply a skewness adjustment to ensure good coverage. Detailed skewness values can be found in supplementary file 6-FOIBootstrapSkewness.xlsx.

A comparison of the performance of the Bayesian formulation and the original *var*coding method based on the simulation output

We compare the performance of the original *var*coding method and its Bayesian formulation against true MOI values using the Cramer-von Mises and Anderson-Darling tests.

MOI estimates from the *var*coding method often differ significantly from true MOI values (pvalue < 0.05). Similarly, there is frequently a statistically significant difference between MOI estimates from the *var*coding method and those from the Bayesian formulation. However, differences between MOI estimates from the Bayesian formulation and true MOI values are generally not statistically significant.

Therefore, the Bayesian formulation improves upon the original *var*coding method, which assumes a constant repertoire size to estimate MOI. This improvement is expected, as the Bayesian formulation accounts for variation in repertoire size due to *var* gene under-sampling.

Both methods perform well in low-transmission settings, where true MOI values are low. In moderate and high-transmission scenarios, the *var*coding method remains reasonably effective, but the Bayesian formulation demonstrates a notable improvement in capturing higher MOI values, where measurement error is more pronounced. This improvement is significant in the high-transmission endemic Bongo District of Ghana, where our empirical MOI estimates were derived.

The documented test results are included in supplementary file 7-BayesianImprovement.xlsx. Overall, both methods tend to underestimate MOI. While the Bayesian formulation provides a more accurate and robust estimation by addressing the imperfect detection of *var* genes, neither method accounts for other factors that may reduce the number of *var* genes detected per individual (see Discussion).

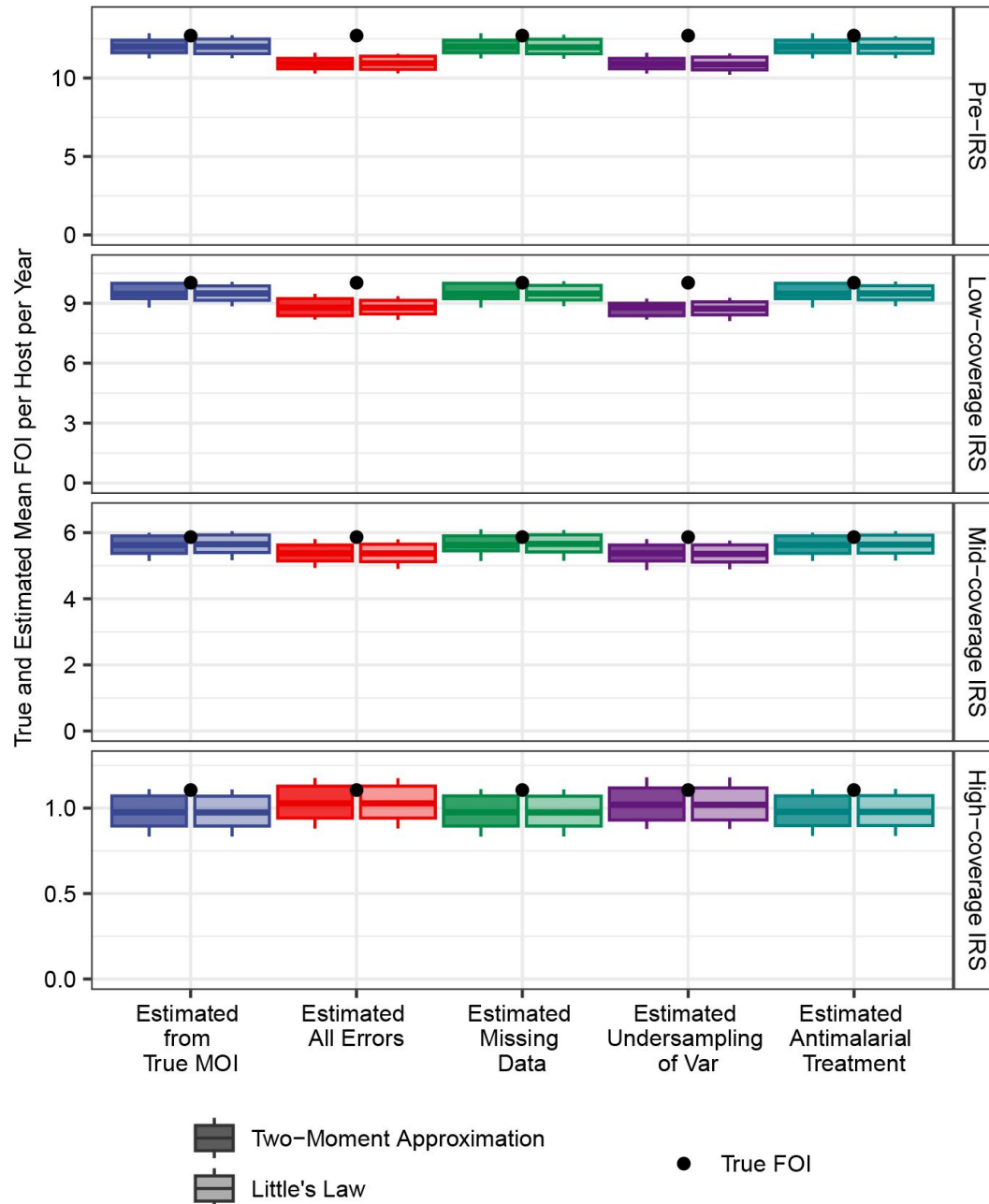


Figure 11.

True and estimated FOI by the two-moment and Little's Law methods for additional simulated scenarios of homogeneous exposure risk.

The times between local transmission events are Gamma-distributed, with non-seasonal transmission in a closed system. The true mean FOI per host per year is calculated by dividing the total number of infections acquired by the population by the total number of hosts in the population. Each boxplot shows minimum, 5% quantile, median, 95% quantile, and maximum values.

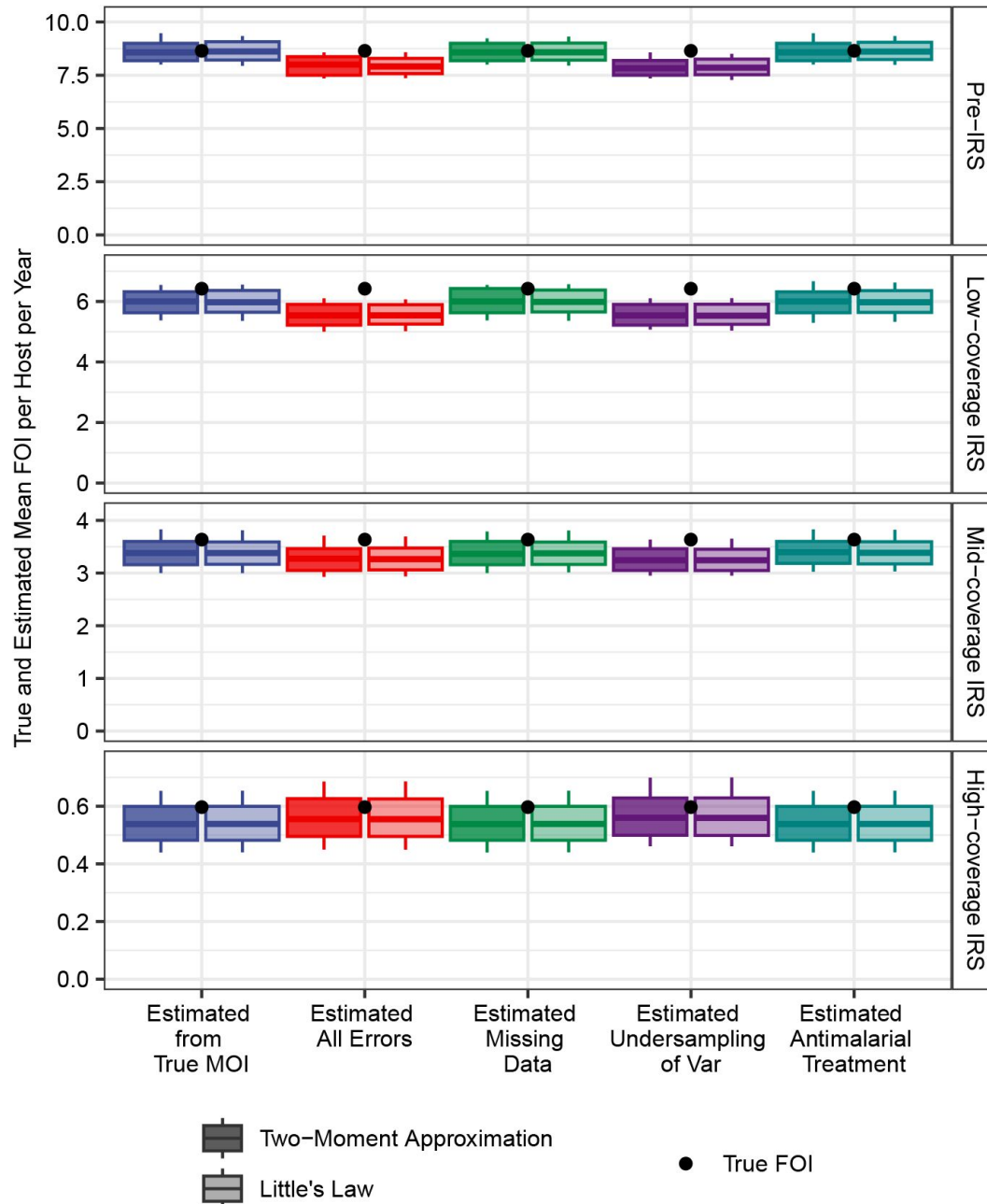


Figure 12.

True and estimated FOI by the two-moment and Little's Law methods for additional simulated scenarios of heterogeneous exposure risk.

The times between local transmission events are Gamma-distributed, with non-seasonal transmission in a semi-open system. The true mean FOI per host per year is calculated by dividing the total number of infections acquired by the population by the total number of hosts in the population. Each boxplot shows minimum, 5% quantile, median, 95% quantile, and maximum values.

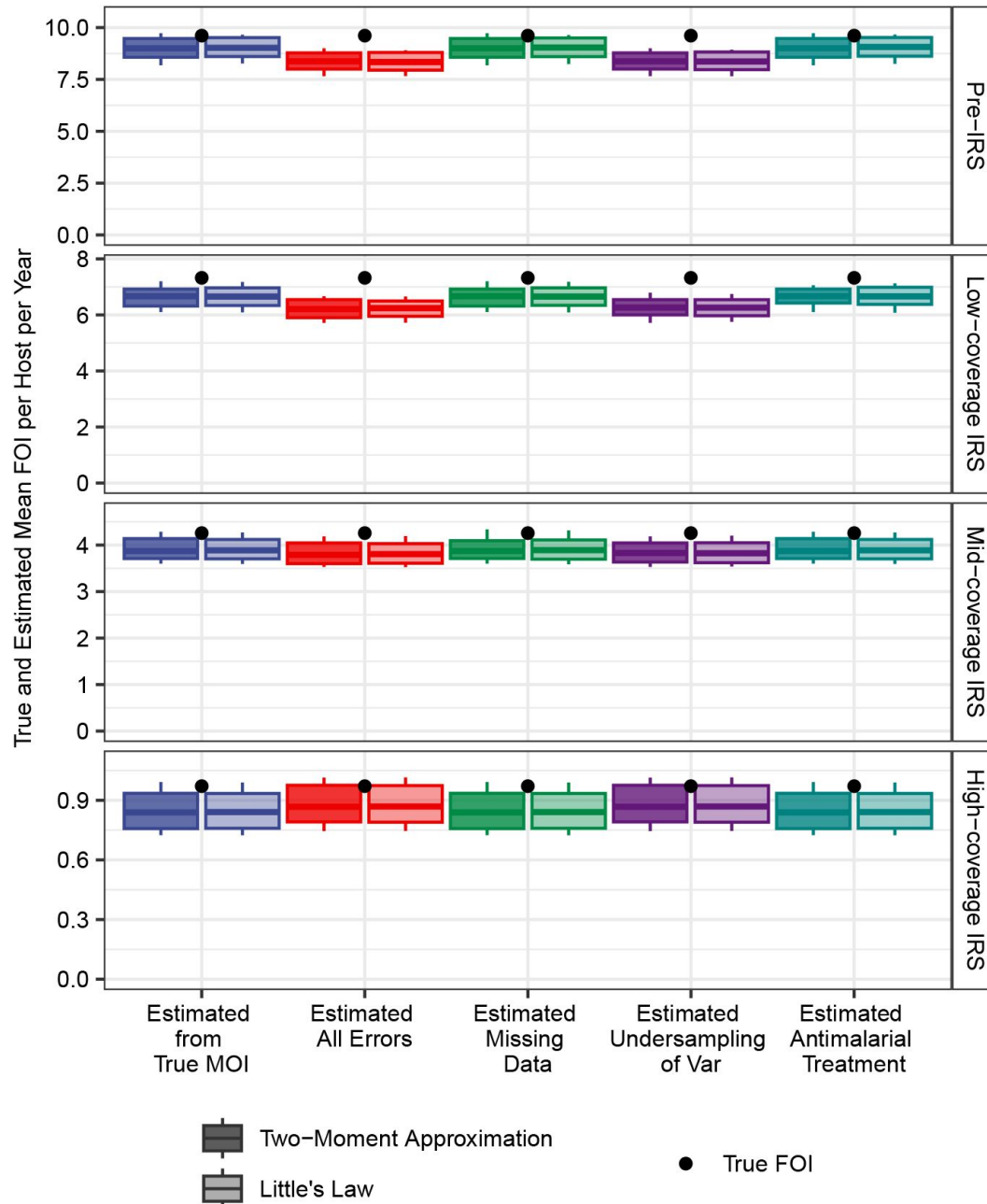


Figure 13.

True and estimated FOI by the two-moment and Little's Law methods for additional simulated scenarios of homogeneous exposure risk.

The times between local transmission events are Gamma-distributed, with seasonal transmission in a regionally-open system. The true mean FOI per host per year is calculated by dividing the total number of infections acquired by the population by the total number of hosts in the population. Each boxplot shows minimum, 5% quantile, median, 95% quantile, and maximum values.

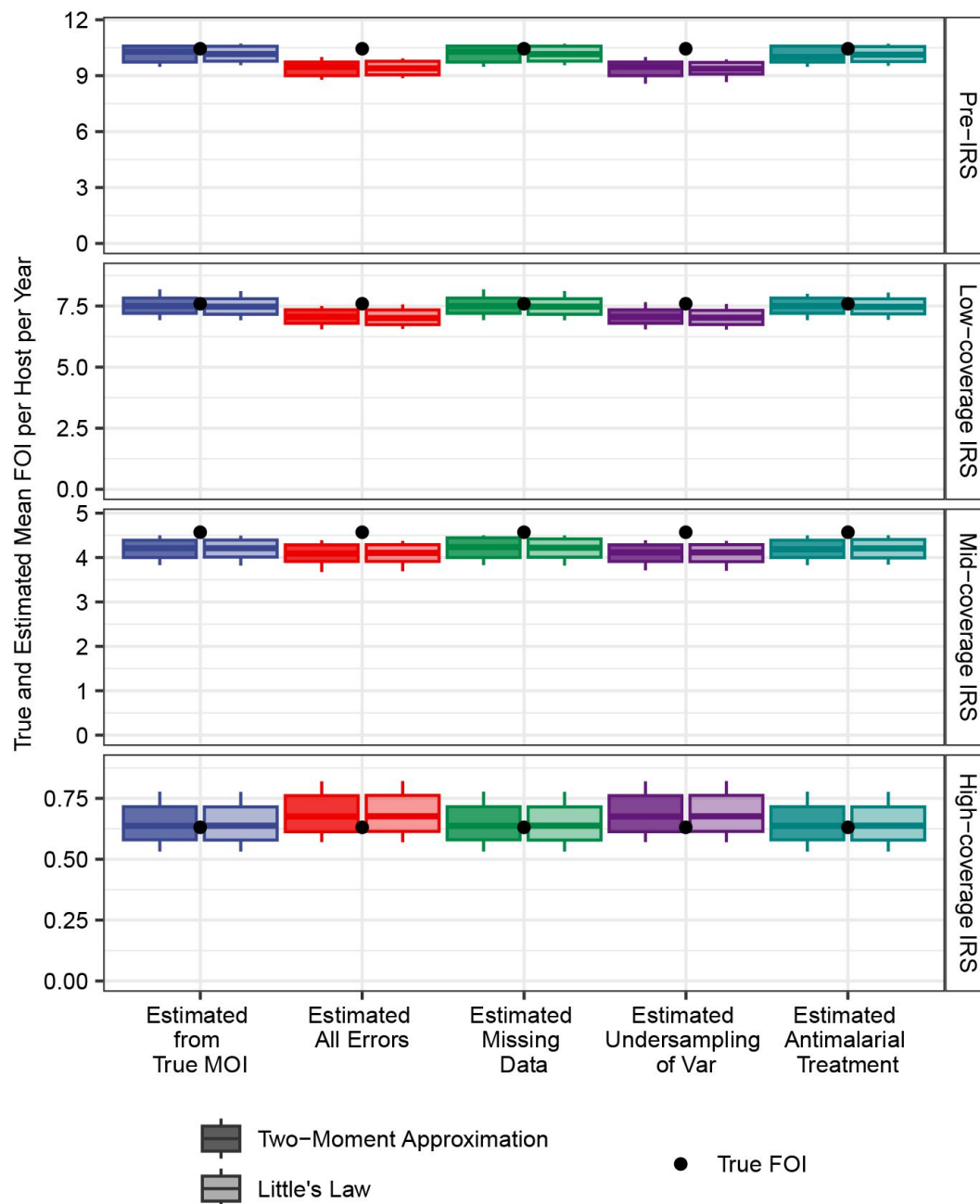


Figure 14.

True and estimated FOI by the two-moment and Little's Law for additional simulated scenarios of homogeneous exposure risk.

The times between local transmission events are Gamma-distributed, with non-seasonal transmission in a regionally-open system. The true mean FOI per host per year is calculated by dividing the total number of infections acquired by the population by the total number of hosts in the population. Each boxplot shows minimum, 5% quantile, median, 95% quantile, and maximum values.

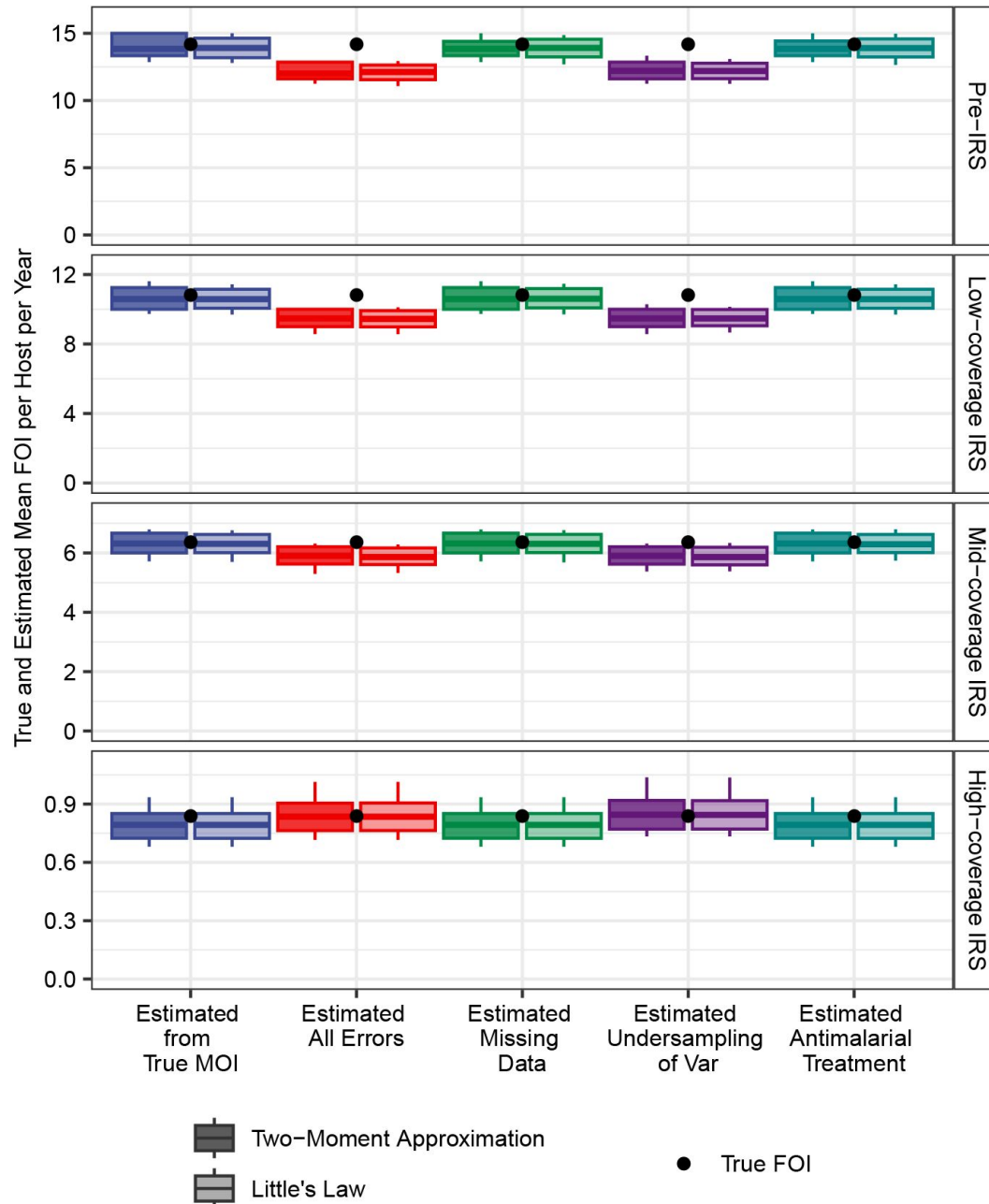


Figure 15.

True and estimated FOI by the two-moment and Little's Law methods for additional simulated scenarios of homogeneous exposure risk.

The times between local transmission events follow an exponential distribution, with seasonal transmission in a closed system. The true mean FOI per host per year is calculated by dividing the total number of infections acquired by the population by the total number of hosts in the population. Each boxplot shows minimum, 5% quantile, median, 95% quantile, and maximum values.

Figure 16.

Confidence intervals for the estimated mean FOI values in Ghana surveys before and immediately after a transient three-round IRS intervention.

(A) The estimated FOI values when excluding these treated individuals from the analysis. (B) The estimated FOI values when discarding the infection status and MOI estimates of treated individuals and sampling from non-treated ones. Each boxplot shows minimum, 5% quantile, median, 95% quantile, and maximum. The value of c is set to 25.

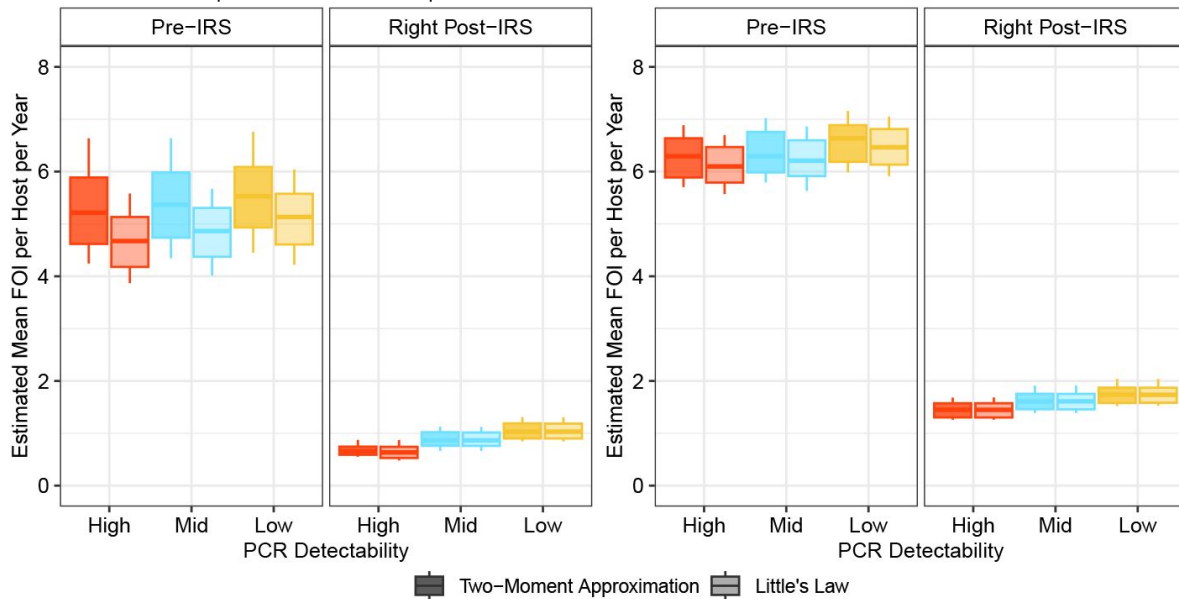
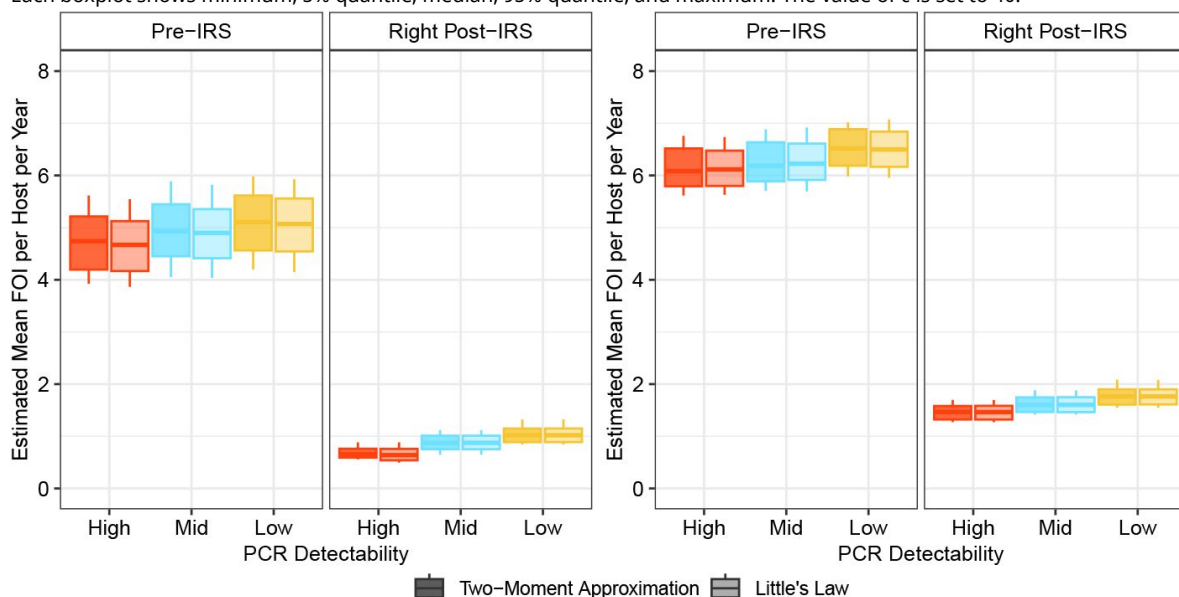


Figure 17.

Confidence intervals for the estimated mean FOI values in Ghana surveys before and immediately after a transient three-round IRS intervention.

(A) The estimated FOI values when excluding these treated individuals from the analysis altogether. (B) The estimated FOI values when discarding the infection status and MOI estimates of treated individuals and sampling from non-treated ones. Each boxplot shows minimum, 5% quantile, median, 95% quantile, and maximum. The value of c is set to 40.



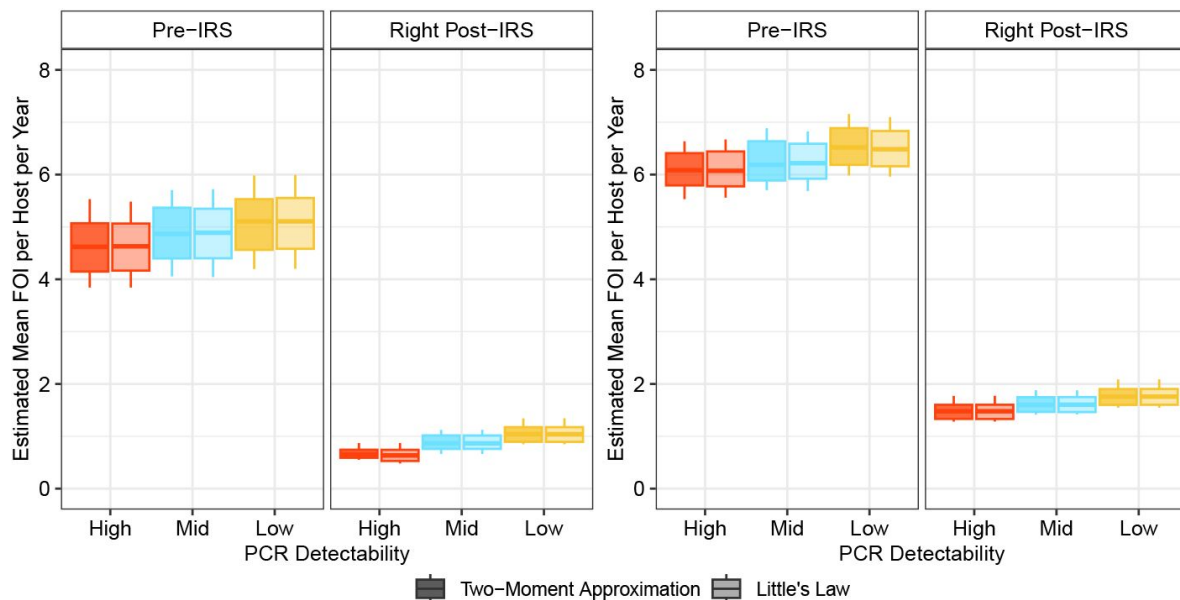


Figure 18.

Confidence intervals for the estimated mean FOI values in Ghana surveys before and immediately after a transient three-round IRS intervention.

(A) The estimated FOI values when excluding these treated individuals from the analysis. (B) The estimated FOI values when discarding the infection status and MOI estimates of treated individuals and sampling from non-treated ones. Each boxplot shows minimum, 5% quantile, median, 95% quantile, and maximum. The value of c is set to 60.

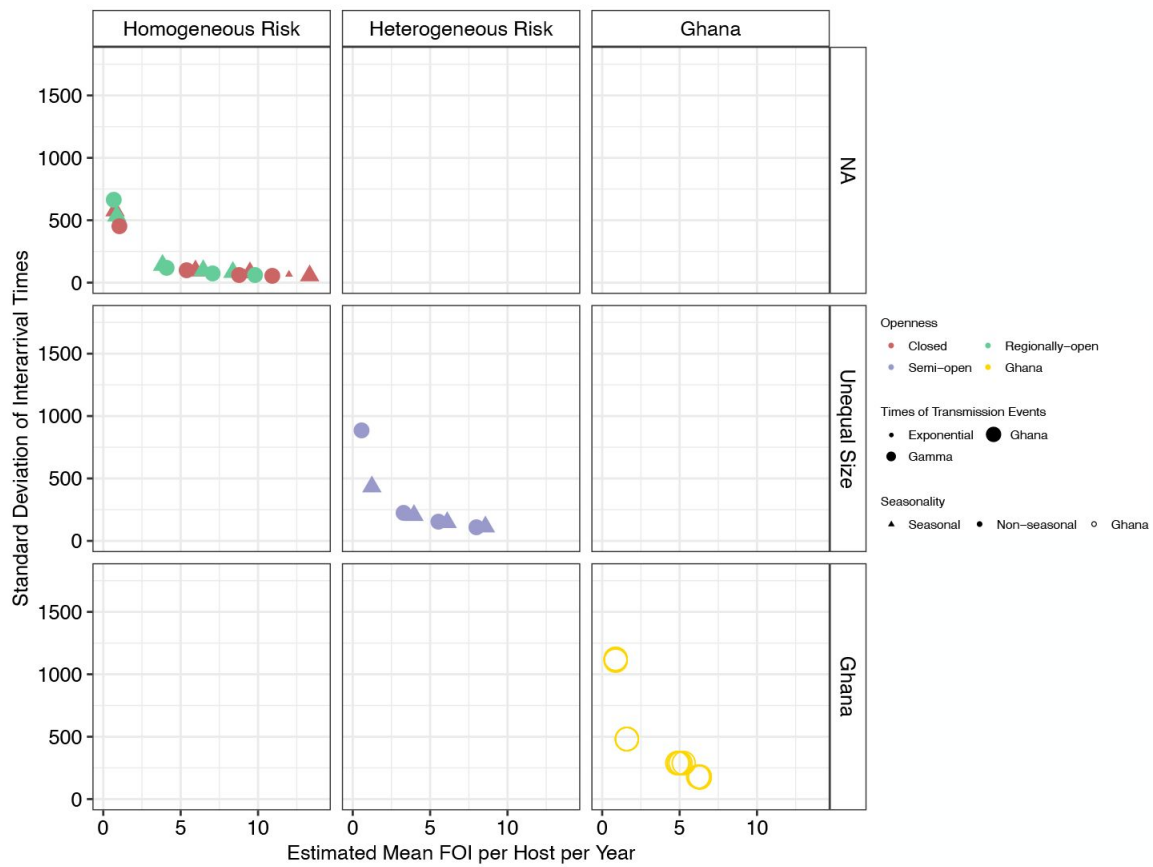


Figure 19.

Estimated standard deviation of the inter-arrival times using the two-moment approximation method across different simulation scenarios and field data from Bongo District, Ghana.

References

- Abukari Z, Okonu R, Nyarko SB, Lo AC, Dieng CC, Salifu SP, Gyan BA, Lo E, Amoah LE (2019) **The Diversity, Multiplicity of Infection and Population Structure of *P. falciparum* Parasites Circulating in Asymptomatic Carriers Living in High and Low Malaria Transmission Settings of Ghana** *Genes (Basel)* <https://doi.org/10.3390/genes10060434>Google Scholar
- Alonso PL, Sacarlal J, Aponte JJ, et al. (2004) **Efficacy of the RTS,S/AS02A vaccine against *Plasmodium falciparum* infection and disease in young African children: randomised controlled trial** *The Lancet* **364**:1411–1420 [https://doi.org/10.1016/S0140-6736\(04\)17223-1](https://doi.org/10.1016/S0140-6736(04)17223-1)Google Scholar
- Anderson TJ, Haubold B, Williams JT, Estrada-Franco JG, Richardson L, Mollinedo R, Bockarie M, Mokili J, Mharakurwa S, French N, Whitworth J, Velez ID, Brockman AH, Nosten F, Ferreira MU, Day KP (2000) **Microsatellite Markers Reveal a Spectrum of Population Structures in the Malaria Parasite *Plasmodium falciparum*** *Molecular Biology and Evolution* **17**:1467–1482 <https://doi.org/10.1093/oxfordjournals.molbev.a026247>Google Scholar
- Andolina C, Rek JC, Briggs J, Okoth J, Musiime A, Ramjith J, Teyssier N, Conrad M, Nankabirwa JI, Lanke K, Rodriguez-Barraquer I, Meerstein-Kessel L, Arinaitwe E, Olwoch P, Rosenthal PJ, Kamya MR, Dorsey G, Greenhouse B, Drakeley C, Staedke SG, et al. (2021) **Sources of persistent malaria transmission in a setting with effective malaria control in eastern Uganda: a longitudinal, observational cohort study** *The Lancet Infectious diseases* [https://doi.org/10.1016/S1473-3099\(21\)00072-4](https://doi.org/10.1016/S1473-3099(21)00072-4)Google Scholar
- Andrade CM, Carrasquilla M, Dabbas U, Briggs J, van Dijk H, Sergeev N, Sissoko A, Niangaly M, Ntalla C, LaVerriere E, Skinner J, Golob K, Richter J, Cisse H, Li S, Hendry JA, Asghar M, Doumtable D, Farnert A, Ruppert T, et al. (2024) **Infection length and host environment influence on *Plasmodium falciparum* dry season reservoir** *EMBO Molecular Medicine* **16**:2349–2375 <https://doi.org/10.1038/s44321-024-00127-w>Google Scholar
- Argyropoulos DC, Tan MH, Adobor C, Mensah B, Labbé F, Tiedje KE, Koram KA, Ghansah A, Day KP (2023) **Performance of SNP barcodes to determine genetic diversity and population structure of *Plasmodium falciparum* in Africa** *Front Genet* <https://doi.org/10.3389/fgene.2023.1071896>Google Scholar
- Arnot D. (1998) **Clone multiplicity of *Plasmodium falciparum* infections in individuals exposed to variable levels of disease transmission** *Transactions of the Royal Society of Tropical Medicine and Hygiene* **92**:580–585 [https://doi.org/10.1016/S0035-9203\(98\)90773-8](https://doi.org/10.1016/S0035-9203(98)90773-8)Google Scholar
- Ashley EA, White NJ (2014) **The duration of *Plasmodium falciparum* infections** *Malar J* <https://doi.org/10.1186/1475-2875-13-500>Google Scholar
- Barry A, Awandu SS, Tiono AB, Grignard L, Bousema T, Collins KA (2021) **Improved Detectability of *Plasmodium falciparum* Clones with Repeated Sampling in Incident and Chronic Infections in Burkina Faso** *Am J Trop Med Hyg* <https://doi.org/10.4269/ajtmh.21-0493>Google Scholar

- Barry AE, Leliwa-Sytek A, Tavul L, Imrie H, Migot-Nabias F, Brown SM, McVean GAV, Day KP (2007) **Population Genomics of the Immune Evasion (var) Genes of Plasmodium falciparum** *PLoS Pathog* <https://doi.org/10.1371/journal.ppat.0030034>Google Scholar
- Baruch DI, Pasloske BL, Singh HB, et al. (1995) **Cloning the P. falciparum gene encoding PfEMP1, a malarial variant antigen and adherence receptor on the surface of parasitized human erythrocytes** *Cell* [https://doi.org/10.1016/0092-8674\(95\)90054-3](https://doi.org/10.1016/0092-8674(95)90054-3)Google Scholar
- Beier JC, Oster CN, Onyango FK, Bales JD, Sherwood JA, Perkins PV, Chumo DK, Koech DV, Whitmire RE, Roberts CR, Diggs CL, Hoffman SL (1994) **Plasmodium falciparum incidence relative to entomologic inoculation rates at a site proposed for testing malaria vaccines in western Kenya** *The American journal of tropical medicine and hygiene* <https://doi.org/10.4269/ajtmh.1994.50.529>Google Scholar
- Bekessy A, Molineaux L, Storey J. (1976) **Estimation of incidence and recovery rates of Plasmodium falciparum parasitaemia from longitudinal data** *Bull World Health Organ* [Google Scholar](https://doi.org/10.15185/wch.1976.1976)
- Bopp SER, Manary MJ, Bright AT, Johnston GL, Dharia NV, Luna FL, et al. (2013) **Mitotic Evolution of Plasmodium falciparum Shows a Stable Core Genome but Recombination in Antigen Families** *PLoS Genet* <https://doi.org/10.1371/journal.pgen.1003293>Google Scholar
- Bretscher MT, Maire N, Chitnis N, Felger I, Owusu-Agyei S, Smith T. (2011) **The distribution of Plasmodium falciparum infection durations** *Epidemics* <https://doi.org/10.1016/j.epidem.2011.03.002>Google Scholar
- Bruce MC, Galinski MR, Barnwell JWea (2000) **Genetic diversity and dynamics of plasmodium falciparum and P. vivax populations in multiply infected children with asymptomatic malaria infections in Papua New Guinea** *Parasitology* <https://doi.org/10.1017/S0031182099006356>Google Scholar
- Buckee CO, Recker M. (2012) **Evolution of the multi-domain structures of virulence genes in the human malaria parasite, Plasmodium falciparum** *PLoS Comput Biol* <https://doi.org/10.1371/journal.pcbi.1002451>Google Scholar
- Carneiro I, Roca-Feltre A, Griffin JT, Smith L, Tanner M, Schellenberg JA, Greenwood B, Schellenberg D. (2010) **Age patterns of malaria vary with severity, transmission intensity and seasonality in sub-Saharan Africa: a systematic review and pooled analysis** *PLoS one* <https://doi.org/10.1371/journal.pone.0008988>Google Scholar
- Chang HH, Childs LM, Buckee CO (2016) **Variation in infection length and superinfection enhance selection efficiency in the human malaria parasite** *Sci Rep* <https://doi.org/10.1038/srep26370>Google Scholar
- Chang HH, Worby CJ, Yeka A, Nankabirwa J, Kamya MR, Staedke SG, Dorsey G, Murphy M, Neafsey DE, Jeffreys AE, Hubbard C, Rockett KA, Amato R, Kwiatkowski DP, Buckee CO, Greenhouse B. (2017) **THE REAL McCOIL: A method for the concurrent estimation of the complexity of infection and SNP allele frequency for malaria parasites** *PLoS Computational Biology* <https://doi.org/10.1371/journal.pcbi.1005348>Google Scholar

- Childs LM, Buckee CO (2014) **Dissecting the determinants of malaria chronicity: why within-host models struggle to reproduce infection dynamics** *Journal of the Royal Society, Interface* <https://doi.org/10.1098/rsif.2014.1379>Google Scholar
- Choi DW, Kim NK, Chae KC (2005) **A Two-Moment Approximation for the GI/G/c Queue with Finite Capacity** *INFORMS Journal on Computing* **17**:75–81 <https://doi.org/10.1287/ijoc.1030.0058>Google Scholar
- Claessens A, Adams Y, Ghumra A, et al. (2012) **A subset of group A-like var genes encodes the malaria parasite ligands for binding to human brain endothelial cells** *Proc Natl Acad Sci USA* <https://doi.org/10.1073/pnas.1120461109>Google Scholar
- Claessens A, Hamilton WL, Kekre M, Faizullahoy A, Rayner JC, Kwiatkowski D. (2014) **Generation of Antigenic Diversity in Plasmodium falciparum by Structured Rearrangement of Var Genes During Mitosis** *PLoS Genet* <https://doi.org/10.1371/journal.pgen.1004812>Google Scholar
- Collins WE, Jeffery GM (1999) **A retrospective examination of sporozoite- and trophozoite-induced infections with Plasmodium falciparum in patients previously infected with heterologous species of Plasmodium: effect on development of parasitologic and clinical immunity** *Am J Trop Med Hyg* <https://doi.org/10.4269/tropmed.1999.61-036>Google Scholar
- Collins WE, Jeffery GM, Skinner JC (1964) **Fluorescent antibody studies in human malaria. II. Development and persistence of antibodies to Plasmodium falciparum** *The American journal of tropical medicine and hygiene* **13**:256–60 <https://api.semanticscholar.org/CorpusID:42309951>Google Scholar
- Collins WE, Skinner JC, Jeffery GM (1968) **Studies on the persistence of malarial antibody response** *American Journal of Epidemiology* **87**:592–598 <https://doi.org/10.1093/oxfordjournals.aje.a120849>Google Scholar
- Daniels R, Volkman SK, Milner DA, et al. (2008) **A general SNP-based molecular barcode for Plasmodium falciparum identification and tracking** *Malar J* <https://doi.org/10.1186/1475-2875-7-223>Google Scholar
- Daubersies P, Sallenave-Sales S, Magne S, Trape JF, Contamin H, Fandeur T, Rogier C, Mercereau-Puijalon O, Druilhe P. (1996) **Rapid turnover of Plasmodium falciparum populations in asymptomatic individuals living in a high transmission area** *The American journal of tropical medicine and hygiene* <https://doi.org/10.4269/ajtmh.1996.54.18>Google Scholar
- Day KP, Artzy-Randrup Y, Tiedje KE, Rougeron V, Chen DS, Rask TS, Rorick MM, Migot-Nabias F, Deloron P, Luty AJF, Pascual M. (2017) **Evidence of strain structure in Plasmodium falciparum var gene repertoires in children from Gabon, West Africa** *Proceedings of the National Academy of Sciences* <https://doi.org/10.1073/pnas.1613018114>Google Scholar
- Deitsch KW, Lukehart SA, Stringer JR (2009) **Common strategies for antigenic variation by bacterial, fungal and protozoan pathogens** *Nature reviews* <https://doi.org/10.1038/nrmicro2145>Google Scholar
- Dietz K, Molineaux L, Thomas A. (1974) **A malaria model tested in the African savannah** *Bull World Health Organ* [Google Scholar](https://doi.org/10.1186/1475-2875-7-223)

- Donovan MJ, Messmore AS, Scrafford DA, Sacks DL, Kamhawi S, McDowell MA (2007) **Uninfected mosquito bites confer protection against infection with malaria parasites** *Infection and immunity* <https://doi.org/10.1128/IAI.01928-06>Google Scholar
- Doolan DL, Dobaño C, Baird JK (2009) **Acquired immunity to malaria** *Clin Microbiol Rev* <https://doi.org/10.1128/CMR.00025-08>Google Scholar
- Doolan DL, Martinez-Alier N. (2006) **Immune response to pre-erythrocytic stages of malaria parasites** *Curr Mol Med* <https://doi.org/10.2174/156652406776055249>Google Scholar
- Drummond DA, Silberg JJ, Meyer MM, Wilke CO, Arnold FH (2005) **On the conservative nature of intragenic recombination** *Proceedings of the National Academy of Sciences* **102**:5380–5385 <https://doi.org/10.1073/pnas.0500729102>Google Scholar
- Earland D, Buchwald AG, Sixpence A, Chimenya M, Damson M, Seydel KB, Mathanga DP, Taylor TE, Laufer MK (2019) **Impact of Multiplicity of Plasmodium falciparum Infection on Clinical Disease in Malawi** *The American Journal of Tropical Medicine and Hygiene* <https://doi.org/10.4269/ajtmh.19-0093>Google Scholar
- Earland D, Buchwald AG, Sixpence A, Chimenya M, Damson M, Seydel KB, Mathanga DP, Taylor TE, Laufer MK (2019) **Impact of Multiplicity of Plasmodium falciparum Infection on Clinical Disease in Malawi** *The American Journal of Tropical Medicine and Hygiene* **101**:412–415 <https://doi.org/10.4269/ajtmh.19-0093>Google Scholar
- Efron B, Tibshirani RJ (1994) **An Introduction to the Bootstrap** Chapman and Hall/CRC <https://doi.org/10.1201/9780429246593>Google Scholar
- Eldh M, Hammar U, Arnot DE, Beck HP, Garcia A, Liljander A, Mercereau-Puijalon O, Migot-Nabias F, Mueller I, Ntoumi F, Ross A, Smith TA, Sondén K, Homann MV, Yman V, Felger I, Färnert A. (2019) **Multiplicity of Asymptomatic Plasmodium falciparum Infections and Risk of Clinical Malaria: A Systematic Review and Pooled Analysis of Individual Participant Data** *The Journal of Infectious Diseases* **221**:775–785 <https://api.semanticscholar.org/CorpusID:203721037>Google Scholar
- Farnert A, Snounou G, Rooth I, Bjorkman A. (1997) **Daily dynamics of Plasmodium falciparum subpopulations in asymptomatic children in a holoendemic area** *The American journal of tropical medicine and hygiene* <https://doi.org/10.4269/ajtmh.1997.56.538>Google Scholar
- Färnert A, Snounou G, Rooth I, Björkman A. (1997) **Daily dynamics of Plasmodium falciparum subpopulations in asymptomatic children in a holoendemic area** *The American journal of tropical medicine and hygiene* **5**:538–47 <https://api.semanticscholar.org/CorpusID:11873696>Google Scholar
- Felger I, Tavul L, Kabintik S, Marshall V, Genton B, Alpers M, Beck HP (1994) **Plasmodium falciparum: Extensive Polymorphism in Merozoite Surface Antigen 2 Alleles in an Area with Endemic Malaria in Papua New Guinea** *Experimental Parasitology* **79**:106–116 <https://doi.org/10.1006/expr.1994.1070>Google Scholar
- Felger I, Maire M, Bretscher MT, Falk N, Tiaden A, Sama W, Beck HP, Owusu-Agyei S, Smith TA (2012) **The Dynamics of Natural Plasmodium falciparum Infections** *PLoS ONE* <https://doi.org/10.1371/journal.pone.0045542>Google Scholar

- Frank M, Kirkman L, Costantini D, Sanyal S, Lavazec C, Templeton TJ, Deitsch KW (2008) **Frequent recombination events generate diversity within the multi-copy variant antigen gene families of *Plasmodium falciparum*** *International journal for parasitology* <https://doi.org/10.1016/j.ijpara.2008.01.010>Google Scholar
- Freitas-Junior LH, Bottius E, Pirrit LA, et al. (2000) **Frequent ectopic recombination of virulence factor genes in telomeric chromosome clusters of *P. falciparum*** *Nature* <https://doi.org/10.1038/35039531>Google Scholar
- Färnert A, Lebbad M, Faraja L, Rooth I. (2008) **Extensive dynamics of *Plasmodium falciparum* densities, stages and genotyping profiles** *Malar J* <https://doi.org/10.1186/1475-2875-7-241>Google Scholar
- Gardner MJ, Hall N, Fung E, White O, Berriman M, Hyman RW, Carlton JM, Pain A, Nelson KE, Bowman S, Paulsen IT, James K, Eisen JA, Rutherford K, Salzberg SL, Craig A, Kyes S, Chan MS, Nene V, Shallom SJ, et al. (2002) **Genome sequence of the human malaria parasite *Plasmodium falciparum*** *Nature* <https://doi.org/10.1038/nature01097>Google Scholar
- Gibson MA, Bruck J. (2000) **Efficient Exact Stochastic Simulation of Chemical Systems with Many Species and Many Channels** *The Journal of Physical Chemistry A* <https://doi.org/10.1021/jp993732q>Google Scholar
- Gillespie DT (1976) **A General Method for Numerically Simulating the Stochastic Time Evolution of Coupled Chemical Reactions** *Journal of Computational Physics* [https://doi.org/10.1016/0021-9991\(76\)90041-3](https://doi.org/10.1016/0021-9991(76)90041-3)Google Scholar
- Gogue C, Wagman J, Tynuv K, et al. (2020) **An observational analysis of the impact of indoor residual spraying in Northern, Upper East, and Upper West Regions of Ghana: 2014 through 2017** *Malar J* <https://doi.org/10.1186/s12936-020-03318-1>Google Scholar
- Guttery DS, Holder AA, Tewari R. (2012) **Sexual development in *Plasmodium*: lessons from functional analyses** *PLoS pathogens* <https://doi.org/10.1371/journal.ppat.1002404>Google Scholar
- He Q, Pilosof S, Tiedje KE, Ruybal-Pesántez S, Artzy-Randrup Y, Baskerville EB, Day KP, Pascual M. (2018) **Networks of genetic similarity reveal non-neutral processes shape strain structure in *Plasmodium falciparum*** *Nature Communications* <https://doi.org/10.1038/s41467-018-04219-3>Google Scholar
- Hergott DEB, Owalla TJ, Staubus WJ, et al. (2024) **Assessing the daily natural history of asymptomatic *Plasmodium* infections in adults and older children in Katakwi, Uganda: a longitudinal cohort study** *The Lancet Microbe* [https://doi.org/10.1016/S2666-5247\(23\)00262-8](https://doi.org/10.1016/S2666-5247(23)00262-8)Google Scholar
- Heyman DP (1985) **Queues and Point Processes (Peter Franken, Dieter König, Ursula Arndt and Volker Schmidt)** *SIAM Review* **27**:275–276 <https://doi.org/10.1137/1027083>Google Scholar
- Hofmann NE, Karl S, Wampfler R, Kiniboro B, Teliki A, Iga J, Waltmann A, Betuela I, Felger I, Robinson LJ, Mueller I. (2017) **The complex relationship of exposure to new *Plasmodium* infections and incidence of clinical malaria in Papua New Guinea** *eLife* <https://doi.org/10.7554/eLife.23708>Google Scholar
- John CC, Moormann AM, Pregibon DC, et al. (2005) **Correlation of high levels of antibodies to multiple pre-erythrocytic *Plasmodium falciparum* antigens and protection from infection**

Am J Trop Med Hyg <https://doi.org/10.4269/ajtmh.2005.73.222>Google Scholar

Kaestli M, Cockburn IA, Cortés A, Baea K, Rowe JA, Beck HP (2006) **Virulence of malaria is associated with differential expression of *Plasmodium falciparum* var gene subgroups in a case-control study** *J Infect Dis* <https://doi.org/10.1086/503776>Google Scholar

Konaté L, Zwetyenga J, Rogier C, Bischoff E, Fontenille D, Tall A, Spiegel A, Trape JF, Mercereau-Puijalon O. (1999) **5. Variation of *Plasmodium falciparum* msp1 block 2 and msp2 allele prevalence and of infection complexity in two neighbouring Senegalese villages with different transmission conditions** *Transactions of The Royal Society of Tropical Medicine and Hygiene* **93**:21 [https://doi.org/10.1016/S0035-9203\(99\)90323-1](https://doi.org/10.1016/S0035-9203(99)90323-1)Google Scholar

Kraemer SM, Kyes SA, Aggarwal G, Springer AL, Nelson SO, Christodoulou Z, Smith LM, Wang W, Levin E, Newbold CI, Myler PJ, Smith JD (2007) **Patterns of gene recombination shape var gene repertoires in *Plasmodium falciparum*: comparisons of geographically diverse isolates** *BMC Genomics* <https://doi.org/10.1186/1471-2164-8-45>Google Scholar

Labbé F, He Q, Zhan Q, Tiedje KE, Argyropoulos DC, Tan MH, Ghansah A, Day KP, Pascual M. (2023) **Neutral vs. non-neutral genetic footprints of *Plasmodium falciparum* multiclonal infections** *PLoS Computational Biology* <https://doi.org/10.1371/journal.pcbi.1010816>Google Scholar

Laishram DD, Sutton PL, Nanda N, Sharma VL, Sobti RC, Carlton JM, Joshi H. (2012) **The complexities of malaria disease manifestations with a focus on asymptomatic malaria** *Malaria journal* <https://doi.org/10.1186/1475-2875-11-29>Google Scholar

Langhorne J, Ndungu FM, Sponaas AM, Marsh K. (2008) **Immunity to malaria: more questions than answers** *Nat Immunol* <https://doi.org/10.1038/ni.f.205>Google Scholar

Larremore DB, Clauset A, Buckee CO (2013) **A network approach to analyzing highly recombinant malaria parasite genes** *PLoS Comput Biol* <https://doi.org/10.1371/journal.pcbi.1003268>Google Scholar

Lavstsen T, Salanti A, Jensen ATR, Arnot DE, Theander TG (2003) **Sub-grouping of *Plasmodium falciparum* 3D7 var genes based on sequence analysis of coding and non-coding regions** *Malaria journal* <https://doi.org/10.1186/1475-2875-2-27>Google Scholar

Lee SA, Yeka A, Nsobya SL, Dokomajilar C, Rosenthal PJ, Talisuna A, Dorsey G. (2006) **Complexity of *Plasmodium falciparum* infections and antimalarial drug efficacy at 7 sites in Uganda** *The Journal of infectious diseases* <https://doi.org/10.1086/501473>Google Scholar

Lindblade KA, Steinhardt L, Samuels A, Kachur SP, Slutsker L. (2013) **The silent threat: asymptomatic parasitemia and malaria transmission** *Expert Rev Anti Infect Ther* <https://doi.org/10.1586/eri.13.45>Google Scholar

Little JDC, Graves SC (2008) **Little's Law** In: Chhajed D, Lowe TJ, editors. *Building Intuition International Series in Operations Research & Management Science* Springer pp. 81–100 <https://doi.org/10.1007/978-0-387-73699-0>Google Scholar

Lloyd-Smith JO, Schreiber SJ, Kopp PE, Getz WM (2005) **Superspreading and the effect of individual variation on disease emergence** *Nature* <https://doi.org/10.1038/nature04153>Google Scholar

- Macdonald G. (1950) **The analysis of malaria parasite rates in infants** *Tropical diseases bulletin* [Google Scholar](#)
- Maire N, Smith T, Ross A, Owusu-Agyei S, Dietz K, Molineaux L. (2006) **A model for natural immunity to asexual blood stages of Plasmodium falciparum malaria in endemic areas** *Am J Trop Med Hyg* <https://doi.org/10.4269/ajtmh.2006.75.19>[Google Scholar](#)
- Molineaux L, Träuble M, Collins WE, Jeffery GM, Dietz K. (2002) **Malaria therapy reinoculation data suggest individual variation of an innate immune response and independent acquisition of antiparasitic and antitoxic immunities** *Transactions of the Royal Society of Tropical Medicine and Hygiene* [https://doi.org/10.1016/s0035-9203\(02\)90308-1](https://doi.org/10.1016/s0035-9203(02)90308-1)[Google Scholar](#)
- Msuya FH, Curtis CF (1991) **Trial of pyrethroid impregnated bednets in an area of Tanzania holoendemic for malaria. Part 4. Effects on incidence of malaria infection** *Acta Trop* [https://doi.org/10.1016/0001-706x\(91\)90035-i](https://doi.org/10.1016/0001-706x(91)90035-i)[Google Scholar](#)
- Mueller I, Schoepflin S, Smith TA, Benton KL, Bretscher MT, Lin E, Kiniboro B, Zimmerman PA, Speed TP, Siba P, Felger I. (2012) **Force of infection is key to understanding the epidemiology of Plasmodium falciparum malaria in Papua New Guinean children** *Proceedings of the National Academy of Sciences of the United States of America* <https://doi.org/10.1073/pnas.1200841109>[Google Scholar](#)
- Muench H. (1959) **Catalytic models in epidemiology** Harvard University Press [Google Scholar](#)
- Mugenyi L, Abrams S, Hens N. (2017) **Estimating age-time-dependent malaria force of infection accounting for unobserved heterogeneity** *Epidemiology and Infection* **145**:2545–2562 <https://doi.org/10.1017/S0950268817000000>[Google Scholar](#)
- Nkhoma SC, Trevino SG, Gorena KM, Nair S, Khoswe S, Jett C, Garcia R, Daniel B, Dia A, Terlouw DJ, Ward SA, Anderson TJC, Cheeseman IH (2020) **Co-transmission of Related Malaria Parasite Lineages Shapes Within-Host Parasite Diversity** *Cell host & microbe* <https://doi.org/10.1016/j.chom.2019.12.001>[Google Scholar](#)
- Okell LC, Bousema T, Griffin JT, Ouedraogo AL, Ghani AC, Drakeley CJ (2012) **Factors determining the occurrence of submicroscopic malaria infections and their relevance for control** *Nature communications* <https://doi.org/10.1038/ncomms2241>[Google Scholar](#)
- Peyerl-Hoffmann G, Jelinek T, Kilian A, Kabagambe G, Metzger WG, von Sonnenburg F. (2001) **Genetic diversity of Plasmodium falciparum and its relationship to parasite density in an area with different malaria endemicities in West Uganda** *Trop Med Int Health* <https://doi.org/10.1046/j.1365-3156.2001.00761.x>[Google Scholar](#)
- Pilosof S, He Q, Tiedje KE, Ruybal-Pesántez S, Day KP, Pascual M. (2019) **Competition for hosts modulates vast antigenic diversity to generate persistent strain structure in Plasmodium falciparum** *PLoS Biology* <https://doi.org/10.1371/journal.pbio.3000336>[Google Scholar](#)
- Piper KP, Roberts DJ, Day KP (1999) **Plasmodium falciparum: analysis of the antibody specificity to the surface of the trophozoite-infected erythrocyte** *Experimental parasitology* <https://doi.org/10.1006/expr.1999.0400>[Google Scholar](#)

Potthoff RF, Whittinghill M. (1966) **Testing for homogeneity. II. The Poisson distribution** *Biometrika* <https://doi.org/10.1093/biomet/53.1-2.183>Google Scholar

Pull JH, Grab B. (1974) **A simple epidemiological model for evaluating the malaria inoculation rate and the risk of infection in infants** *Bulletin of the World Health Organization* [Google Scholar](#)

Rask TS, Hansen DA, Theander TG, Gorm Pedersen A, Lavstsen T. (2010) **Plasmodium falciparum Erythrocyte Membrane Protein 1 Diversity in Seven Genomes – Divide and Conquer** *PLoS Computational Biology* 6:e1000933 <https://doi.org/10.1371/journal.pcbi.1000933>Google Scholar

Rask TS, Hansen DA, Theander TG, Pedersen AG, Lavstsen T. (2010) **Plasmodium falciparum erythrocyte membrane protein 1 diversity in seven genomes—divide and conquer** *PLoS computational biology* <https://doi.org/10.1371/journal.pcbi.1000933>Google Scholar

de Roos AM, He Q M P. (2023) **An immune memory-structured SIS epidemiological model for hyperdiverse pathogens** *Proc Natl Acad Sci USA* <https://doi.org/10.1073/pnas.2218499120>Google Scholar

Rottmann M, Lavstsen T, Mugasa JP, et al. (2006) **Differential expression of var gene groups is associated with morbidity caused by Plasmodium falciparum infection in Tanzanian children** *Infect Immun* <https://doi.org/10.1128/IAI.02073-05>Google Scholar

Ruybal-Pesántez S, Tiedje KE, Pilosof S, Tonkin-Hill G, He Q, Rask TS, Amenga-Etego L, Oduro AR, Koram KA, Pascual M, Day KP (2022) **Age-specific patterns of DBLα var diversity can explain why residents of high malaria transmission areas remain susceptible to Plasmodium falciparum blood stage infection throughout life** *International Journal for Parasitology* <https://doi.org/10.1016/j.ijpara.2021.12.00>Google Scholar

Ruybal-Pesántez S, Tiedje KE, Tonkin-Hill G, Rask TS, Kamya MR, Greenhouse B, Dorsey G, Duffy MF, Day KP (2017) **Population genomics of virulence genes of Plasmodium falciparum in clinical isolates from Uganda** *Sci Rep* <https://doi.org/10.1038/s41598-017-11814-9>Google Scholar

Shaukat AM, Breman JG, McKenzie FE (2010) **Using the entomological inoculation rate to assess the impact of vector control on malaria parasite transmission and elimination** *Malaria journal* <https://doi.org/10.1186/1475-2875-9-122>Google Scholar

Simpson JA, Aarons L, Collins WE, Jeffery GM, White NJ (2002) **Population dynamics of untreated Plasmodium falciparum malaria within the adult human host during the expansion phase of the infection** *Parasitology* <https://doi.org/10.1017/S0031182001001202>Google Scholar

Smith DL, Drakeley CJ, Chiyaka C, Hay SI (2010) **A quantitative analysis of transmission efficiency versus intensity for malaria** *Nature Communication* <https://doi.org/10.1038/ncomms1107>Google Scholar

Smith JD, Chitnis CE, Craig AG, et al. (1995) **Switches in expression of Plasmodium falciparum var genes correlate with changes in antigenic and cytoadherent phenotypes of infected erythrocytes** *Cell* [https://doi.org/10.1016/0092-8674\(95\)90056-x](https://doi.org/10.1016/0092-8674(95)90056-x)Google Scholar

Smith T, Vounatsou P. (2003) **Estimation of infection and recovery rates for highly polymorphic parasites when detectability is imperfect, using hidden Markov models**

Statistics in medicine <https://doi.org/10.1002/sim.1274>Google Scholar

Sondo P, Derra K, Lefevre T, et al. (2019) **Genetically diverse *Plasmodium falciparum* infections, withinhost competition and symptomatic malaria in humans** *Sci Rep* **9** <https://doi.org/10.1038/s41598-018-36493-y>Google Scholar

Sondo P, Derra K, Rouamba T, Diallo SN, Taconet P, Kazienga A, Ilboudo H, Tahita MC, Valéa I, Sorgho H, Lefèvre T, Tinto H. (2020) **Determinants of *Plasmodium falciparum* multiplicity of infection and genetic diversity in Burkina Faso** *Parasites Vectors* <https://doi.org/10.1186/s13071-020-04302-z>Google Scholar

Su XZ, Heatwole VM, Wertheimer SP, et al. (1995) **The large diverse gene family var encodes proteins involved in cytoadherence and antigenic variation of *Plasmodium falciparum*-infected erythrocytes** *Cell* [https://doi.org/10.1016/0092-8674\(95\)90055-1](https://doi.org/10.1016/0092-8674(95)90055-1)Google Scholar

Tan MH, Shim H, Chan Y, Day KP (2023) **Unravelling var complexity: Relationship between DBLα types and var genes in *Plasmodium falciparum*** *Front Parasitol* <https://doi.org/10.3389/fpara.2022.1006341>Google Scholar

Tiedje KE, Oduro AR, Agongo G, Anyorigiya T, Azongo D, Awine T, Ghansah A, Pascual M, Koram KA, Day KP (2017) **Seasonal Variation in the Epidemiology of Asymptomatic *Plasmodium falciparum* Infections across Two Catchment Areas in Bongo District, Ghana** *Am J Trop Med Hyg* <https://doi.org/10.4269/ajtmh.16-0959>Google Scholar

Tiedje KE, Oduro AR, Bangre O, Amenga-Etego L, Dadzie SK, Appawu MA, Frempong K, Asoala V, Ruybal-Pésantez S, Narh CA, Deed SL, Argyropoulos DC, Ghansah A, Agyei SA, Segbaya S, Desewu K, Williams I, Simpson JA, Malm K, Pascual M, et al. (2022) **Indoor residual spraying with a non-pyrethroid insecticide reduces the reservoir of *Plasmodium falciparum* in a high-transmission area in northern Ghana** *PLOS Global Public Health* <https://doi.org/10.1371/journal.pgph.0000285>Google Scholar

Tiedje KE, Zhan Q, Ruybal-Pésantez S, Tonkin-Hill G, He Q, Tan MH, Argyropoulos DC, Deed SL, Ghansah A, Bangre O, Oduro AR, Koram KA, Pascual M, Day KP (2023) **Measuring changes in *Plasmodium falciparum* var census population size and structure in response to sequential malaria control interventions** *eLife* <https://doi.org/10.7554/eLife.91411.1>Google Scholar

Tran TM, Li S, Doumbo S, Doumtabe D, Huang CY, Dia S, Bathily A, Sangala J, Kone Y, Traore A, Niangaly M, Dara C, Kayentao K, Ongoiba A, Doumbo OK, Traore B, Crompton PD (2013) **An Intensive Longitudinal Cohort Study of Malian Children and Adults Reveals No Evidence of Acquired Immunity to *Plasmodium falciparum* Infection** *Clinical Infectious Diseases* **57**:40–47 <https://doi.org/10.1093/cid/cit174>Google Scholar

White MT, Griffin JT, Churcher TS, Ferguson NM, Basáñez MG, Ghani AC (2011) **Modelling the impact of vector control interventions on *Anopheles gambiae* population dynamics** *Parasites Vectors* <https://doi.org/10.1186/1756-3305-4-153>Google Scholar

Wong W, Griggs AD, Daniels RF, Schaffner SF, Ndiaye D, Bei AK, Deme AB, MacInnis B, Volkman SK, Hartl DL, Neafsey DE, Wirth DF (2017) **Genetic relatedness analysis reveals the cotransmission of genetically related *Plasmodium falciparum* parasites in Thiès, Senegal** *Genome Med* <https://doi.org/10.1186/s13073-017-0398-0>Google Scholar

Wong W, Volkman S, Daniels R, Schaffner S, Sy M, Ndiaye YD, Badiane AS, Deme AB, Diallo MA, Gomis J, Sy N, Ndiaye D, Wirth DF, Hartl DL (2022) **RH: a genetic metric for measuring**

intra-host *Plasmodium falciparum* relatedness and distinguishing cotransmission from superinfection *PNAS Nexus* <https://doi.org/10.1093/pnasnexus/pgac187> Google Scholar

World Health Organization (2015) **Indoor Residual Spraying: An Operational Manual for Indoor Residual Spraying (IRS) for Malaria Transmission Control and Elimination** World Health Organization [Google Scholar](#)

World Health Organization (2023) **World malaria report 2023** World Health Organization [Google Scholar](#)

Zhan Q, He Q, Tiedje K, Day KP, Pascual M. (2024) **Hyper-diverse antigenic variation and resilience to transmission-reducing intervention in *falciparum* malaria** *medRxiv* <https://doi.org/10.1101/2024.02> Google Scholar

Zhan Q, He Q, Tiedje KE, Day KP, Pascual M. (2024) **Hyper-diverse antigenic variation and resilience to transmission-reducing intervention in *falciparum* malaria** *Nat Commun* <https://doi.org/10.1038/s414024-51468-6> Google Scholar

Zhang X, Deitsch KW (2022) **The mystery of persistent, asymptomatic *Plasmodium falciparum* infections** *Curr Opin Microbiol* <https://doi.org/10.1016/j.mib.2022.102231> Google Scholar

Zhong D, Koepfli C, Cui L, Yan G. (2018) **Molecular approaches to determine the multiplicity of *Plasmodium* infections** *Malaria Journal* <https://doi.org/10.1186/s12936-018-2322-5> Google Scholar

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Reviewer #1 (Public review):

Summary

In their paper Zhan et al. have used Pf genetic data from simulated data and Ghanaian field samples to elucidate a relationship between multiplicity of infection (MOI) (the number of distinct parasite clones in a single host infection) and force of infection (FOI). Specifically, they use sequencing data from the var genes of Pf along with Bayesian modeling to estimate MOI individual infections and use these values along with methods from queueing theory that rely on various assumptions to estimate FOI. They compare these estimates to known FOIs in a simulated scenario and describe the relationship between these estimated FOI values and another commonly used metric of transmission EIR (entomological inoculation rate).

This approach does fill an important gap in malaria epidemiology, namely estimating force of infection, which is currently complicated by several factors including superinfection, unknown duration of infection, and highly genetically diverse parasite populations. The authors use a new approach borrowing from other fields of statistics and modeling and make extensive efforts to evaluate their approach under a range of realistic sampling scenarios. However, the write-up would greatly benefit from added clarity both in the description of methods, and in the presentation of the results. Without these clarifications, rigorously evaluating whether the author's proposed method of estimating FOI is sound remains difficult. Additionally, there are several limitations that call into question the stated generalizability of this method that should at minimum be further discussed by authors and in some cases require a more thorough evaluation.

Major comments:

(1) Description and evaluation of FOI estimation procedure.

a. The methods section describing the two-moment approximation and accompanying appendix is lacking several important details. Equations on line 891 and 892 are only a small part of the equations in Choi et al. and do not adequately describe the procedure notably several quantities in those equations are never defined some of them are important to understand the method (e.g. A, S as the main random variables for inter-arrival times and service times, aR and bR which are the known time average quantities, and these also rely on the squared coefficient of variation of the random variable which is also never introduced in the paper). Without going back to the Choi paper to understand these quantities, and to understand the assumptions of this method it was not possible to follow how this works in the paper. At minimum, all variables used in the equations should be clearly defined.

b. Additionally, the description in the main text of how queueing procedure can be used to describe malaria infections would benefit from a diagram currently as written it's very difficult to follow.

c. Just observing the box plots of mean and 95% CI on a plot with the FOI estimate (Figures 1, 2 and 10-14) is not sufficient to adequately assess the performance of this estimator. First, it is not clear whether authors are displaying the bootstrapped 95% CIs or whether they are just showing the distribution of the mean FOI taken over multiple simulations, and then it seems that they are also estimating mean FOI per host on an annual basis. Showing a distribution of those per host estimates would also be helpful. Second, a more quantitative assessment of the ability of the estimator to recover the truth across simulations (e.g. proportion of simulations where the truth is captured in the 95% CI or something like this) is important in many cases it seems that the estimator is always underestimating the true FOI and may not even contain the true value in the FOI distribution (e.g. figure 10, figure 1 under the mid IRS panel). But it's not possible to conclude on way or the other based on this visualization. This is a major issue since it calls into question whether there is in fact data to support that these methods give good and consistent FOI estimates.

d. Furthermore authors state in the methods that the choice of mean and variance (and thus second moment) parameters for inter arrival times are varied widely, however, it's not clear what those ranges are there needs to be a clear table or figure caption showing what combinations of values were tested and which results are produced from them, this is an essential component of the method and it's impossible to fully evaluate its performance without this information. This relates to the issue of selecting the mean and variance values that maximize the likelihood of observing a given distribution of MOI estimates, this is very unclear since no likelihoods have been written down in the methods section of the main text, which likelihood are the authors referring to, is this the probability distribution of the steady state queue length distribution? At other places the authors refer to these quantities as Maximum Likelihood estimators, how do they know they have found the MLE? There are no derivations in the manuscript to support this. The authors should specify and likelihood and include in an appendix why their estimation procedure is in fact maximizing this likelihood preferably with evidence of the shape of the likelihood, and how fine the grid of values they tested are for their mean and variance since this could influence the overall quality of the estimation procedure.

(2) Limitation of FOI estimation procedure.

a. The authors discuss the importance of duration of infection to this problem. While I agree that empirically estimating this is not possible, there are other options besides assuming that all 1-5 year olds have the same duration of infection distribution as naïve adults co-infected with syphilis. E.g. it would be useful to test a wide range of assumed infection duration and assess their impact on the estimation procedure. Furthermore, if the authors are going to stick to the described method for duration of infection, the potentially limited generalizability of this method needs to be further highlighted in both the introduction, and the discussion. In particular, for an estimated mean FOI of about 5 per host per year in the pre-IRS season as estimated in Ghana (Figure 3) it seems that this would not translate to 4 year old being immune naïve, and certainly this would not necessarily generalize well to a school-aged child population or an adult population.

b. The evaluation of the capacity parameter c seems to be quite important, and is set at 30, however, the authors only describe trying values of 25 and 30, and claim that this does not impact FOI inference, however it is not clear that this is the case. What happens if carrying capacity is increased substantially? Alternatively, this would be more convincing if the authors provided a mathematical explanation of why the carrying capacity increasing will not influence the FOI inference, but absent that, this should be mentioned and discussed as a limitation.

Comments on revisions:

The authors have adequately responded to all comments.

<https://doi.org/10.7554/eLife.100076.2.sa3>

Reviewer #2 (Public review):

Summary:

The authors combine a clever use of historical clinical data on infection duration in immunologically naive individuals and queuing theory to infer the force of infection (FOI) from measured multiplicity of infection (MOI) in a sparsely sampled setting. They conduct extensive simulations using agent based modeling to recapitulate realistic population dynamics and successfully apply their method to recover FOI from measured MOI. They then go on to apply their method to real world data from Ghana before and after an indoor residual spraying campaign.

Strengths:

- The use of historical clinical data is very clever in this context
- The simulations are very sophisticated with respect to trying to capture realistic population dynamics
- The mathematical approach is simple and elegant, and thus easy to understand

Weaknesses:

- The assumptions of the approach are quite strong, and the authors have made clear that applicability is constrained to individuals with immune profiles that are similar to malaria naive patients with neurosyphilis. While the historical clinical data is a unique resource and likely directionally correct, it remains somewhat dubious to use the exact estimated values as inputs to other models without extensive sensitivity analysis.

<https://doi.org/10.7554/eLife.100076.2.sa2>

Reviewer #3 (Public review):

Summary:

It has been proposed that the FOI is a method of using parasite genetics to determine changes in transmission in areas with high asymptomatic infection. The manuscript attempts to use queuing theory to convert multiplicity of infection estimates (MOI) into estimates of the force of infection (FOI), which they define as the number of genetically distinct blood-stage strains. They look to validate the method by applying them to simulated results from a previously published agent based model. They then apply these queuing theory methods to previously published and analysed genetic data from Ghana. They then compare their results to previous estimates of FOI.

Strengths:

It would be great to be able to infer FOI from cross sectional surveys which are easier and cheaper than current FOI estimates which require longitudinal studies. This work proposes a method to convert MOI to FOI for cross sectional studies. They attempt to validate this process using a previously published agent based model which helps us understand the complexity of parasite population genetics.

Weaknesses:

(1) I fear that the work could be easily over-interpreted as no true validation was done as no field estimates of FOI (I think considered true validation) were measured. You have developed a method of estimating FOI from MOI which makes a number of biological and structural assumptions. I would not call being able to recreate model results that were generated using a model that makes its own (probably similar) defined set of biological and structural assumptions acts as a validation of what is going on in the field. The authors claim this at times (for example, Line 153) and I feel it would be appropriate to differentiate this in the discussion.

(2) Another aspect of the paper is adding greater realism to the previous agent based model, by including assumptions on missing data and under sampling. This takes prominence in the figures and results section, but I would imagine is generally not as interesting to the less specialised reader. The apparent lack of impact of drug treatment on MOI is interesting and counterintuitive, though it is not really mentioned in the results or discussion sufficiently to allay my confusion. I would have been interested in understanding the relationship between MOI and FOI as generated by your queuing theory method and the model. It isn't clear to me why these more standard results are not presented, as I would imagine they are outputs of the model (though happy to stand corrected - it isn't entirely clear to me what the model is doing in this manuscript alone).

(3) I would suggest that outside of malaria geneticists, the force of infection is considered to be the entomological inoculation rate, not the number of genetically distinct blood-stage strains. I appreciate that FOI has been used to explain the later before by others, though the authors could avoid confusion by stating this clearly throughout the manuscript. For example, the abstract says FOI is "the number of new infections acquired by an individual host over a given time interval" which suggests the former, please consider clarifying.

(4) Line 319 says "Nevertheless, overall, our paired EIR (directly measured by the entomological team in Ghana (Tiedje et al., 2022)) and FOI values are reasonably consistent with the data points from previous studies, suggesting the robustness of our proposed methods". I would agree that the results are consistent, given that there is huge variation in Figure 4 despite the transformed scales, but I would not say this suggests a robustness of the method.

(5) The text is a little difficult to follow at times, and sometimes requires multiple reads to understand. Greater precision is needed with the language in a few situations and some of the assumptions made in the modelling process are not referenced, making it unclear whether it is a true representation of the biology.

Comments on revisions:

I think the authors gave a robust but thorough response to our reviews and made some important changes to the manuscript which certainly clarify things for me.

<https://doi.org/10.7554/eLife.100076.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public Review):

Summary:

In their paper, Zhan et al. have used Pf genetic data from simulated data and Ghanaian field samples to elucidate a relationship between multiplicity of infection (MOI) (the number of distinct parasite clones in a single host infection) and force of infection (FOI). Specifically, they use sequencing data from the var genes of Pf along with Bayesian modeling to estimate MOI individual infections and use these values along with methods from queueing theory that rely on various assumptions to estimate FOI. They compare these estimates to known FOIs in a simulated scenario and describe the relationship between these estimated FOI values and another commonly used metric of transmission EIR (entomological inoculation rate).

This approach does fill an important gap in malaria epidemiology, namely estimating the force of infection, which is currently complicated by several factors including superinfection, unknown duration of infection, and highly genetically diverse parasite populations. The authors use a new approach borrowing from other fields of statistics and modeling and make extensive efforts to evaluate their approach under a range of realistic sampling scenarios. However, the write-up would greatly benefit from added clarity both in the description of methods and in the presentation of the results. Without these clarifications, rigorously evaluating whether the author's proposed method of estimating FOI is sound remains difficult. Additionally, there are several limitations that call into question the stated generalizability of this method that should at minimum be further discussed by authors and in some cases require a more thorough evaluation.

Major comments:

(1) Description and evaluation of FOI estimation procedure.

a. The methods section describing the two-moment approximation and accompanying appendix is lacking several important details. Equations on lines 891 and 892 are only a small part of the equations in Choi et al. and do not adequately describe the procedure notably several quantities in those equations are never defined some of them are important to understand the method (e.g. A, S as the main random variables for inter-arrival times and service times, aR and bR which are the known time average quantities, and these also rely on the squared coefficient of variation of the random variable which is also never introduced in the paper). Without going back to the Choi paper to understand these quantities, and to understand the assumptions of this method it was not possible to follow how this works in the paper. At a minimum, all variables used in the equations should be clearly defined.

We thank the reviewer for this useful comment. We have clarified the method and defined all relevant variables in the revised manuscript (Line 537-573). The reviewer correctly pointed out additional sections and equations in Choi et al., including the derivation of an exact expression for the steady-state queue-length distribution and the two-moment approximation. Since our work directly utilized the two-moment approximation, our previous manuscript included only material on that section. However, we agree that providing additional details on the derivation of the exact expression would benefit readers. Therefore, we have summarized this derivation in the revised manuscript (Line 561-564). Additionally, we clarified the method's assumptions, particularly those involved in transitioning from the exact expression to the two-moment approximation (Line 565-570).

b. Additionally, the description in the main text of how the queueing procedure can be used to describe malaria infections would benefit from a diagram currently as written it's very difficult to follow.

We thank the reviewer for this suggestion. In the revised manuscript, we included a diagram illustrating the connection between the queueing procedure and malaria transmission

(Appendix 1-Figure 8).

c. Just observing the box plots of mean and 95% CI on a plot with the FOI estimate (Figures 1, 2, and 10-14) is not sufficient to adequately assess the performance of this estimator. First, it is not clear whether the authors are displaying the bootstrapped 95% CIs or whether they are just showing the distribution of the mean FOI taken over multiple simulations, and then it seems that they are also estimating mean FOI per host on an annual basis. Showing a distribution of those per-host estimates would also be helpful. Second, a more quantitative assessment of the ability of the estimator to recover the truth across simulations (e.g. proportion of simulations where the truth is captured in the 95% CI or something like this) is important in many cases it seems that the estimator is always underestimating the true FOI and may not even contain the true value in the FOI distribution (e.g. Figure 10, Figure 1 under the mid-IRS panel). But it's not possible to conclude one way or the other based on this visualization. This is a major issue since it calls into question whether there is in fact data to support that these methods give good and consistent FOI estimates.

There seems to be some confusion on what we display in some key figures. Figures 1-2 and 10-14 (labeled as Figure 1-2 and Appendix 1-Figure 11-15 in the revised manuscript) display bootstrapped distributions including the 95% CIs, not the distribution of the mean FOI taken over multiple simulations. To estimate the mean FOI per host on an annual basis, the two proposed methods require either the steady-state queue length distribution (MOI distribution) or the moments of this distribution. Obtaining such a steady-state queue length distribution necessitates either densely tracked time-series observations per host or many realizations at the same sampling time per host. However, under the sparse sampling schemes, we only have two one-time-point observations per host: one at the end of wet/high-transmission and another at the end of dry/low-transmission. This is typically the case for empirical data, although numerical simulations could circumvent this limitation and generate such output. Nonetheless, we have a population-level queue length distribution from both simulation outputs and empirical data by aggregating MOI estimates across all sampled individuals. We use this population-level distribution to represent and approximate the steady-state queue length distribution at the individual level, not explicitly considering any individual heterogeneity due to transmission. The estimated FOI is per host in the sense of representing the FOI experienced by an individual host whose queue length distribution is approximated from the collection of all sampled individuals. The true FOI per host per year in the simulation is the total FOI of all hosts per year divided by the number of hosts. Therefore, our estimator, combined with the demographic information on population size, estimates the total number of *Plasmodium falciparum* infections acquired by all individual hosts in the population of interest per year. We clarified this point in the revised manuscript in the subsection of the Materials and Methods, entitled 'Population-level MOI distribution for approximating time-series observation of MOI per host or many realizations at the same sampling time per host' (Line 623-639).

We evaluated the impact of individual heterogeneity due to transmission on FOI inference using simulation outputs (Line 157-184, Figure 1-2 and Appendix 1-Figure 11-15). Even with significant heterogeneity among individuals (2/3 of the population receiving approximately 94% of all bites whereas the remaining 1/3 receives the rest of the bites), our methods performed comparably to scenarios with homogeneous transmission. Furthermore, our methods demonstrated similar performance for both non-seasonal and seasonal transmission scenarios.

Regarding the second point, we quantitatively assessed the ability of the estimator to recover the truth across simulations and included this information in a supplementary table in the revised manuscript (supplementary file 3-FOImethodsPerformance.xlsx). Specifically, we indicated whether the truth lies within the bootstrap distribution and provided a measure of

relative deviation, which is defined as the true FOI value minus the median of the bootstrap distribution for the estimate, normalized by the true FOI value $(\frac{FOI_{true} - FOI_{Est_{median}}}{FOI_{true}})$. This

assessment is a valuable addition which enhances clarity, but please note that our previous graphical comparisons do illustrate the ability of the methods to estimate “sensible” values, close to the truth despite multiple sources of errors. “Close” here is relative to the scale of variation of FOI in the field and to the kind of precision that would be useful in an empirical context. From a practical perspective based on the potential range of variation of FOI, the graphical results already illustrate that the estimated distributions would be informative.

We also thank the reviewer for highlighting instances where our proposed methods for FOI inference perform sub-optimally (e.g. Figure 10, Figure 1 under the mid-IRS panel in the previous manuscript). This feedback prompted us to examine these instances more closely and identify the underlying causes related to the stochastic impact introduced during various sampling processes. These include sampling the host population and their infections at a specific sampling depth in the simulated output, matching the depth used for collecting empirical data. In addition, previously, we imputed MOI estimates for treated individuals by sampling only once from non-treated individuals. This time, we conducted 200 samplings and used the final weighted MOI distribution for FOI inference. By doing so, we reduced the impact of extreme single-sampling efforts on MOI distribution and FOI inference. In other words, some of these suboptimal instances correspond to the scenarios where the one-time sampled MOIs from non-treated individuals do not fully capture the MOI distribution of non-treated individuals. We added a section titled ‘Reducing stochastic impact in sampling processes’ to Appendix 1 on this matter (Line 841-849).

The reviewer correctly noted that our proposed methods tend to underestimate FOI (Figure 1-2, 10-14, ‘Estimated All Errors’ and ‘Estimated Undersampling of Var’ panels in the previous manuscript, corresponding to Figure 1-2 and Appendix 1-Figure 11-15 in the revised manuscript). This underestimation arises from the underestimation of MOI. The Bayesian formulation of the *varcoding* method does not account for the limited overlap between co-infecting strains, an additional factor that reduces the number of *var* genes detected per individual. We have elaborated on this matter in the Results and Discussion sections of the revised manuscript (Line 142-149, 252-256).

d. Furthermore the authors state in the methods that the choice of mean and variance (and thus second moment) parameters for inter-arrival times are varied widely, however, it's not clear what those ranges are there needs to be a clear table or figure caption showing what combinations of values were tested and which results are produced from them, this is an essential component of the method and it's impossible to fully evaluate its performance without this information. This relates to the issue of selecting the mean and variance values that maximize the likelihood of observing a given distribution of MOI estimates, this is very unclear since no likelihoods have been written down in the methods section of the main text, which likelihood are the authors referring to, is this the probability distribution of the steady state queue length distribution? At other places the authors refer to these quantities as Maximum Likelihood estimators, how do they know they have found the MLE? There are no derivations in the manuscript to support this. The authors should specify the likelihood and include in an appendix an explanation of why their estimation procedure is in fact maximizing this likelihood, preferably with evidence of the shape of the likelihood, and how fine the grid of values they tested is for their mean and variance since this could influence the overall quality of the estimation procedure.

We thank the reviewer for pointing out these aspects of the work that can be further clarified. In response, we maximized the likelihood of observing the population-level MOI

distribution in the sampled population (see our responses to your previous comment c), given queue length distributions, derived from the two-moment approximation method for various mean and variance combinations of inter-arrival times. We added a new section to the Materials and Methods in the revised manuscript with an explicit likelihood formulation (Line 574-585).

Additionally, we specified the ranges for the mean and variance parameters for inter-arrival times and provided the grid of values tested in a supplementary table (supplementary file 4-meanVarianceParams.xlsx). Example figures illustrating the shape of the likelihood have also been included in Appendix 1-Figure 9. We tested the impact of different grid value choices on estimation quality by refining the grid to include more points, ensuring the FOI inference results are consistent. The results of the test are documented in the revised manuscript (Line 587-593, Appendix 1-Figure 10).

(2) Limitation of FOI estimation procedure.

a. The authors discuss the importance of the duration of infection to this problem. While I agree that empirically estimating this is not possible, there are other options besides assuming that all 1-5-year-olds have the same duration of infection distribution as naïve adults co-infected with syphilis. E.g. it would be useful to test a wide range of assumed infection duration and assess their impact on the estimation procedure. Furthermore, if the authors are going to stick to the described method for duration of infection, the potentially limited generalizability of this method needs to be further highlighted in both the introduction, and the discussion. In particular, for an estimated mean FOI of about 5 per host per year in the pre-IRS season as estimated in Ghana (Figure 3) it seems that this would not translate to 4-year-old being immune naïve, and certainly this would not necessarily generalize well to a school-aged child population or an adult population.

We thank the reviewer for this useful comment. The reviewer correctly noted the challenge in empirically measuring the duration of infection for 1-5-year-olds and comparing it to that of naïve adults co-infected with syphilis. We nevertheless continued to use the described method for the duration of infection, while more thoroughly acknowledging and discussing the limitations this aspect of the method introduces. We have highlighted this potential limitation in the Abstract, Introduction, and Discussion sections of the revised manuscript (Line 26-28, 99-103, 270-292). It is important to note that the infection duration from the historical clinical data we have relied on has been used, and is still used, in the malaria modeling community as a credible source for this parameter in untreated natural infections of malaria-naïve individuals in endemic settings of Africa (e.g. in the agent-based model OpenMalaria, see 1).

To reduce misspecification in infection duration and fully utilize our proposed methods, future data collection and sampling could prioritize subpopulations with minimal prior infections and an immune profile similar to naïve adults, such as infants and toddlers. As these individuals are also the most vulnerable, prioritizing them aligns with the priority of all intervention efforts in the short term, which is to monitor and protect the most vulnerable individuals from severe symptoms and death. We discuss this aspect in detail in the Discussion section of the revised manuscript (Line 287-292).

In the pre-IRS phase of Ghana surveys, an estimated mean FOI of about 5 per host per year indicates that a 4-year-old child would have experienced around 20 infections, which could suggest they are far from naïve. The extreme diversity of circulating *var* genes (2) implies, however, that even after 20 infections, a 4-year-old may have only developed immunity to a small fraction of the variant surface antigens (PfEMP1, *Plasmodium falciparum* erythrocyte membrane protein 1) encoded by this important gene family. Consequently, these children are not as immunologically experienced as it might initially seem. Moreover, studies have

shown that long-lived infections in older children and adults can persist for months or even years, including through the dry season. This persistence is driven by high antigenic variation of *var* genes and associated incomplete immunity. Additionally, parasites can skew PfEMP1 expression to produce less adhesive erythrocytes, enhancing splenic clearance, reducing virulence, and maintaining sub-clinical parasitemia (3, 4, 5). The impact of immunity on infection duration with age for falciparum malaria remains a challenging open question.

Lastly, the FOI for naïve hosts is a key basic parameter for epidemiological models of complex infectious diseases like falciparum malaria, in both agent-based and equation-based formulations. This is because FOI for non-naïve hosts is typically a function of their immune status, body size, and the FOI of naïve hosts. Thus, knowing the FOI of naïve hosts helps parameterize and validate these models by reducing degrees of freedom.

b. The evaluation of the capacity parameter c seems to be quite important and is set at 30, however, the authors only describe trying values of 25 and 30, and claim that this does not impact FOI inference, however it is not clear that this is the case. What happens if the carrying capacity is increased substantially? Alternatively, this would be more convincing if the authors provided a mathematical explanation of why the carrying capacity increase will not influence the FOI inference, but absent that, this should be mentioned and discussed as a limitation.

Thank you for this question. This parameter represents the carrying capacity of the queuing system, or the maximum number of blood-stage strains with which an individual human host can be co-infected. Empirical evidence, estimated using the *var*coding method, suggests this value is 20 (2), providing a lower bound for parameter c . However, the *var*coding method does not account for the limited overlap between co-infecting strains, which reduces the number of *var* genes detected in an individual, thereby affecting the basis of MOI estimation. Additional factors, such as the synchronicity of clones in their 48-hour life cycle on alternate days (6) and within-host competition of strains leading to low-parasitemia levels (7, 8), contribute to under-sampling of strains and are not accounted for in MOI estimation (9). To address these potential under-sampling issues, we previously tested values of 25 and 30.

This time, we systematically investigated a wider range of values, including substantially higher ones: 25, 30, 40, and 60. We found that the FOI inference results are similar across these values. Figure 3 in the main text and supplementary figures (Appendix 1-Figure 16-18) illustrates these findings.

The parameter c influences the steady-state queue length distribution based on the two-moment approximation with specific mean and variance combinations, primarily affecting the distribution's tail when customer or infection flows are high. Smaller values of c lower the maximum possible queue length, making the system more prone to “overflow”. In such cases, customers or infections may find no space available upon their arrival, hence not incrementing the queue length.

Empirical MOI distributions for high-transmission endemic regions center around 4 or 5, mostly remaining below 10, with only a small fraction between 15-20 (2). These distributions do not support parameter combinations resulting in frequent overflow for a system with c equal to 25 or 30. As one increases the value of c further, these parameter combinations would cause the MOI distributions to shift to larger values inconsistent with the empirical MOI distributions. We therefore do not expect substantially higher values for parameter c to noticeably change either the relative shape of the likelihood or the MLE.

We have included a subsection on parameter c in the Materials and Methods section of the revised manuscript (Line 596-612).

Reviewer #2 (Public Review):

Summary:

The authors combine a clever use of historical clinical data on infection duration in immunologically naïve individuals and queuing theory to infer the force of infection (FOI) from measured multiplicity of infection (MOI) in a sparsely sampled setting. They conduct extensive simulations using agent-based modeling to recapitulate realistic population dynamics and successfully apply their method to recover FOI from measured MOI. They then go on to apply their method to real-world data from Ghana before and after an indoor residual spraying campaign.

Strengths:

- (1) The use of historical clinical data is very clever in this context.*
- (2) The simulations are very sophisticated with respect to trying to capture realistic population dynamics.*
- (3) The mathematical approach is simple and elegant, and thus easy to understand.*

Weaknesses:

- (1) The assumptions of the approach are quite strong and should be made more clear. While the historical clinical data is a unique resource, it would be useful to see how misspecification of the duration of infection distribution would impact the estimates.*

We thank the reviewer for bringing up the limitation of our proposed methods due to their reliance on a known and fixed duration of infection distribution from historical clinical data. Please see our response to Reviewer 1, Comment 2a, for a detailed discussion on this matter.

- (2) Seeing as how the assumption of the duration of infection distribution is drawn from historical data and not informed by the data on hand, it does not substantially expand beyond MOI. The authors could address this by suggesting avenues for more refined estimates of infection duration.*

We thank the reviewer for pointing out a potential improvement to our work. We acknowledge that FOI is inferred from MOI and thus depends on the information contained in MOI. However, MOI by definition is a number and not a rate parameter. FOI for naïve hosts is a fundamental parameter for epidemiological models of complex infectious diseases like falciparum malaria, in both agent-based and equation-based formulations. FOI of non-naïve hosts is typically a function of their immune status, body size, and the FOI of naïve hosts. Thus, knowing the FOI of naïve hosts helps parameterize and validate these models by reducing degrees of freedom. In this sense, we believe the transformation from MOI to FOI is valuable.

Measuring infection duration is challenging, making the simultaneous estimation of infection duration and FOI an attractive alternative, as the referee noted. This, however, would require closely monitored cohort studies or densely sampled cross-sectional surveys to reduce issues like identifiability. For instance, a higher arrival rate of infections paired with a shorter infection duration could generate a similar MOI distribution to a lower arrival rate with a longer infection duration. In some cases, incorrect combinations of rate and duration might even produce an MOI distribution that appears closer to the targeted distribution. Such cohort studies and densely sampled cross-sectional surveys have not been and will not be widely available across different geographical locations and times. This work utilizes more readily available data from sparsely sampled single-time-point cross-sectional surveys, which

precludes more sophisticated derivation of time-varying average arrival rates of infections and lacks the resolution to simultaneously estimate arrival rates and infection duration. In the revised manuscript, we have elaborated on this matter and added a paragraph in the Discussion section (Line 306-309).

(3) It is unclear in the example how their bootstrap imputation approach is accounting for measurement error due to antimalarial treatment. They supply two approaches. First, there is no effect on measurement, so the measured MOI is unaffected, which is likely false and I think the authors are in agreement. The second approach instead discards the measurement for malaria-treated individuals and imputes their MOI by drawing from the remaining distribution. This is an extremely strong assumption that the distribution of MOI of the treated is the same as the untreated, which seems unlikely simply out of treatment-seeking behavior. By imputing in this way, the authors will also deflate the variability of their estimates.

We thank the reviewer for pointing out aspects of the work that can be further clarified. Disentangling the effect of drug treatment on measurements like infection duration is challenging. Since our methods rely on the known and fixed distribution of infection duration from historical data of naïve patients with neurosyphilis infected with malaria as a therapy, drug treatment can potentially violate this assumption. In the previous manuscript, we did not attempt to directly address the impact of drug treatment. Instead, we considered two extreme scenarios that bound reality, well summarized by the reviewer. Reality lies somewhere in between these two extremes, with antimalarial treatment significantly affecting measurements in some individuals but not in others. Nonetheless, the results of FOI inference do not differ significantly across both extremes.

The impact of the drugs likely depends on their nature, efficiency, and duration. We note that treatment information was collected via a routine questionnaire, with participant self-reporting that they had received an antimalarial treatment in the previous two-weeks before the surveys (i.e., participants that reported they were sick, sought treatment, and were provided with an antimalarial treatment). No confirmation through hospital or clinic records was conducted, as it was beyond the scope of the study. Additionally, many of these sick individuals seek treatment at local chemists, which may limit the relevance of hospital or clinic records, if they are even available. Consequently, information on the nature, efficiency, and duration of administered drugs was incomplete or lacking. As this is not the focus of this work, we do not elaborate on the impact of drug treatment in the revised manuscript.

The reviewer correctly noted that this imputation might not add additional information and could reduce MOI variability. Therefore, in the revised manuscript, we reported FOI estimates with drug-treated 1-5-year-olds excluded. Additionally, we discarded the infection status and MOI values of treated individuals and sampled their MOI from non-treated microscopy-positive individuals, imputing a positive MOI for treated and uninfected individuals. We also reported FOI estimates based on these MOI values. This scenario provides an upper bound for FOI estimates. Note that we do not assume that the MOI distribution for treated individuals is the same as that for untreated individuals. Rather, we aim to estimate what their MOI would have been, and consequently, determine what the FOI per individual per year in the combined population would be, had these individuals not received antimalarial treatment. The results of FOI inference do not differ significantly between these two approaches. They can serve as general solutions to antimalarial treatment issues for others applying our FOI inference methods. These details can be found in the revised manuscript (Line 185-210, 462-484).

- For similar reasons, their imputation of microscopy-negative individuals is also questionable, as it also assumes the same distributions of MOI for microscopy-positive and negative individuals.

We thank the reviewer for this comment. The reviewer correctly noted that we imputed the MOI values for microscopy-negative but PCR-positive 1-5-year-olds by sampling from the microscopy-positive 1-5-year-olds, under the assumption that both groups have similar MOI distributions. This approach was motivated by the analysis of our Ghana surveys, which shows no clear relationship between MOI (or the number of *var* genes detected within an individual host, on the basis of which our MOI values were estimated) and the parasitemia levels of those hosts. Parasitemia levels underlie the difference in detection sensitivity between PCR and microscopy.

In the revised manuscript, we elaborated on this issue and included formal regression tests showing the lack of a relationship between MOI/the number of *var* genes detected within an individual host and the parasitemia levels of those hosts (Line 445-451, Appendix 1-Figure 7). We also described potential reasons or hypotheses behind this observation (Line 452-461).

Reviewer #3 (Public Review):

Summary:

It has been proposed that the FOI is a method of using parasite genetics to determine changes in transmission in areas with high asymptomatic infection. The manuscript attempts to use queuing theory to convert multiplicity of infection estimates (MOI) into estimates of the force of infection (FOI), which they define as the number of genetically distinct blood-stage strains. They look to validate the method by applying it to simulated results from a previously published agent-based model. They then apply these queuing theory methods to previously published and analysed genetic data from Ghana. They then compare their results to previous estimates of FOI.

Strengths:

It would be great to be able to infer FOI from cross-sectional surveys which are easier and cheaper than current FOI estimates which require longitudinal studies. This work proposes a method to convert MOI to FOI for cross-sectional studies. They attempt to validate this process using a previously published agent-based model which helps us understand the complexity of parasite population genetics.

Weaknesses:

(1) I fear that the work could be easily over-interpreted as no true validation was done, as no field estimates of FOI (I think considered true validation) were measured. The authors have developed a method of estimating FOI from MOI which makes a number of biological and structural assumptions. I would not call being able to recreate model results that were generated using a model that makes its own (probably similar) defined set of biological and structural assumptions a validation of what is going on in the field. The authors claim this at times (for example, Line 153) and I feel it would be appropriate to differentiate this in the discussion.

We thank the reviewer for this comment, although we think there is a mis-understanding on what can and cannot be practically validated in the sense of a “true” measure of FOI that would be free from assumptions for a complex disease such as malaria. We would not want the results to be over-interpreted, and we have extended the discussion of what we have done to test the methods in the revised manuscript (Line 314-328). Performance evaluation via simulation output is common and often necessary for statistical methods. These simulations can come from dynamical or descriptive models, each making their own assumptions to simplify reality. Our stochastic agent-based model (ABM) of malaria transmission, used in this study, has successfully replicated several key patterns from high-

transmission endemic regions in the field, including aspects of strain diversity not represented and captured by simpler models (10).

In what sense this ABM makes a set of biological and structural assumptions that are “probably similar” to those of the queuing methods we present is not clear to us. We agree that using models with different structural assumptions from the method being tested is ideal. Our FOI inference methods based on queuing theory require the duration of infection distribution and the MOI distribution among sampled individuals. However, these FOI inference methods are agnostic to the specific biological mechanisms governing these distributions.

Another important point raised by this comment is what would be the “true” FOI value against which to validate our methods. Empirical MOI-FOI pairs from cohort studies tracking FOI directly are still lacking. Direct FOI measurements are prone to errors because differentiating new infections from the temporary absence of an old infection in the peripheral blood and its subsequent re-emergence remains challenging. Reasons for this challenge include the low resolution of the polymorphic markers used in cohort studies, which cannot fully differentiate hyper-diverse antigenic strains, and the complexity of within-host dynamics and competitive interaction of co-infecting strains (6, 8, 9). Alternative approaches also do not provide a “true” FOI estimation free from assumptions. These approaches involve fitting simplified epidemiological models to densely sampled/repeated cross-sectional surveys for FOI inference. In this case, no FOI is measured directly, and thus, there are no FOI values available for benchmarking against fitted FOI values. The evaluation or validation of these model-fitting approaches is typically based on their ability to capture other epidemiological quantities that are easier to sample or measure, such as prevalence or incidence, with criteria such as the Akaike information criterion (AIC). This type of evaluation is similar to the one done in this work. We selected FOI values that maximize the likelihood of observing the given MOI distribution. Furthermore, we paired our estimated FOI values for Ghana surveys with the independently measured EIR (Entomological Inoculation Rate), a common field measure of transmission intensity. We ensured that our resulting FOI-EIR points align with existing FOI-EIR pairs and the relationship between these quantities from previous studies. We acknowledge that, like model-fitting approaches, our validation for the field data is also indirect and further complicated by high variance in the relationship between EIR and FOI from previous studies.

Prompted by the reviewer’s comment, we elaborated on these points in the revised manuscript, emphasizing the indirect nature and existing constraints of our validation with field data in the Discussion section (Line 314-328). Additionally, we clarified certain basic assumptions of our agent-based model in Appendix 1-Simulation data.

(2) Another aspect of the paper is adding greater realism to the previous agent-based model, by including assumptions on missing data and under-sampling. This takes prominence in the figures and results section, but I would imagine is generally not as interesting to the less specialised reader. The apparent lack of impact of drug treatment on MOI is interesting and counterintuitive, though it is not really mentioned in the results or discussion sufficiently to allay my confusion. I would have been interested in understanding the relationship between MOI and FOI as generated by your queuing theory method and the model. It isn't clear to me why these more standard results are not presented, as I would imagine they are outputs of the model (though happy to stand corrected - it isn't entirely clear to me what the model is doing in this manuscript alone).

We thank the reviewer for this comment. Please refer to our response to Reviewer 2, comment (3), as we made changes in the revised manuscript regarding antimalarial drug treated individuals. We reported two sets of FOI estimates. In the first, we excluded these

treated individuals from the analysis as suggested by Reviewer 2. In the second, we discarded their infection status and MOI estimates and sampling from non-treated individuals.

The reviewer correctly noted the surprising lack of impact of antimalarial treatment on MOI estimates. This pattern is indeed interesting and counterintuitive. The impact of the drugs likely depends on their nature, efficiency, and duration. We note that treatment information was collected via a routine questionnaire, with participant self-reporting that they had received an antimalarial treatment in the previous two-weeks before the surveys (i.e., participants that reported they were sick, sought treatment, and were provided with an antimalarial treatment). No confirmation through hospital or clinic or pharmacy records was conducted, as it was beyond the scope of the study. Additionally, many of these sick individuals seek treatment at local chemists, which may limit the relevance of hospital or clinic records, if they are even available. Consequently, information on the nature, efficiency, and duration of administered drugs was incomplete or lacking. As this is not the focus of this work, we do not elaborate on the impact of drug treatment in the revised manuscript.

Regarding the last point of the reviewer, on understanding the relationship between MOI and FOI, we are not fully clear about what was meant. We are also confused about the statement on what the “model is doing in this manuscript alone”. We interpret the overall comment as the reviewer suggesting a better understanding of the relationship between MOI and FOI generated by the two-moment approximation method and the agent-based model. This could involve exploring the relationship between the moments of their distributions, possibly by fitting models such as simple linear regression models. Although this approach is in principle possible, it falls outside the focus of our work. Moreover, it would be challenging to evaluate the performance of this alternative approach given the lack of MOI-FOI pairs from empirical settings with directly measured FOI values (from large cohort studies). Nonetheless, we note that the qualitative relationship between the two quantities is intuitive. Higher FOI values should correspond to higher MOI values. Less variable FOI values should result in more narrow or concentrated MOI distributions, whereas more variable FOI values should lead to more spread-out MOI distributions. We described this qualitative relationship between MOI and FOI in the revised manuscript (Line 499-502).

As mentioned in the response to the reviewer’s previous point (1), we hope that our clarification of the basic assumptions underlying our agent-based model in Appendix 1-Simulation data helps the reviewer gain a better sense of the model. We appreciate agent-based models involve more assumptions and parameters than typical equation-based models in epidemiology, and their description can be difficult to follow. We have extended this description to rely less on previous publications. As for other ABMs, the population dynamics of the disease is followed over time by tracking individual hosts and strains. This allows us to implement specific immune memory to the large number of strains arising from the *var* multigene family. There is no equation-based formulation of the transmission dynamics that can incorporate immune memory in the presence of such large variation as well as recombination of the strains. We rely on this model because large strain diversity at high transmission underlies superinfection of individual hosts, and therefore, MOI values larger than one. We relied on the estimation of MOI with a method based on *var* gene sampling, and therefore, simulated such sampling for individual hosts (which requires an ABM and one that represents such genes and resulting strains explicitly).

(3) I would suggest that outside of malaria geneticists, the force of infection is considered to be the entomological inoculation rate, not the number of genetically distinct blood-stage strains. I appreciate that FOI has been used to explain the latter before by others, though the authors could avoid confusion by stating this clearly throughout the manuscript. For example, the abstract says FOI is "the number of new infections acquired by an individual host over a given time interval" which suggests the former, please consider clarifying.

We thank the reviewer for this helpful comment, as it is crucial to avoid any confusion regarding basic definitions. EIR, the entomological inoculation rate, is closely related to the FOI, force of infection, but they are not equivalent. EIR focuses on the rate of arrival of infectious bites and is measured as such by focusing on the mosquito vectors that are infectious and arrive to bite a given host. Not all these bites result in actual infection of the human host. Epidemiological models of malaria transmission clearly make this distinction, as FOI is defined as the rate at which a host acquires infection. This definition comes from more general models of the population dynamics of infectious diseases. For simpler diseases without super-infection, the typical SIR models define FOI as the rate at which a susceptible individual becomes infected. In the context of malaria, FOI refers to the number of new infections acquired by an individual host over a given time interval. This distinction between EIR and FOI is the reason why studies have investigated their relationship, with the nonlinearity of this relationship reflecting the complexity of the underlying biology and how host immunity influences the outcome of an infectious bite.

We added “blood-stage strains” to the definition of FOI in the previous manuscript, as pointed out by the reviewer, for the following reason. After an individual host acquires an infection/strain from an infectious mosquito bite, the strain undergoes a multi-stage life cycle within the host, including the liver stage and asexual blood stage. Liver-stage infections can fail to advance to the blood stage due to immunity or exceeding the blood-stage carrying capacity. Only active blood-stage infections are detectable in all direct measures of FOI. Quantities used in indirect model-fitting approaches for estimating FOI are also based on or reflect these blood-stage strains/infections. Only these blood-stage strains/infections are transmissible to other individuals, impacting disease dynamics. Ultimately, the FOI we seek to estimate is the one defined as specified above, as well as in both the previous and revised manuscripts, consistent with the epidemiological literature. We expanded on this point in the revised manuscript (Line 641-656).

(4) Line 319 says "Nevertheless, overall, our paired EIR (directly measured by the entomological team in Ghana (Tiedje et al., 2022)) and FOI values are reasonably consistent with the data points from previous studies, suggesting the robustness of our proposed methods". I would agree that the results are consistent, given that there is huge variation in Figure 4 despite the transformed scales, but I would not say this suggests a robustness of the method.

We thank the reviewer for this comment and have modified the relevant sentences to use “consistent” instead of “robust” (Line 229-231).

(5) The text is a little difficult to follow at times and sometimes requires multiple reads to understand. Greater precision is needed with the language in a few situations and some of the assumptions made in the modelling process are not referenced, making it unclear whether it is a true representation of the biology.

We thank the reviewer for this comment. As mentioned in the response to Reviewer 1 and in response to your previous points, we have shortened, reorganized and rewritten parts of the text in the revised manuscript to improve clarity and readability.

Reviewer #1 (Recommendations For The Authors):

Minor comments:

Bar graphs in Figures 6 and 7 are not an appropriate way to rigorously compare whether your estimated MOI (under different approaches) is comparable to your true MOIs. Particularly in Figure 6 it is very difficult to clearly compare what is going on. If

anything in Figure 7 it looks like as MOI gets higher, Bayesian methods and barcoding are overestimating relative to the truth. The large Excel file that shows KS statistics could be better summarized (and include p-values not in a separate table) and further discussion of how these methods perform on metrics other than the mean value would be important given that MOI distributions can be heavily right skewed and these high MOI values contain a large proportion of genetic diversity which can be highly informative for the purposes of this estimation.

We appreciate the reviewer's comment. It appears there may have been some misinterpretation of the pattern in Figure 7 in the previous manuscript. We believe the reviewer meant "as MOI gets higher, Bayesian methods and varcoding are UNDERESTIMATING relative to the truth" rather than "OVERESTIMATING".

We agree with the reviewer that the comparison of MOI distributions can be improved. To better quantify the difference between the MOI distribution from the original varcoding method and its Bayesian formulation relative to true MOIs, we replaced the KS test conducted in the previous manuscript with two alternative, more powerful tests: the Cramer-von Mises Test and the Anderson-Darling Test. The Cramer-von Mises Test quantifies the sum of the squared differences between the two cumulative distribution functions, while the Anderson-Darling Test, a modification of the Cramer-von Mises Test, gives more weight to the tails of the distribution, as noted by the reviewer. We have summarized the results, including test statistics and their associated p-values, in a supplementary table (Line 135-149, Line 862-883, supplementary file 1-MOImethodsPerformance.xlsx and supplementary file 7-BayesianImprovement.xlsx).

Throughout the text the authors use "consistent" to describe their estimation of FOI, I know this is meant in the colloquial use of the word but consider changing this word to replicable or something similar. When talking about estimators, usually, consistency implies asymptotic convergence in probability which we do not know whether the proposed estimator does.

We thank the reviewer for this suggestion. We changed "consistent" to "replicable" in the revised manuscript.

I think there is an issue with the numbering of the figures, they are just numbered continuously between the main text and appendix between 1 and 15, but in the text, there is a different numbering system between the main text and appendix figures.

We thank the reviewer for this comment. We have double-checked to ensure that the numbering of the figures is consistent with the text in the revised manuscript. Figures are numbered continuously between the main text and the appendix. When referring to these figures in the text, we provide a prefix (i.e., Appendix 1) indicating whether the figure is in the main text or Appendix 1, followed by the figure number.

The description of the bootstrap for 95% CI is a bit sparse, did bootstrap distributions look symmetric? If not did authors use a skewness adjustment to ensure good coverage? Also, is the bootstrap unit of resampling at the individual level, the simulation scenario level, population level?

We checked the bootstrap distributions and calculated their skewness. The majority fall within the range of -0.5 to 0.5, with a few exceptions falling within the range of 0.5-0.75 (supplementary file 6-FOIBootstrapSkewness.xlsx). We considered them as fairly symmetric and thus did not use a skewness adjustment.

In Figures 8 and 9 the x-axes seem to imply there are both the true and estimated MOI distributions on the plot but only 1 color of grey is clearly visible. If there are 2 distributions the color or size needs to be changed or if not consider re-labeling the x-axis.

We thank the reviewer for this comment. There was a mistake in the x-axis labels in Figure 8 and 9. Only the estimated MOI distributions were shown because the true ones are not available for the Ghana field surveys. The labels should simply be “Estimated MOIvar”.

Reviewer #2 (Recommendations For The Authors):

(1) Throughout the results section there are lots of vague statements such as "differ only slightly", "exhibit a somewhat larger, but still small, difference", etc. Please include the exact values and ranges within the text where appropriate because it can be difficult to discern from the figure.

We thank the reviewer for this useful comment. In the revised manuscript, we have provided exact values and ranges where appropriate (supplementary file 1- MOImethodsPerformance.xlsx, supplementary file 3- FOImethodsPerformance.xlsx, and supplementary file 7-BayesianImprovement.xlsx).

(2) Truncate decimals to 2 places.

We thank the reviewer for this comment. In the revised manuscript, we have truncated decimals to two places where applicable.

(3) The queueing theory notation in the methods section is unfamiliar, specifically things like "M/M/c/k", please define the variables used.

We thank the reviewer for this useful comment. In the revised manuscript, we have defined all the variables used. Please refer to our responses to Reviewer 1 Point (1) a.

Reviewer #3 (Recommendations For The Authors):

(1) The work takes many of the models and data from a previous paper published in eLife in 2023 (the 4 most senior authors of this previous manuscript are the 4 authors of the current manuscript). This previous paper introduced some new terminology "census population" which was highlighted as being potentially confusing by 2 of the 3 reviewers of the original article. This was somewhat rebuffed by the authors, though their response was ambiguous about whether the terminology would be changed in any potential future revision. The census population terminology does not appear in this manuscript, though the same data is being used. Publication of similar papers with the same data and different terminology could generate confusion, so I would encourage authors to be consistent and make sure the two papers are in line. To this end, it feels like this paper would be better suited to be classified as a "Research Advances" on this original manuscript and linked, which is a nice functionality that eLife offers.

We thank the reviewer for this comment, but we do not think our work would fall under the criteria of “Research Advances” based on our previous paper pointed out by the reviewer. The reviewer correctly noted that the current work and the previous paper used the same datasets. However, they have different goals and are not related in terms of content.

The previous paper examined how epidemiological quantities and diversity measurements of the local parasite population change following the initiation of effective control interventions

and subsequently as this control wanes. These quantities included MOI and census population size (MOI was estimated using the Bayesian formulation of the *varcoding* method, and the census population size was derived from summing *MOIvar* across individuals in the human population). In contrast, our current work focused on a different goal: inferring FOI based on MOI. We proposed two methods from queuing theory and illustrated them with MOI estimates obtained with the Bayesian formulation of the "*varcoding*" method. Although the method applied to estimate MOI is indeed the same as that of the paper mentioned by the reviewer, the proposed methods should be applicable to MOI estimates obtained in any other way, as stated in the Abstract in the previous manuscript. That is, the methods we present in the current paper are independent from the way the MOI estimation has been carried out. Our results are not about the MOI values themselves but rather on an illustration of the methods for converting those MOI values to FOI. In fact, there are different ways to obtain MOI estimates for *Plasmodium falciparum* (9). The most common approach for determining MOI involves size-polymorphic antigenic markers, such as *msh1*, *msh2*, *msh3*, *glurp*, *ama1*, and *csp*. Similarly, microsatellites, also termed simple sequence repeat (SSR), are another type of size-polymorphic marker that can be amplified to estimate MOI by determining the number of alleles detected. Combinations of genome-wide single nucleotide polymorphisms (SNPs) have also been used to estimate MOI.

The result section of the current manuscript begins by evaluating how different kinds of errors/sampling limitations affect the estimation of MOI using the Bayesian formulation of the *varcoding* method. Only that brief section, which is not the core or primary objective of the manuscript, could be considered an extension and an advancement related to the other paper. We considered the effect of these errors on the resulting estimates of FOI.

We further note that, as the reviewer pointed out, the census population size is not utilized at all in our current work. We are unclear on why this quantity is mentioned here. Our previous paper has been revised and can be found in eLife as such. We have not changed this terminology and have provided a clear explanation for why we chose it. The reviewer seems to have read the previous response to version 1 posted on December 28, 2023 (Note that version 2 and the associated response was posted on November 20, 2024). Regardless, this is not the place for a discussion on another paper on a quantity that is irrelevant to the current work being reviewed.

We understand that the reviewer's impression may have been influenced by the previous emphasis on the Bayesian formulation of the *varcoding* method in our manuscript. With the reorganization and rewriting of parts of the manuscript, we hope the revised version will clearly convey the central goal of our work.

(2) Similar statements that could be toned down. 344 "... two-moment approximation approach and Little's law are shown to provide consistent and good FOI estimates,.....", 374 "Thus, the flexibility and generality of these two proposed methods allow robust estimation of an important metric for malaria transmission"

We thank the reviewer for this comment. We have modified the descriptive terms for the performance of our methods. Please also refer to our responses to Reviewer 1, Point (1) c and your previous Point (1).

(3) Various assumptions seem to have been made which are not justified. For example, heterogeneous mixing is defined as 2/3rd of the population receives 90% of the bites. A reference for this would be good.

In this work, we considered heterogenous transmission arising from 2/3 of the population receiving approximately 94% of all bites, because we believe this distribution introduces a

reasonable and sufficient amount of heterogeneity in exposure risk across individuals. We are not aware of field studies justifying this degree of heterogeneity.

(4) The work assumes children under 5 have no immunity (Line 648 says "It is thus safe to consider negligible the impact of immune memory accumulated from previous infections on the duration of a current infection."). Is there supporting evidence for this and what would happen if this wasn't the case?

We thank the reviewer for this helpful comment. Please refer to our responses to Reviewer 1 Point (2) a.

(5) Similarly, there are a few instances of a need for more copy-editing. The text says "We continue with the result of the heterogeneous exposure risk scenarios in which a high-risk group (2/3 of the total population) receives around 94% of all bites whereas a low-risk group (1/3 of the total population) receives the remaining bites (Appendix 1-Figure 5C)." whereas the referenced caption says "For example, heterogeneous mixing is defined as 2/3rd of population receives 90% of the bites."

We believe there was a misinterpretation of the legend caption. In the referenced caption, we stated “2/3rd of population receives MORE THAN 90% of the bites”, which aligns with “around 94% of all bites”. Nonetheless, to maintain consistency in the revised manuscript, we have updated the description to uniformly state “approximately 94% of all bites” throughout.

(6) The term "measurement error" is used to describe the missing potential under-sampling of var genes. Given this would only go one way isn't the term "bias" more appropriate?

We understand that, in general English, “bias” might seem more precise for describing a deviation in one direction. However, in malaria epidemiology and in models for malaria and other infectious diseases, “measurement error” is a general term that describes deviations introduced in the process of measurement and sampling, which can confound or add noise to the true values being collected. This term is commonly used, and we have adhered to it in the revised manuscript.

(7) Line 739 "Though FOI and EIR both reflect transmission intensity, the former refers directly to detectable blood-stage infections whereas the latter concerns human-vector contact rates." In my mind this is not true, the EIR is the number of potentially invading parasites (a contact rate between parasites in mosquitoes and humans if you will). The human-vector contact rate is the human biting rate.

We thank the reviewer for this comment. We have clarified the definition regarding FOI and EIR in our response to your previous comment (3) and in the revised manuscript. We agree that the term “human-vector contact rates” was not precise enough for EIR. We intended “human-infectious vector contact rates”, and we have updated the text to reflect this change (Line 644-645).

References and Notes

- (1) Maire, N. et al. A model for natural immunity to asexual blood stages of *Plasmodium falciparum* malaria in endemic areas. *Am J Trop Med Hyg.*, 75(2 Suppl):19-31 (2006).
- (2) Tiedje, K. E. et al. Measuring changes in *Plasmodium falciparum* census population size in response to sequential malaria control interventions. *eLife*, 12 (2023).

- (3) Andrade C. M. et al. Infection length and host environment influence on *Plasmodium falciparum* dry season reservoir. *EMBO Mol Med.*,16(10):2349-2375 (2024).
- (4) Zhang X. and Deitsch K. W. The mystery of persistent, asymptomatic *Plasmodium falciparum* infections, *Current Opinion in Microbiology*, 70:102231 (2022).
- (5) Tran, T. M. et al. An Intensive Longitudinal Cohort Study of Malian Children and Adults Reveals No Evidence of Acquired Immunity to *Plasmodium falciparum* Infection, *Clinical Infectious Diseases*, 57(1):40–47 (2013).
- (6) Farnert, A., Snounou, G., Rooth, I., Bjorkman, A. Daily dynamics of *Plasmodium falciparum* subpopulations in asymptomatic children in a holoendemic area. *Am J Trop Med Hyg.*, 56(5):538-47 (1997).
- (7) Read, A. F. and Taylor, L. H. The Ecology of Genetically Diverse Infections, *Science*, 292:1099-1102 (2001).
- (8) Sondo, P. et al. Genetically diverse *Plasmodium falciparum* infections, within-host competition and symptomatic malaria in humans. *Sci Rep* 9(127) (2019).
- (9) Labbe, F. et al. Neutral vs. non-neutral genetic footprints of *Plasmodium falciparum* multiclonal infections. *PLoS Comput Biol*, 19(1) (2023).
- (10) He, Q. et al. Networks of genetic similarity reveal non-neutral processes shape strain structure in *Plasmodium falciparum*. *Nat Commun* 9(1817) (2018).

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